



## **Technical Documentation Version 9.6**

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# **Automation Tools**

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**Center for Advanced Decision Support  
for Water and Environmental Systems  
UNIVERSITY OF COLORADO BOULDER**

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# Chapter 1

## Script Management

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Scripts allow you to organize and run sequences of actions. This allows you to automate many of the tasks involved in operating a model. This section presents key Script features and provides some examples of how they might be used.

### Topics

- [About Script Management, page 1](#)
- [Script Management Uses, page 3](#)
- [Managing Scripts, page 3](#)
- [Editing a Script, page 10](#)
- [Action Type Settings Reference, page 36](#)
- [Script Dashboard, page 84](#)
- [Script Diagnostics, page 92](#)

## About Script Management

Like the model itself, a script is often created and configured once and then executed many times. In our discussion of Scripts, we use the term *modeler* to refer to the person who defines and configures the script and the term *user* for the person who runs the script.

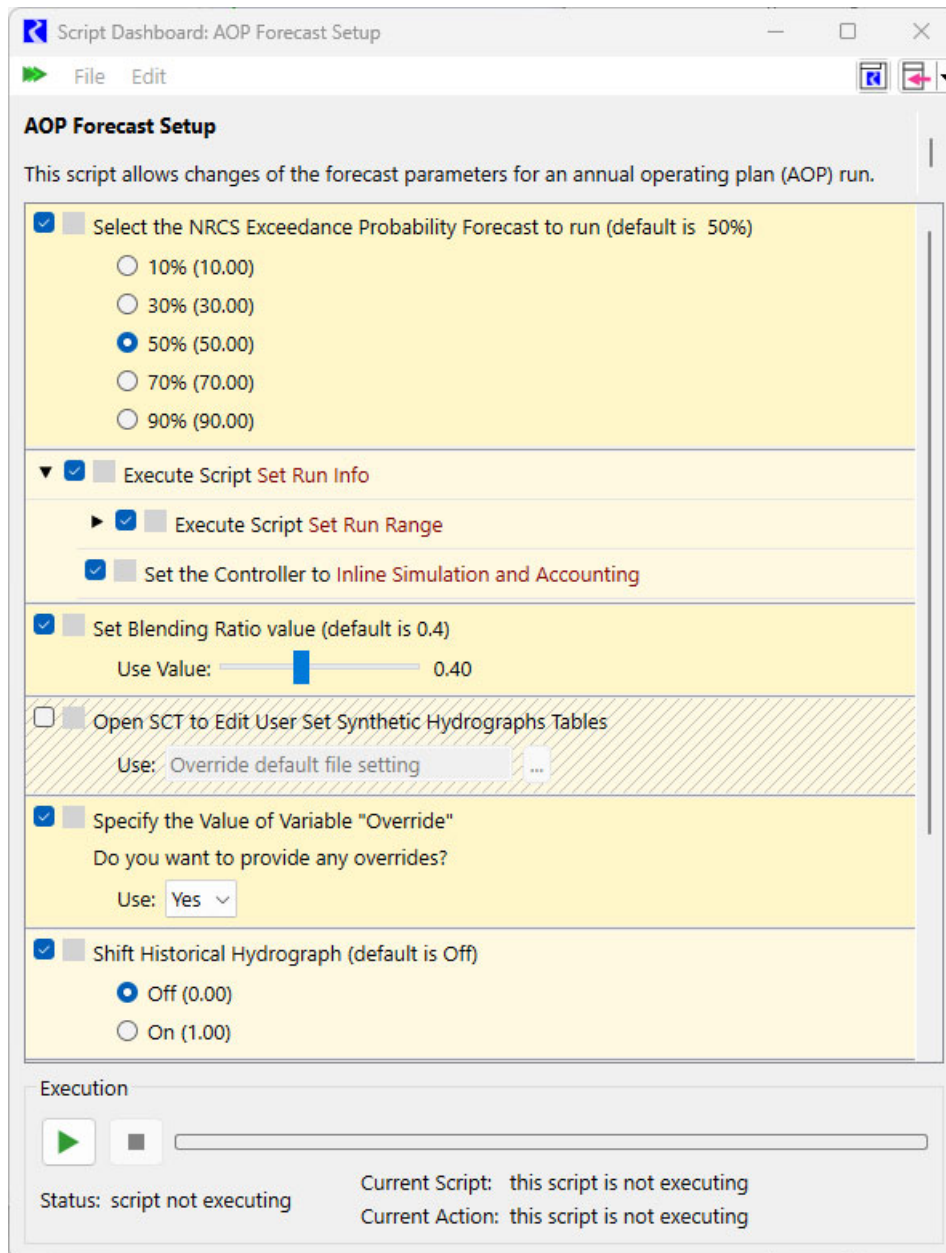
- The modeler creates scripts in the Script Manager and then configures each script in its Script Editor.
- Scripts are composed of a series of *Actions*, each of which represents a single step to be taken by the script, such as setting the run controller, setting a slot value, running a DMI or loading a ruleset.
- After an action has been added to a script in the Script Editor, the precise behavior of the action can be controlled by editing its settings.
- Any action that is not of relevant to the user can be configured as hidden by the modeler. Users can also temporarily hide actions that are not needed.
- The Script Dashboard lists the associated script's visible actions and allows the user to override key configuration settings before running the script. Convenient controls such as radio buttons, sliders, and toggles, are provided for manipulating configuration settings.
- Scripts are saved to the model file but can be exported/imported to other models.
- From both the Script Editor and Script Dashboard, right-click context menus allow you to directly open the slot or object dialogs when the action refers to slots or objects.

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### Script Management

- A script can execute another script (using the **Execute Script** action). Only five levels of a script calling another script is allowed.
- A script action can be dependent or conditional on another action using script variables.
- Scripts in a single Script Group can be automatically run when the model opens.
- Execute a script from batch mode using the following RCL command: **ExecuteScript <Script Name>**

Following is a sample script.



Script Dashboard: AOP Forecast Setup

File Edit

### AOP Forecast Setup

This script allows changes of the forecast parameters for an annual operating plan (AOP) run.

- ☒ Select the NRCS Exceedance Probability Forecast to run (default is 50%)
  - ☐ 10% (10.00)
  - ☐ 30% (30.00)
  - ☒ 50% (50.00)
  - ☐ 70% (70.00)
  - ☐ 90% (90.00)
- ☒ Execute Script Set Run Info
  - ☒ Execute Script Set Run Range
  - ☒ Set the Controller to Inline Simulation and Accounting
- ☒ Set Blending Ratio value (default is 0.4)  
Use Value:
- ☐ Open SCT to Edit User Set Synthetic Hydrographs Tables  
Use:
- ☒ Specify the Value of Variable "Override"  
Do you want to provide any overrides?  
Use:
- ☒ Shift Historical Hydrograph (default is Off)
  - ☒ Off (0.00)
  - ☐ On (1.00)

Execution

Status: script not executing      Current Script: this script is not executing  
Current Action: this script is not executing



# Script Management Uses

Now that you know some of the features of scripts, why might you use them? Following are some applications or use scenarios.

- Update a model to a new time range. For example, at the beginning of the year, advance the model's run range, import new data, and set desired operation triggers.
- Prepare a model for the current day's run. In one click of a button, clear yesterday's proposed operations, import observed data, load the necessary rulesets and goal sets so the model is ready to run.
- Change an after-the-fact accounting model into an operations or planning model. Run a script to change the controller, load an alternative ruleset, modify object methods, and set slot values.
- Provide the model to stakeholders who can use the Script Manager to specify their desired inputs. They then run a script that executes the model.

This document describes the Script Manager, Script Editor, and Script Dashboard including all of the action types and settings. See [“Script Dashboard”](#) on page 84 for details on running scripts.

## Managing Scripts

The Script Manager is used to create, export/import, and delete scripts.

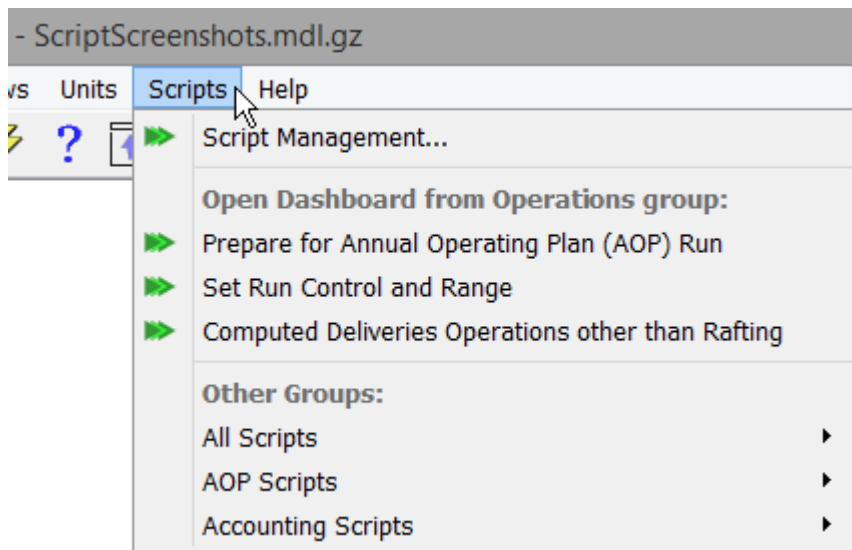
## Accessing the Script Manager

To access the Script Manager, from the workspace, use the **Scripts**, then **Script Management** menu.

The names of the defined scripts are presented in the Scripts menu. Those scripts in the default group are listed first followed by a cascading menu of the other groups. For more information on specifying the default group, see [“Tour of the Script Manager”](#) on page 4.

From the Scripts menu, select the name of a script to open the Script Dashboard.

**Figure 1.1** Screenshot of the Scripts menu.

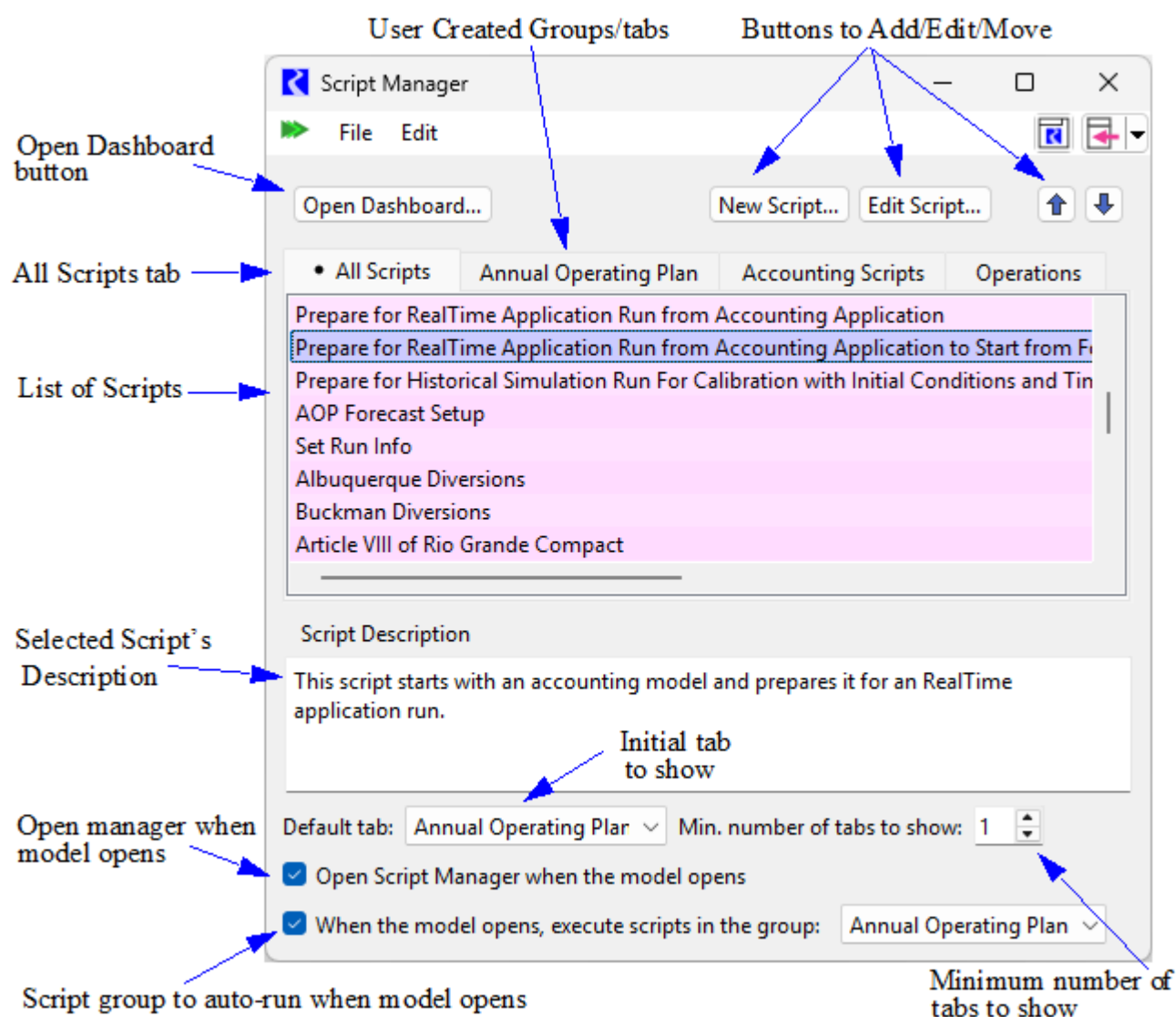


**Note:** In this screenshot, the “Operations” group is the default group for this model. There are three other groups as well listed in the Other Groups section the internal “All Scripts” group, “AOP Scripts” group, and “Accounting Scripts” group.

## Tour of the Script Manager

The Script Manager contains lists of the defined scripts in the model. These scripts are all shown on the **All Scripts** tab, as shown in [Figure 1.2](#). In addition, you can create your own tabs or groups; see [“Working With Script Groups”](#) on page 6 for details.

**Figure 1.2** Screenshot of the Script Manager



Within any of the tabs, use the **Edit Script** button to open the **Script Editor** for the selected scripts; see “[Editing a Script](#)” on page 10 for details. Select the **Open Dashboard** button or double-click a script to open the Script Dashboard for the selected scripts; see “[Script Dashboard](#)” on page 84 for details. Use the up and down arrows to reorder the selected scripts.

The **All Scripts** tab contains the complete list of scripts, and it is the only group from which new scripts can be created or deleted. The following options are available when on the All Scripts tab

- To Create a new Script, use the **New Script** button or the **Edit** and then **New Script** menu.
- To duplicate the selected scripts, use the **Edit** and then **Duplicate Script** menu. A copy will be created with the Name “Copy of <script>”. A number (for example “\_01” will be added to the end of the new script name if the name already exists.

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### Script Management

- To delete existing scripts, use the **Edit** and then **Delete Script** menu.

**Tip:** Use a right-click context menu on the list of scripts to perform many of the above operations on the selected scripts.

The lower portion of the Script Manager shows the last selected script's description. This allows you to quickly see the description to know what the script does or why you would want to use it. The descriptions are edited as part of the script; see [“Using the Script Editor”](#) on page 12 for details.

Finally, there are four independent controls at the bottom:

- Specify which script group tab is initially displayed by selecting a script from the **Default tab** menu. That tab is shown when the Script Manager window is initially opened (either upon model load or manually) within a RiverWare session.
- Specify the minimum number of tabs or groups to show when the Script Manager is first opened. The default is 1. Specify a large number to show all tabs/groups. Note, the Script Manager will never open wider than the current display.
- The Script Manager can be configured to automatically open when the model opens by checking the **Open Script Manager when the model opens** checkbox. This opens the script manager and makes it ready for the user to interact with the scripts.
- The Script Manager can be configured to automatically run the scripts in the specified group, when the model opens. Check the **When the model opens, execute scripts in the group** checkbox and specify the group.

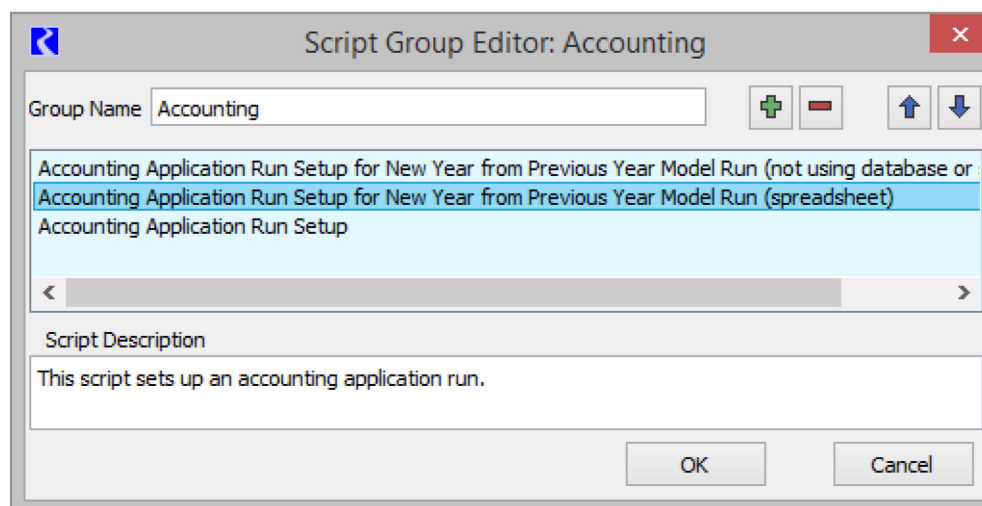
When the model finishes opening, the scripts in the designated script group run automatically in the order specified in the group, see [“Working With Script Groups”](#) on page 6. As each script runs, the script's dashboard opens and the progress indicators are updated. Pauses via the Memo action are supported in the script dashboard, or ignored if RiverWare is running from batch mode. In the event of script failure, a message box and diagnostic message indicate the failure and processing of the script group is halted.

Automatic execution can run a group of scripts that get your model ready for some usage. For example, the scripts could open and load a ruleset and an SCT, thus further automating this process when the model opens.

## Working With Script Groups

To create a user-defined script group, use the Script Manager's **Edit**, then **New Group** menu. To modify an existing user-defined group, select the tab and then use the **Edit**, then **Edit Group** menu item. Both of these menu items open the Script Group Editor, shown in [Figure 1.3](#).

Figure 1.3

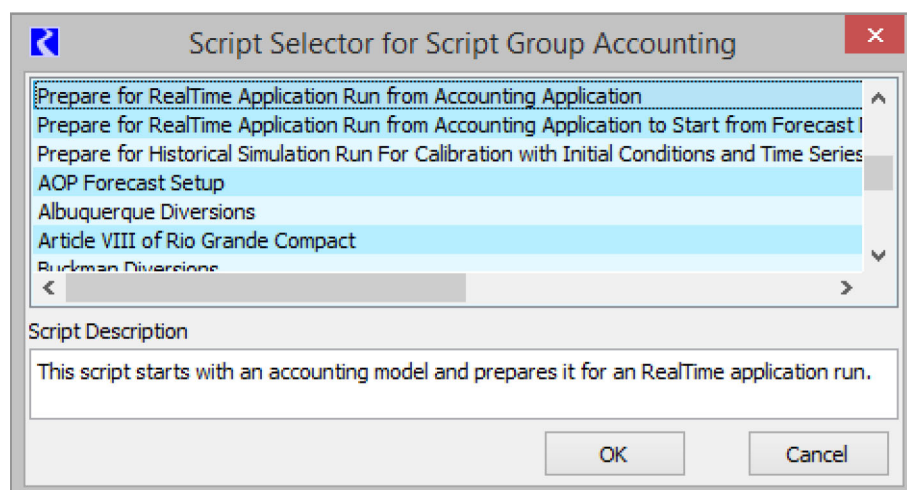


The Script Group Editor lists the scripts within the group and you can add, remove and reorder the scripts within the group.

To add a script to the group, select the green **Plus** button. This opens the Script Selector, which lists those scripts not already included in the script group. Select one or more scripts and select **OK** to add the scripts to the group.

Within the Script Manager, you can rearrange the tab groups by dragging a tab to a new position.

**Note:** A script group does not *own* the scripts listed as within the group. Instead, the group is simply a set of references to existing scripts. Thus, a script can be shared by multiple groups, and the effort of developing a new script need not be duplicated in order to immediately include and use a script in other groups.

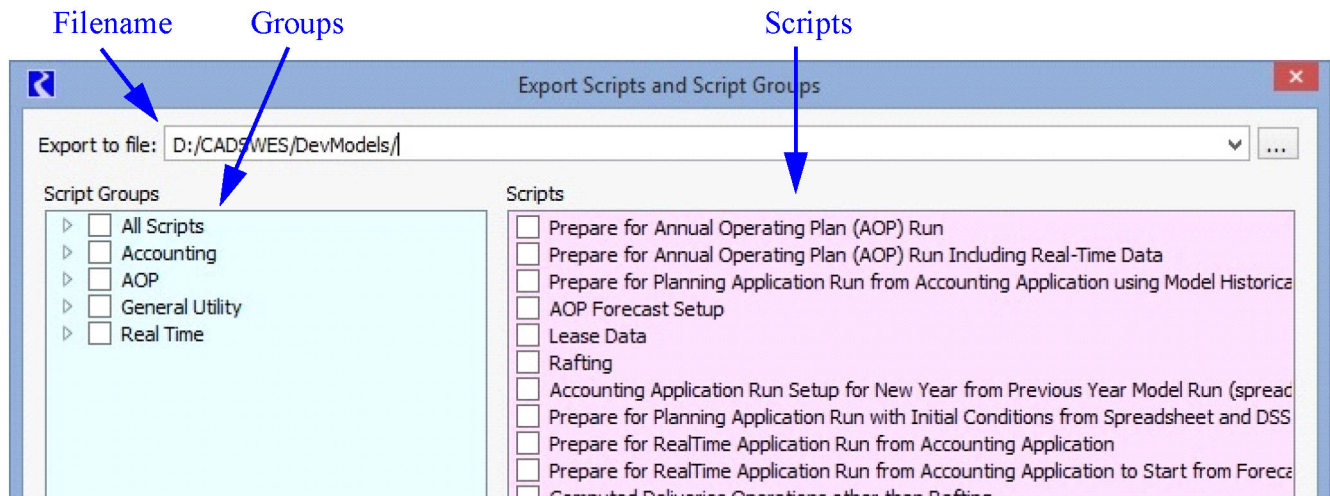


## Importing and Exporting Scripts

One or more Scripts or Script Groups can be exported from the model and then imported into another model. This section describes this process.

### Exporting Scripts

From the Script Manager, use the **File**, then **Export Selected Script** to export one or more scripts or groups to a file. This opens the Export Scripts and Script Groups.



First, specify a file for export. Either select from the menu, type in a file name or select the **More** button to choose a file on the system.

Then specify the scripts or groups to export.

The dialog contains two sections, one on the left, listing Script Groups and one on the right, listing all Scripts. The lower panel shows the description of the highlighted (not checked) script.

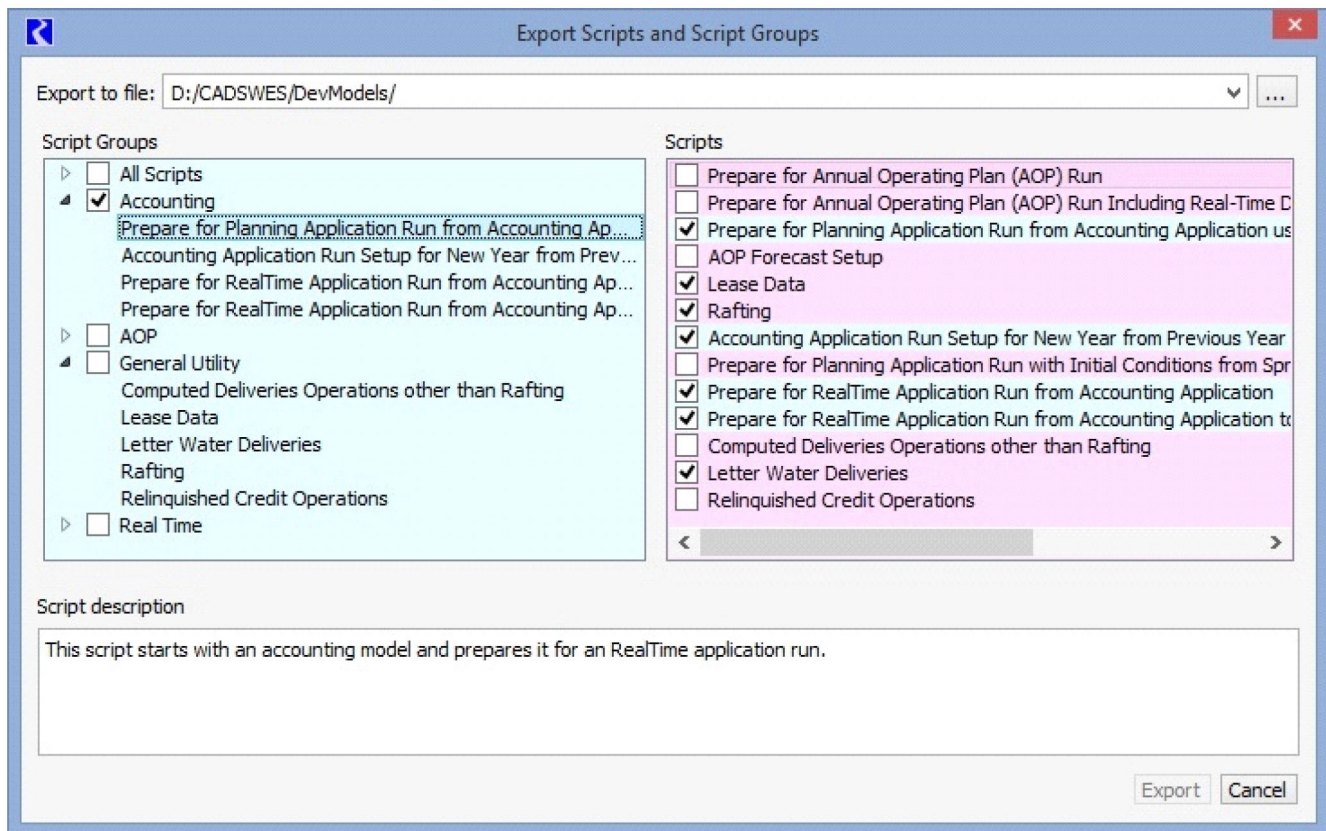
To export one or more Script Groups and the scripts referenced by the group, select the checkboxes in the left panel. All constituent scripts will be exported. Use the tree view arrows to see which scripts are part of the group. Also, as you select a group, the referenced scripts will also be checked and the background color will change in the Scripts panel on the right, as shown in [Figure 1.4](#).

**Note:** You cannot deselect scripts that have been selected for export that are part of a group. The following screenshot shows that one group containing four scripts (blue) will be exported as well as three additional scripts (pink).

**Note:** To export all of the scripts in the model, simply select the **All Scripts** checkbox.

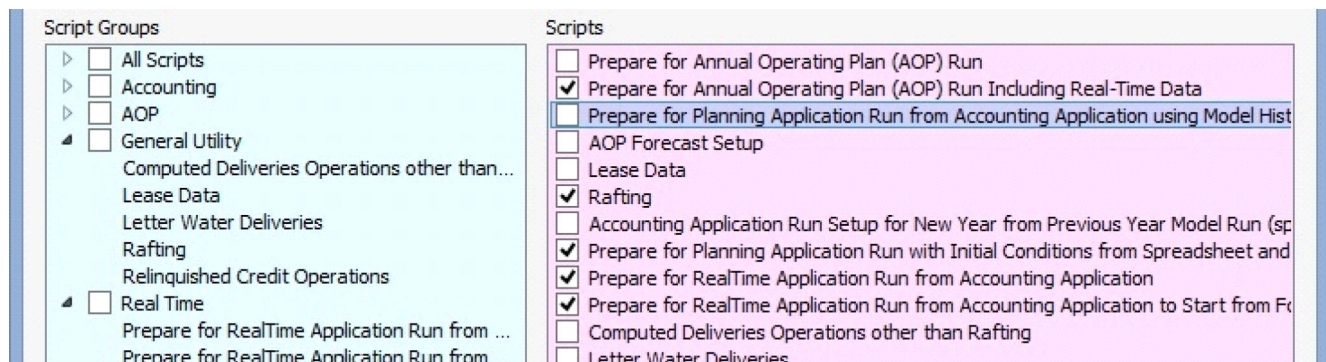


**Figure 1.4**



To select one or more scripts to export that are not part of a group, select them from the right panel. In [Figure 1.5](#), five scripts will be exported.

**Figure 1.5**



Once you have checked the desired groups/scripts, make sure to specify a file name at the top of the dialog. Then select **Export** to export the file.

## Importing Scripts

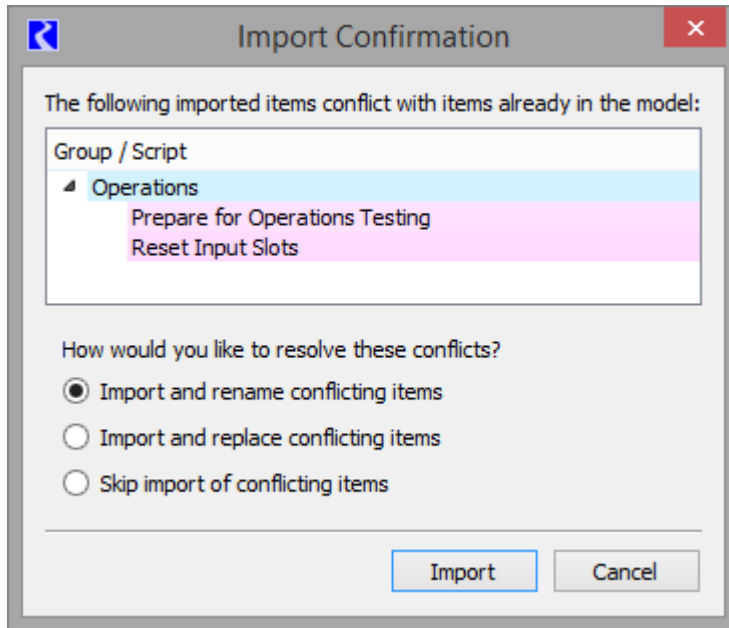
Use the **File**, then **Import Script** to import previously exported scripts. Choose the file name and select **Open**.

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### Script Management

If there are no conflicts between imported items and existing items, then the items are imported and the Script Manager is shown.

If there is a conflict between the names of imported scripts or groups, you are presented with the following dialog to specify what to do.



The top panel shows the conflicting items. The bottom portion asks how to handle those conflicts. Select either:

- Import and rename conflicting items: import the items and rename with a unique name (the default). The newly imported items are renamed by appending "\_NN" to the existing name, where NN is a two digit unique integer.
- Import and replace conflicting items: import the items and replace the items in the model.
- Skip Import of conflicting items: skip the conflicting items but import any new items.

Tool tips on the radio buttons gives more information.

Select **Import** to bring in the items. After the import has completed, a notification dialog informs you of the actions taken during the import. The possible messages are:

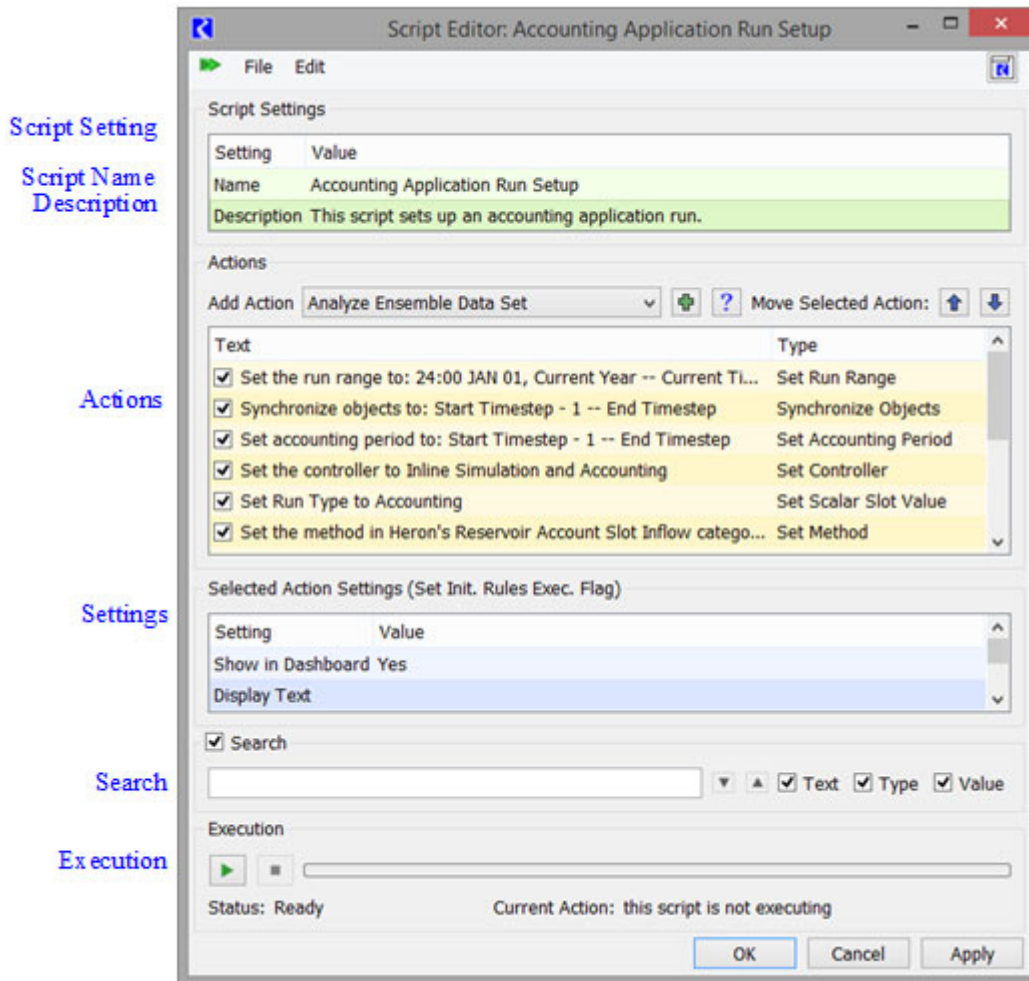
- Renamed to
- Replaced the script in the model
- Skipped

## Editing a Script

The Script Editor is where you define the Actions that the script performs. In addition, you define how the script will appear in the Script Dashboard and what values can be edited by the script user. To open the Script Editor, select a script in the Script Manager and select **Edit**. [Figure 1.6](#) identifies the sections of the Script Editor.



**Figure 1.6** Screenshot of the Script Editor



## Script Settings

This set of items applies to the entire script and includes the following configuration:

- Script Name. The unique name of the script appears here and in the list of scripts in the Script Manager.
- Description Editor. Enter a description, if desired, that will appear in the Script Manager.

## Actions panel

This is an ordered sequence of actions, each of which has one of the defined action types. Each item has either a default or user defined label and its type. E.g., Text: “Set the run range to:” Type: “Set Run Range”. Each action can be disabled (or enabled) using the checkbox.

Select one or more action from the panel using drag, Ctrl+Click and Shift+Click. Select multiple actions to move, copy, or delete multiple actions. Select a single action to edit the settings.

## Settings panel for the Selected Action

Each action type has two or more associated parameters, each of which has an associated setting. For example, an action of type Set Method has two standard parameters, Show in Manager and Display Text, and three specific setting, Object Name, Category, and Method. These are configured in the settings panel for the selected action.

**Note:** If you select more than one action, then you can't edit the settings. Only one action's settings may be edited at a time.

## Search

Search for actions that match the specified text. Select which action attributes (Text, Type, and/or Value of settings) in which to search.

## Execution panel

When creating and debugging your script, it may be more convenient to run the script from here than opening the Script Dashboard. Use the **Start** button to execute the script.

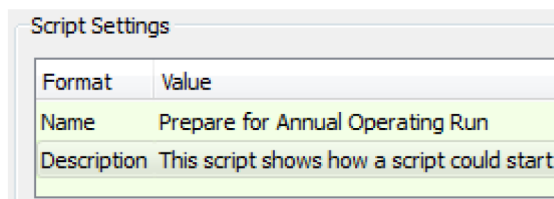
**Tip:** Alternative, use the context menus to execute a single action, the script up through that action or the script from that action onward. See [“Executing Actions From the Script Editor”](#) on page 16 for details.

The ability to run the script from the Editor is strictly for convenience while creating and editing the script. The interface and feedback for running a script is much better on the Script Dashboard; see [“Script Dashboard”](#) on page 84 for details.

# Using the Script Editor

This section describes how to use the Script Editor to define the script. This assumes you have created a new Script from the Script Manager. The Script Editor opens automatically for new scripts or select **Edit** to open an existing script.

1. Enter a meaningful name in the Name field.
2. Enter a description if desired.

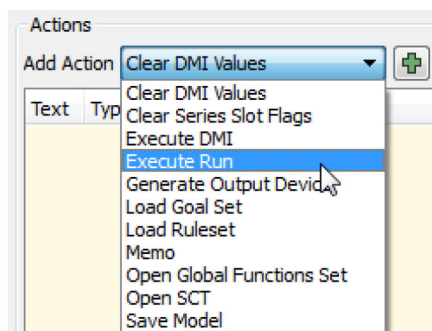


The screenshot shows a 'Script Settings' panel with a table containing two rows of settings. The first row is 'Name' with the value 'Prepare for Annual Operating Run'. The second row is 'Description' with the value 'This script shows how a script could start'.

| Format      | Value                                      |
|-------------|--------------------------------------------|
| Name        | Prepare for Annual Operating Run           |
| Description | This script shows how a script could start |

3. Add an Action to the script; see [“About Action Types”](#) on page 16 for a complete list of the available actions.

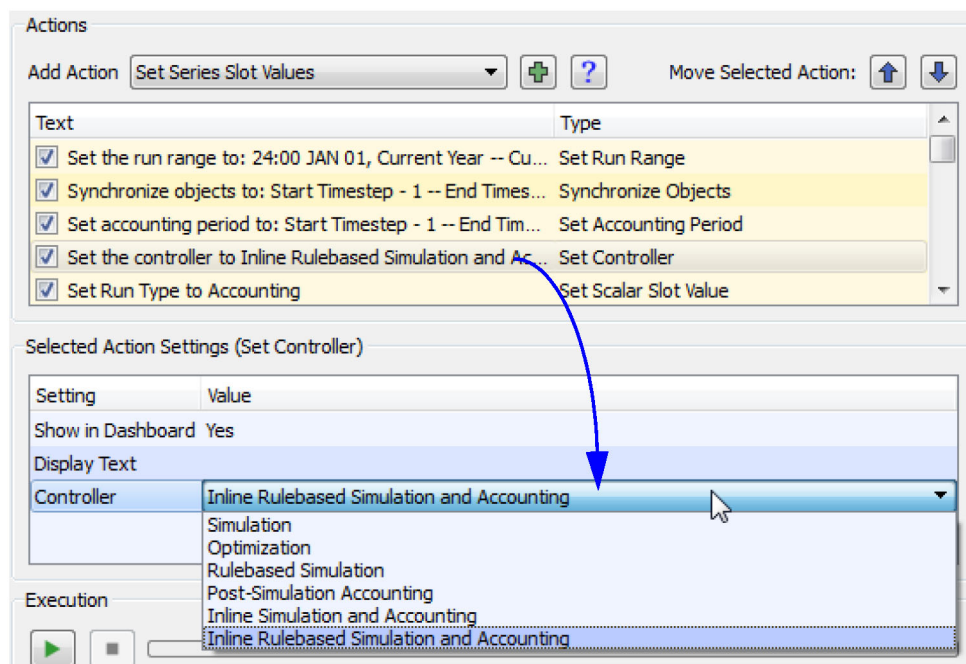
Select **Help**  for a quick description of each action. Use the **Add Action** menu and select the desired action. When you select the action it will automatically add it to the script and select it.



**Tip:** Copy and Paste of actions is available from the Edit menu, context menu, or using the Ctrl-C and Ctrl-V keyboard accelerators. See [“Copy and Paste of Actions”](#) on page 14.

4. Configure that action. With the single action selected, the Settings panel shows the available parameters. For example, in [Figure 1.7](#), the Set Controller action is selected, so now the controller can be chosen from the list in the settings panel.

**Figure 1.7**

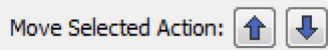


Each action has two or more settings that can be configured. For many of the settings, you provide a numeric value but can also specify that it is editable in the Script Manager. For example, you may use the Set Scalar Slot Value action to set a model trigger. You can specify that the value should default to 0 in the script but allow the

## Chapter 1 Script Management

user to override the default and use a slider bar in the Script Manager to change the value. See [“Configuring User Control on the Script Dashboard”](#) on page 27 for details on this type of configuration.

- Repeat [Step 3.](#) and [Step 4.](#) until you have fully defined the script.



- Use the arrows to rearrange the actions as necessary. They will execute in order from top to bottom.
- Once you have the script defined, consider what text to be show on the Script Manager. Use the **Show in Manager** and **Display Text**, to modify whether the setting is shown and if so, the text that is displayed, respectively. See [“Common Settings for Action Types”](#) on page 21 for details. In addition, you can disable actions using the check boxes. This is enabled/disabled setting is shown on the dashboard but can be re-enabled there too. See [“Common Settings for Action Types”](#) on page 21 for details.
- Select **OK** to apply the changes and close the dialog.

## Copy and Paste of Actions

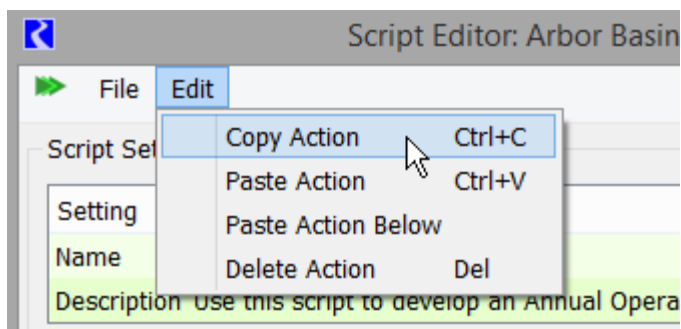
You can copy one or more actions from a script and paste them in any other script. Standard multi-select controls allow you to select multiple actions: Drag or Shift+Click to select a contiguous block of actions, and Ctrl+Click to select non-contiguous actions.

With one or more script actions selected in the Script Editor, use the following operations in the Script Editor in the actions panel:

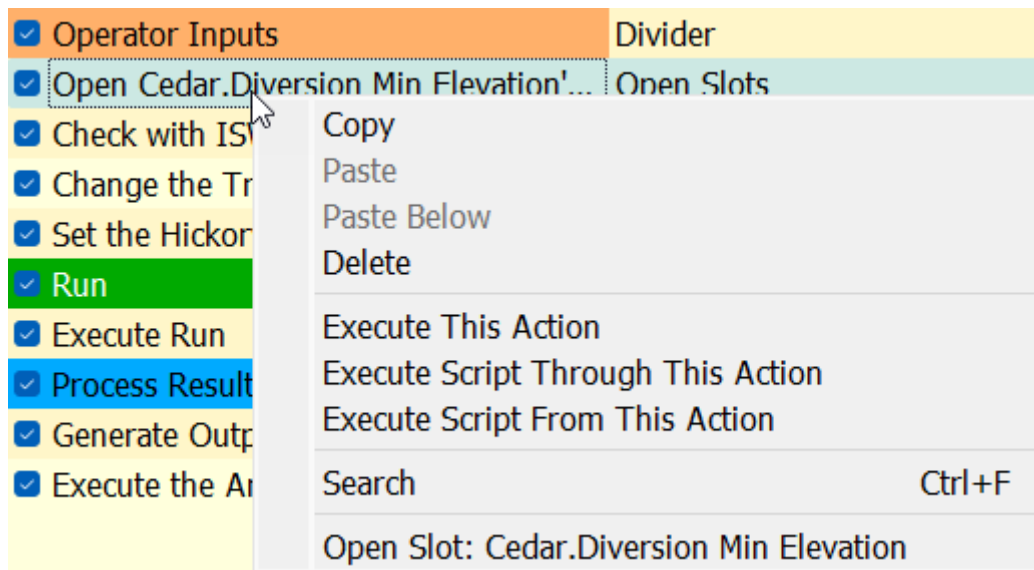
- Copy Action: Copy the selected action.
- Paste Action: Paste the copied action over the selected action(s) with a confirmation. Pasting over multiple actions only works with a contiguous selection.
- Paste Action Below: Paste the copied action below the last selected action.

To access these operations, use the **Edit** menu as illustrated in [Figure 1.8](#) or from the context menu for the action list as shown in [Figure 1.9](#). Keyboard accelerators are also available: Copy: Ctrl+C and Paste: Ctrl+V. The context menu also gives access to slots referenced by the action, where applicable.

**Figure 1.8** Script Editor Edit menu operations



**Figure 1.9** Script Editor context operations



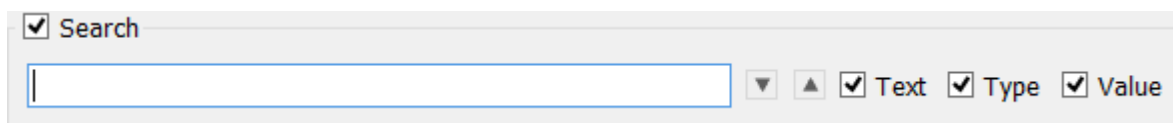
Also, on the script dashboard, the right-click context menu provides access to copy an action. These can then be pasted into a Script Editor as described above

**Tip:** To copy an entire script, from the Script Manager, use the **Edit** and then **Duplicate Script** operation.

## Searching for Actions

Use the Search panel to find an action by text, type, or value in the script editor. This can be very useful if you have a script that issues an error as the message says the type of action and its name. Use the search to find that action in the editor. To open the **Search** panel, use the **Search** check box, use **Edit** menu and then **Search** or use the Ctrl+F keyboard shortcut.

**Figure 1.10** Screenshot of Script Editor Search Panel



Then, type in the desired text and select the attributes in which to search. The attributes correspond to columns in the Script Editor's action panel and settings panel:

- **Text:** Search in the action **Text** column. The default text or any specified display text is searched.
- **Type:** Search in the action **Type** column.
- **Value:** Search in the **Value** column for each setting. This is particularly useful as you can search for slots in slot selections, objects in object selections, datetimes or any other value in the settings.

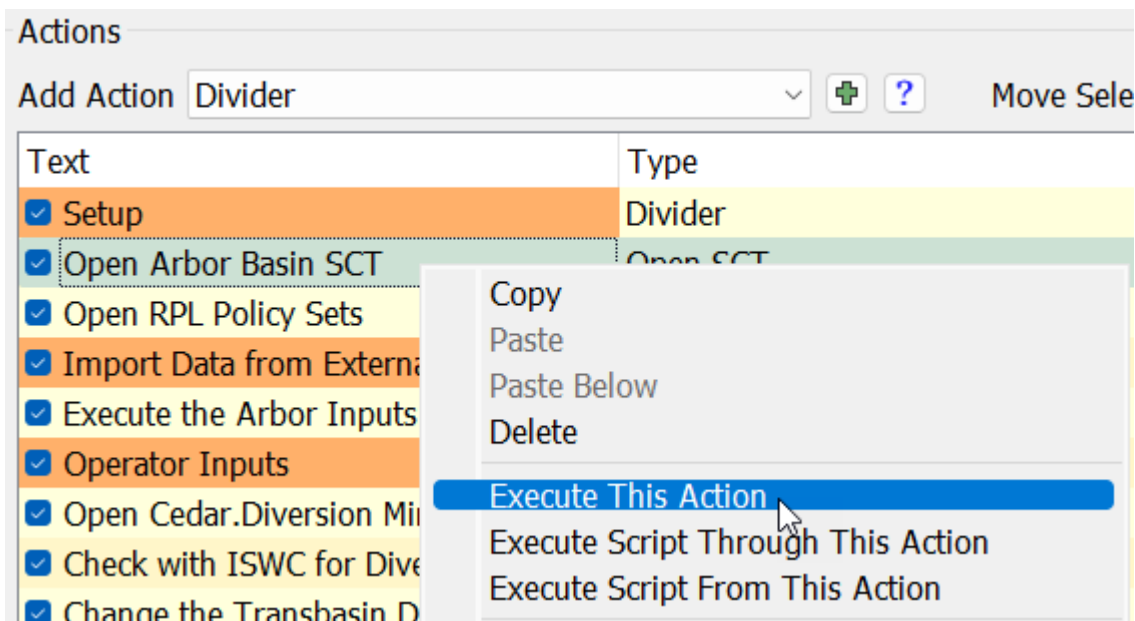
To perform the search, select Enter or use the up or down arrows. When a matching action is found, that action is selected in the Action panel. The search will wrap when it meets the end of the script.

**Tip:** This tool finds actions within scripts. You can also search for scripts that perform actions on a specified slots using the **Find Slot References** tool. See [“Find Slot References Dialog” in User Interface](#).

## Executing Actions From the Script Editor

When creating and debugging your script, it may be convenient to run the script from the editor than opening the Script Dashboard. There are three ways to execute actions from the editor:

- Use the **Start** button to execute the entire script. 
- Use the **Execute This Action** context menus to execute a single action.
- Use the **Execute Through This Action** context menu to start the script at the beginning and run up to and including this action.
- Use the **Execute Script From This Action** context menu to execute a single action and the rest of the script.



These operations execute the actions but do not give status or display information on progress. As a result, running actions is best performed from the Script Dashboard; see [“Script Dashboard”](#) on page 84 for details.

## About Action Types

[Table 1.1](#) describes the types of actions. Within the Script Editor, select **Help** for a quick reference of the actions. Each action type also has a link to additional information on its settings. See [“Action Type Settings Reference”](#) on page 36 for details about the settings.

**Table 1.1** Descriptions of the available actions

| Action Type                    | Description                                                                                                                                                                                                           | Link                                                     |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Activate Unit Scheme           | Activate the specified Unit Scheme.                                                                                                                                                                                   | <a href="#">Activate Unit Scheme</a> , page 37           |
| Analyze Ensemble Data Set      | Compute the specified statistic for an open ensemble data set. Statistics are computed across traces, resulting in values for each slot and date/time in the data set.                                                | <a href="#">Analyze Ensemble Data Set</a> , page 37      |
| Apply Window Layout            | Apply a window layout created with the Window Manager.                                                                                                                                                                | <a href="#">Apply Window Layout</a> , page 41            |
| Clear DMI Values               | Clear the values set by a DMI (by using the DMI invocation manager functionality)                                                                                                                                     | <a href="#">Clear DMI Values</a> , page 41               |
| Clear Scalar Slot Values       | Clear the values on a set of Scalar Slots (i.e. set them to NaN).                                                                                                                                                     | <a href="#">Clear Scalar Slot Values</a> , page 41       |
| Clear Table Slot Values        | Clear one or more values on a set of Table slots (i.e. set them to NaN).                                                                                                                                              | <a href="#">Clear Table Slot Values</a> , page 42        |
| Close RPL Set                  | Close the selected RPL set from the model. An RBS Ruleset or Optimization Goal set is unloaded and closed, a Global Function Set is closed, and the Initialization Rules Set is cleared.                              | <a href="#">Close RPL Set</a> , page 42                  |
| Compute Duration Curve         | Compute a duration curve for specified slots for an open ensemble data set across traces and timesteps.                                                                                                               | <a href="#">Compute Duration Curve</a> , page 43         |
| Compute Scalar Regression      | Compute a multiple linear regression for an open ensemble data set. The regression is computed across traces, resulting in a set of regression coefficients and statistical outputs regarding the regression quality. | <a href="#">Compute Scalar Regression</a> , page 44      |
| Compute Series Regression      | Compute a multiple linear regression for series data set. The regression is computed using time series data, resulting in a set of regression coefficients and statistical outputs regarding the regression quality.  | <a href="#">Compute Series Regression</a> , page 48      |
| Configure Excel Dataset        | For the specified Database DMI Excel Dataset, configure various aspects including the workbook, run name type, run name, and run suffix.                                                                              | <a href="#">Configure Excel Dataset</a> , page 51        |
| Configure MRM Index Sequential | For the specified MRM configuration, modify the configuration in terms of Index Sequential.                                                                                                                           | <a href="#">Configure MRM Index Sequential</a> , page 52 |
| Configure MRM Input DMI        | For the specified MRM configuration, modify the configuration in terms of Input DMIs, particularly the repeat count.                                                                                                  | <a href="#">Configure MRM Input DMI</a> , page 53        |
| Configure MRM Output           | For the specified MRM configuration, modify the configuration in terms of the Output.                                                                                                                                 | <a href="#">Configure MRM Output</a> , page 53           |
| Configure MRM Traces           | For the specified MRM configuration, modify the first trace and number of traces.                                                                                                                                     | <a href="#">Configure MRM Traces</a> , page 53           |
| Configure Trace Directory DMI  | For the specified Trace Directory DMI, specify the Trace Number to use if executed outside of MRM.                                                                                                                    | <a href="#">Configure Trace Directory DMI</a> , page 54  |
| Create Ensemble Data Set       | Read an RDF file and add it to the list of data sets managed by the Data Analysis Tool.                                                                                                                               | <a href="#">Create Ensemble Data Set</a> , page 54       |
| Create Slot Cache              | Create a slot cache of workspace series slot values.                                                                                                                                                                  | <a href="#">Create Slot Cache</a> , page 54              |



## Chapter 1

### Script Management

**Table 1.1** (Continued) Descriptions of the available actions

| Action Type                        | Description                                                                                                                                                           | Link                                                         |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Create Snapshot                    | Create a Snapshot of the specified slots.                                                                                                                             | <a href="#">Create Snapshot</a> , page 54                    |
| Delete Custom Slots                | Delete the selected custom slots.                                                                                                                                     | <a href="#">Delete Custom Slots</a> , page 55                |
| Delete Ensemble Data Set           | Remove an ensemble data set, freeing up the memory required to represent those data.                                                                                  | <a href="#">Delete Ensemble Data Set</a> , page 55           |
| Delete Link                        | Delete a link between two slots.                                                                                                                                      | <a href="#">Delete Link</a> , page 56                        |
| Delete Objects                     | Delete the selected objects.                                                                                                                                          | <a href="#">Delete Objects</a> , page 56                     |
| Delete Optimization Restore Points | Delete all optimization restore points currently in existence                                                                                                         | <a href="#">Delete Optimization Restore Points</a> , page 56 |
| Delete Snapshots                   | Delete all snapshot objects older than a specified number of days relative to the current system time                                                                 | <a href="#">Delete Snapshots</a> , page 56                   |
| Divider                            | Create a different colored row in the Script Editor and Script Dashboard to visually separate actions.                                                                | <a href="#">Divider</a> , page 57                            |
| Enable Dispatching                 | Enable or disable dispatching of the specified objects.                                                                                                               | <a href="#">Enable Dispatching</a> , page 57                 |
| Enable MRM Distributed Runs        | For the specified MRM configuration, specify whether to distribute concurrent runs to multiple processors on the same machine.                                        | <a href="#">Enable MRM Distributed Runs</a> , page 57        |
| Enable RPL Item                    | Enable or disable a RPL item (policy group, rule, method, or goal)                                                                                                    | <a href="#">Enable RPL Item</a> , page 57                    |
| Evaluate Expression Slots          | Evaluate a set of expression slots.                                                                                                                                   | <a href="#">Evaluate Expression Slots</a> , page 58          |
| Execute DMI                        | Invoke a DMI or DMI group.                                                                                                                                            | <a href="#">Execute DMI</a> , page 58                        |
| Execute MRM Run                    | Execute the runs for the specified MRM configuration.                                                                                                                 | <a href="#">Execute MRM Run</a> , page 58                    |
| Execute Run                        | Run the model.                                                                                                                                                        | <a href="#">Execute Run</a> , page 59                        |
| Execute Script                     | Execute a script from within a script.                                                                                                                                | <a href="#">Execute Script</a> , page 60                     |
| Export Database DMI                | Export Database Data Management Interfaces (DMIs) to a file and optionally include name maps and datasets.                                                            | <a href="#">Export DMIs</a> , page 60                        |
| Export Object                      | Export Objects to a file and optionally include links, object attributes, slot notes and accounting items. Note, exporting clusters is not supported within a script. | <a href="#">Export Objects</a> , page 60                     |
| Export Optimization Restore Point  | Export the specified optimization restore point to a file                                                                                                             | <a href="#">Export Optimization Restore Point</a> , page 61  |
| Export Table Slot as CSV           | Export the selected table slot to a comma-separated values (CSV) file.                                                                                                | <a href="#">Export Table Slot as CSV</a> , page 61           |
| Generate Output Device             | Generate the specified Output Device. This could be a file based device like a Model Report or a graphical device like a Plot Page.                                   | <a href="#">Generate Output Device</a> , page 62             |
| Global Time Scroll                 | Scroll time and series displays to the specified symbolic date/time.                                                                                                  | <a href="#">Global Time Scroll</a> , page 62                 |
| Import DMI                         | Import Data Management Interfaces (DMIs) from a file and specify how to resolve conflicts.                                                                            | <a href="#">Import DMIs</a> , page 62                        |



**Table 1.1** (Continued) Descriptions of the available actions

| Action Type                                | Description                                                                                                                                                                                                                                                                                                 | Link                                                                 |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Import Object                              | Import Objects from a file and specify how to import object attributes, slot notes and object clusters, and how to locate the imported objects.                                                                                                                                                             | <a href="#">Import Objects</a> , page 62                             |
| Import Optimization Restore Point          | Import an optimization restore point from the specified file                                                                                                                                                                                                                                                | <a href="#">Import Optimization Restore Point</a> , page 63          |
| Link Slots                                 | Create a link between two slots.                                                                                                                                                                                                                                                                            | <a href="#">Link Slots</a> , page 63                                 |
| Load Goal Set                              | Open and load a goal set (for Optimization runs).                                                                                                                                                                                                                                                           | <a href="#">Load Goal Set</a> , page 64                              |
| Load Ruleset                               | Open and load a ruleset (for Rulebased Simulation runs)                                                                                                                                                                                                                                                     | <a href="#">Load Ruleset</a> , page 64                               |
| Memo                                       | Pause script execution and optionally open a confirmation dialog with your user specified message. While script execution is paused, you can interact with RiverWare or other applications in ways not directly supported by scripts. Select the script <b>Start</b> button to resume the script execution. | <a href="#">Memo</a> , page 64                                       |
| Move Slots to Object                       | Move custom slots on one object to another object.                                                                                                                                                                                                                                                          | <a href="#">Move Slots to Object</a> , page 65                       |
| Open Data Analysis Tool                    | Open the Data Analysis Tool and bring it to the top layer of windows.                                                                                                                                                                                                                                       | <a href="#">Open Data Analysis Tool</a> , page 65                    |
| Open File                                  | Open the specified external file, for example an Excel file or Word document.                                                                                                                                                                                                                               | <a href="#">Open File</a> , page 65                                  |
| Open Global Functions Set                  | Open a global function set.                                                                                                                                                                                                                                                                                 | <a href="#">Open Global Functions Set</a> , page 66                  |
| Open Object                                | Open the specified objects.                                                                                                                                                                                                                                                                                 | <a href="#">Open Objects</a> , page 66                               |
| Open SCT                                   | Open an SCT with the specified filename.                                                                                                                                                                                                                                                                    | <a href="#">Open SCT</a> , page 66                                   |
| Open Slots                                 | Open the specified slots.                                                                                                                                                                                                                                                                                   | <a href="#">Open Slots</a> , page 66                                 |
| Reorder RPL Set                            | Reorder the items (rules, goals, statements) in a RPL set according to priority or index defined on the specified table slot.                                                                                                                                                                               | <a href="#">Reorder RPL Set</a> , page 66                            |
| Replace Initialization Rules Set from File | Replace the existing Initialization Rules Set from a file.                                                                                                                                                                                                                                                  | <a href="#">Replace Initialization Rules Set From File</a> , page 67 |
| Save Model                                 | Save the model to a file.                                                                                                                                                                                                                                                                                   | <a href="#">Save Model</a> , page 67                                 |
| Set Account Method                         | Set the selected method in one of an account's method categories.                                                                                                                                                                                                                                           | <a href="#">Set Account Method</a> , page 67                         |
| Set Accounting Period                      | Set the time range of the accounting system (i.e., the Begin Accounting Period and End Accounting Period dates).                                                                                                                                                                                            | <a href="#">Set Accounting Period</a> , page 67                      |
| Set Controller                             | Set the current controller.                                                                                                                                                                                                                                                                                 | <a href="#">Set Controller</a> , page 67                             |
| Set Dataset File                           | Set the DSS file for a DSS dataset, or the Excel workbook for an Excel dataset.                                                                                                                                                                                                                             | <a href="#">Set Dataset File</a> , page 68                           |
| Set Evaluation Time for Expression Slots   | Set the time at which the selected expression slots will evaluate. Options include beginning of run, end of run, beginning of timestep, and end of timestep.                                                                                                                                                | <a href="#">Set Evaluation Time for Expression Slots</a> , page 68   |

## Chapter 1

### Script Management

**Table 1.1** (Continued) Descriptions of the available actions

| Action Type                         | Description                                                                                                                                              | Link                                                           |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Set Excel Dataset Run Name          | For the specified Database DMI Excel Dataset, set the run name type and/or specify a new single run name.                                                | <a href="#">Set Excel Dataset Run Name</a> , page 68           |
| Set Init. Rules Exec. Flag          | Set the “Execute Initialization Rules” run control flag. This flag controls whether or not the initialization rules are executed as part of each run.    | <a href="#">Set Init. Rules Exec. Flag</a> , page 69           |
| Set Method                          | Set the selected method in the specified category for a set of objects.                                                                                  | <a href="#">Set Method</a> , page 69                           |
| Set MRM Descriptor                  | For the specified MRM configuration, set the specified keyword, value pair MRM Descriptor.                                                               | <a href="#">Set MRM Descriptor</a> , page 69                   |
| Set MRM Ruleset                     | In a given MRM configuration, set a new file path to a ruleset.                                                                                          | <a href="#">Set MRM Ruleset</a> , page 70                      |
| Set MRM Run Range                   | Set the time range of the run (i.e. the Run Start and End dates) for the specified MRM configuration.                                                    | <a href="#">Set MRM Run Range</a> , page 70                    |
| Set Optimization Parameter          | Set the value of any integer or real number optimization parameter                                                                                       | <a href="#">Set Optimization Parameter</a> , page 70           |
| Set Periodic Slot Values            | Set one or more Periodic Slot’s value at the specified row(s) and column(s).                                                                             | <a href="#">Set Periodic Slot Value</a> , page 71              |
| Set Run Range                       | Set the time range of the run (i.e. the Run Start and End dates).                                                                                        | <a href="#">Set Run Range</a> , page 73                        |
| Set Run Timestep                    | Set the timestep size of the run (E.g 1 Hour, 1 Day, 1 Month, 1 Year). Options are included to specify how to synchronize and optionally aggregate data. | <a href="#">Set Run Timestep</a> , page 73                     |
| Set Scalar Slot Value               | Set one or more Scalar Slots values.                                                                                                                     | <a href="#">Set Scalar Slot Value</a> , page 74                |
| Set Series Slot Flags               | Set the specified flag (input or output) on a set of series slots within the specified time range.                                                       | <a href="#">Set Series Slot Flags</a> , page 75                |
| Set Series Slot Input Mode          | Set the Input Mode for one or more Series Slots with Periodic Input slot.                                                                                | <a href="#">Set Series Slot Input Mode</a> , page 76           |
| Set Series Slot With Periodic Input | Set the periodic values of a Series Slot with Periodic Input slot.                                                                                       | <a href="#">Set Series Slot With Periodic Inputs</a> , page 76 |
| Set Series Slot Values              | Set Series Slot’s values within a time range.                                                                                                            | <a href="#">Set Series Slot Values</a> , page 77               |
| Set Table Slot Value                | Set one or more Table Slot’s value at the specified row(s) and column(s).                                                                                | <a href="#">Set Table Slot Value</a> , page 78                 |
| Specify Script Variable Value       | Specify a script variable and the default value; other actions can be dependent on the value of the variable.                                            | <a href="#">Specify Script Variable Value</a> , page 79        |
| Synchronize Objects                 | Set the time range for the specified objects.                                                                                                            | <a href="#">Synchronize Objects</a> , page 81                  |
| Synchronize Slots                   | Set the time range for the specified slots.                                                                                                              | <a href="#">Synchronize Slots</a> , page 81                    |

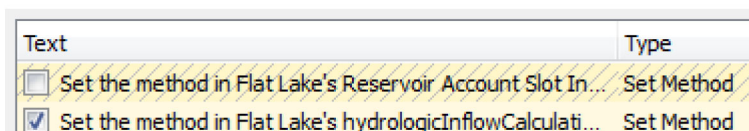
**Table 1.1** (Continued) Descriptions of the available actions

| Action Type                          | Description                                                                                                                                                                                                                       | Link                                                           |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Visualize Ensemble Data Set          | Visualize data for an open ensemble data set. Currently, this action will create a plot of line curves. The plot can be shown in the Data Analysis Tool and/or in a Plot Page.                                                    | <a href="#">Visualize Ensemble Data Set</a> , page 82          |
| Visualize Series Regression Data Set | Visualize data for a series regression results data set. Currently, this action will create a plot of line curves including the fitted and dependent data. The plot can be shown in the Data Analysis Tool and/or in a Plot Page. | <a href="#">Visualize Series Regression Data Set</a> , page 83 |

## Common Settings for Action Types

An action is controlled by user settings. Actions of all types have the following settings:

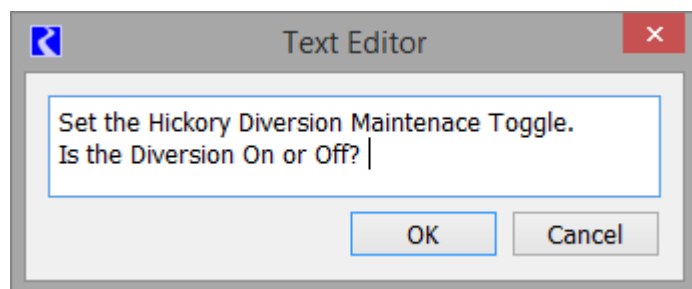
- **Enabled/Disabled.** Each action has a check box that determines whether it is enabled or disabled in the script editor. This setting is also used to indicate whether the action starts as enabled or disabled in the script dashboard, although it can be overridden in the dashboard.



- **Show In Dashboard.** This is a Yes/No value that controls whether the action is listed in the Script Dashboard.
- **Display Text.** This single line of text identifies the action. This text is shown in the dashboard. If the field is empty, default text is automatically generated based on the action type and settings.
- **Description.** This multi-line text provides a description, comment, or notes on the action. On the dashboard, the text is displayed below the Display Text. Select the description value to start the edit and then enter the text in the text box or use the  button to open a text editor.



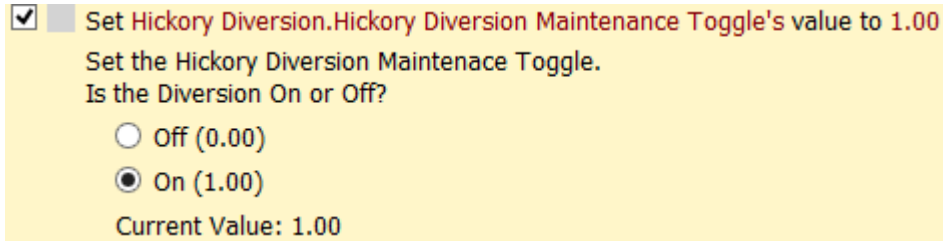
Within the text editor, type or paste in your description. Enter line breaks as necessary. A sample editor is shown.



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### Script Management

For this sample Set Scalar Slot action, the resulting description would show on the dashboard as two lines below the display text above any override controls:

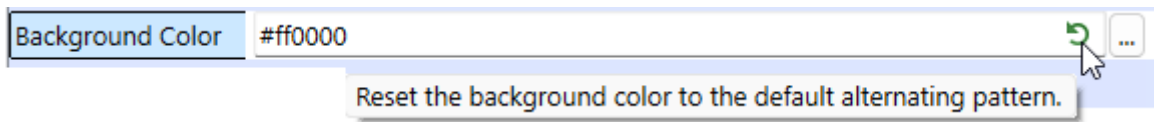


- **Dependencies.** All actions have three settings to define if it is dependent on a Script Variable. See [“Specify Dependency Settings for other Actions”](#) on page 33 for more information.

- **Dependency Type** - Specify whether the action is dependent on a variable or not.
- **Dependency Variable Name** - If dependent, specify the variable name on which the action depends.
- **Dependency Variable Value** - If dependent, specify the variable value on which the action depends.

See [“Specify Dependency Settings for other Actions”](#) on page 33 for more information.

- **Background Color.** Specify the background color to use in the Script Editor and on the Script Dashboard. Use the selector to choose the desired color. Use the reset button highlighted below to clear the color and instead use the default alternating yellow pattern.



## Setting Information

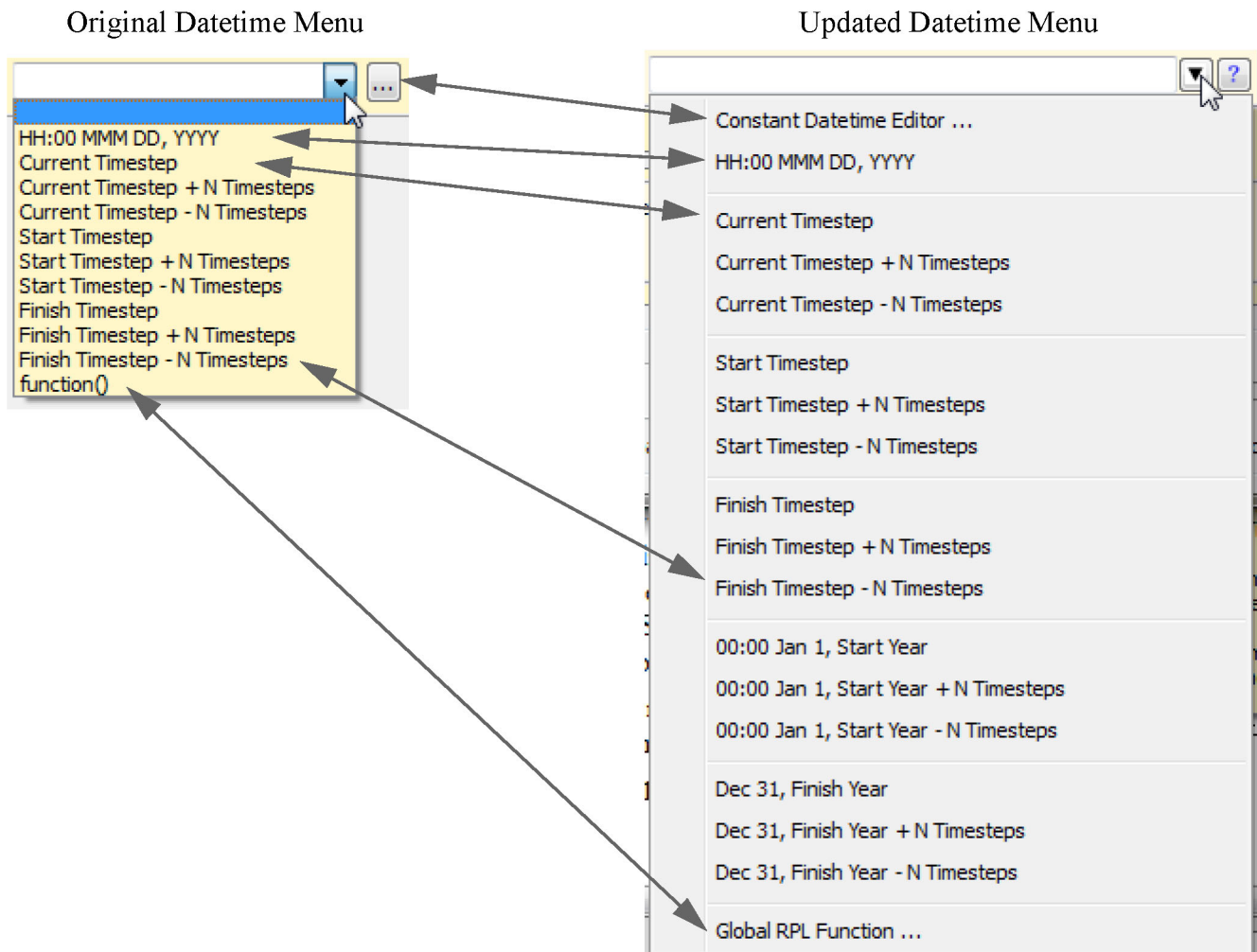
The following sections provide more information on certain settings or on settings that are shared by many actions that have additional controls.

### Datetime Information

Datetimes in a script can be specified either as absolute values or a symbolic values. There are two slightly different interfaces, but both have essentially the same functionality. Depending on the action and whether it is the editor or dashboard, one of the following interfaces will be used.

**Note:** The current menu is being transitioned to the updated menu. Common functionality has been highlighted in [Figure 1.11](#).

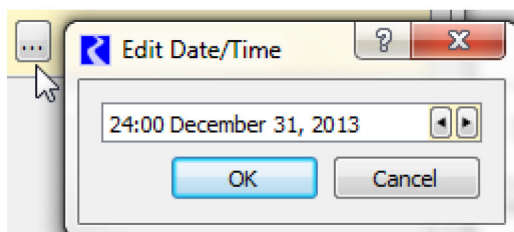
Figure 1.11



Following is additional information on each of these ways to specify the datetime.

- **Choose a Datetime.** Use the datetime spinner by selecting the **More** button or the **Constant Datetime Editor** option. This opens the Date/Time selector.

**Note:** This uses the model timestep size.



- **Type a Datetime.** Enter a date by typing. Use the HH:00 MMM DD, YYYY as a guide: Hour (H), Month (M), Day (D), Year (Y).

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- **Specify the Datetime Symbolically:** Use one of the options shown and specify any of the pieces necessary like integers “N”, Hour (H), Month (M), Day (D), Year (Y). On the updated menu, select **Help** for details on the symbolical datetimes. Basically, any fully specified datetime can be used. See [“DATETIME” in RiverWare Policy Language \(RPL\)](#) for details.
  - Note:** No @ or quotes “ ” are necessary when specifying the datetimes. In this context, “Current Timestep” means the machine/operating system time, rounded up to a controller timestep. So if you are running the script at noon on the 4th of July, 2014 in a model with a 1 Day timestep, “Current Timestep” would return 24:00 July 4th, 2014.
- **Use a Global Function:** To allow additional flexibility specifying datetimes, you can specify a RPL function from an opened Global Function Set; see [“Global RPL Functions” in RiverWare Policy Language \(RPL\)](#) for details. You can type in the name of the function (original menu) of the form function() (see note below) or choose it using a selector (updated menu). This function must return a fully specified DATETIME variable
  - Note:** Datetimes stored on slots cannot be set to NaN as an override on the dashboard, but can be configured to be set to NaN from the Script Editor.
  - Note:** Originally, you could set a datetime to a RPL function using the function() notation on the Script Editor. Functions specified this way are still supported. But for new scripts or any edits, functions should be specified by changing the “Set by” setting to Function and choosing the function from the selector.

## Filename Information

Filenames can be entered manually or chosen using the file chooser. You can specify files using environment variables using the \$VARIABLE syntax.

## RPL Reorder Set Action

This action allows you to reorder items (rules, goals, groups, or even statements within the specified set. The new order is specified on a table slot. Following is an example that shows the use of this action. [Figure 1.12](#) shows the ruleset before the script is executed.

**Note:** Statements are shown using the **View** menu.

Figure 1.12

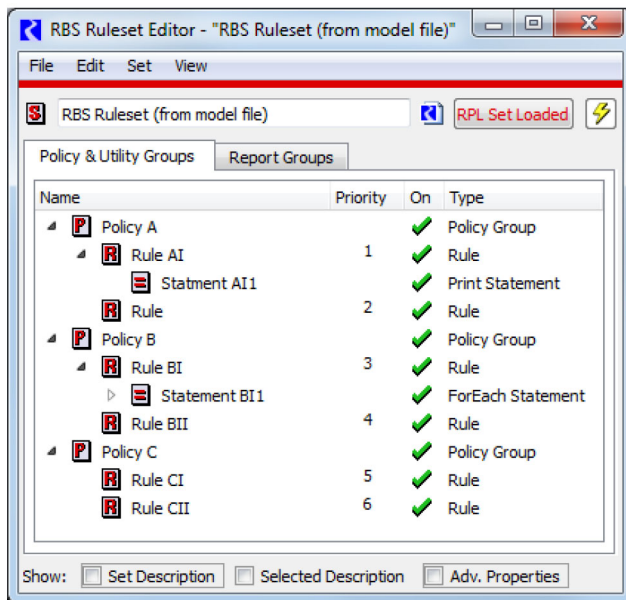


Figure 1.13 shows the specified table that describes the desired ordering. The row labels must match the names of the items in the set.

Figure 1.13

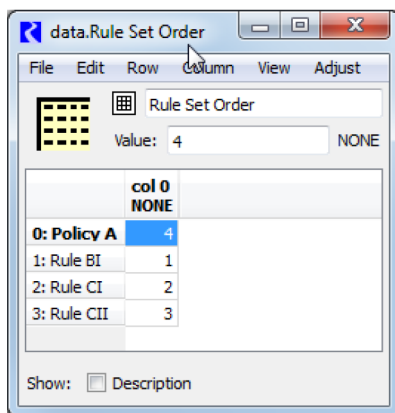


Figure 1.14 shows a text description of this reordering.

Figure 1.14

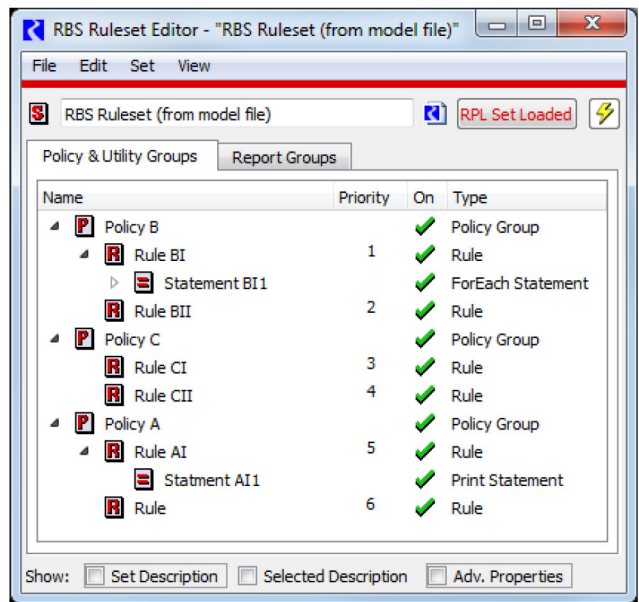
```
Before
1. Policy A: desired order = 4, index in set = 2
2. Rule BI: desired order = 1, index in set = 3
3. Rule CI: desired order = 2, index in set = 5
4. Rule CII: desired order = 3, index in set = 6

Reordering:
Rule BI: leaving in place.
Rule CI: leaving in place.
Rule CII: leaving in place.
Policy A: moving to after Rule CII

After
1. Rule BI: desired order = 1, index in set = 1
2. Rule CI: desired order = 2, index in set = 3
3. Rule CII: desired order = 3, index in set = 4
4. Policy A: desired order = 4, index in set = 6
```

Figure 1.15 shows the ruleset after reordering.

Figure 1.15



# Setting Slot Values

This section contains important information on the following actions that set slot values in the model:

- [“Set Scalar Slot Value”](#) on page 74
- [“Set Series Slot With Periodic Inputs”](#) on page 76
- [“Set Table Slot Value”](#) on page 78



“[Configuring User Control on the Script Dashboard](#)” describes how to configure user controls on the dashboard. Then, see “[Setting Slot Values Using a Global Function](#)” on page 29 for more information on setting numeric values using a global function.

## Configuring User Control on the Script Dashboard

This section applies when the “Set by” setting is using “Values” For actions that set values in the model (Set Scalar Slot Value, Set Series Slot Values, Set Table Slot Value, Set Periodic Slot Values), you can either specify the value to set or give a default, but allow the user of the script to override that value. This section describes configuring the action so that the user can modify the value using either a value editor, radio buttons or a slider. These are controlled by the settings shown in [Figure 1.16](#).

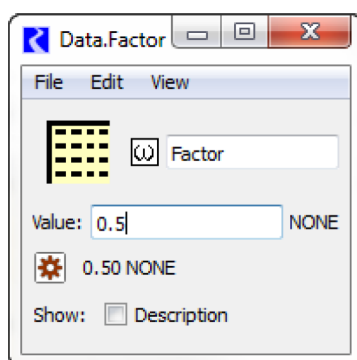
**Figure 1.16**

|                         |                    |
|-------------------------|--------------------|
| Allow Editing           | Yes: Editor        |
| Minimum Edit Value      | No                 |
| Maximum Edit Value      | Yes: Editor        |
| Encoding TableSlot Name | Yes: Radio Buttons |
|                         | Yes: Slider        |

The following sections describe the three “Yes” cases. In each case, an example will be shown where a script is setting a scalar slot called Data.Factor to a value between 0.1 and 1.5. Perhaps this factor is then referenced by a rule to compute demands or inflows. For reference, [Figure 1.17](#) shows this scalar slot.

For all of the Yes selections for Allow Editing, you can also choose to Show Current Value. When Yes, the current value (for a single scalar slot only) is shown on the Dashboard below the editing controls.

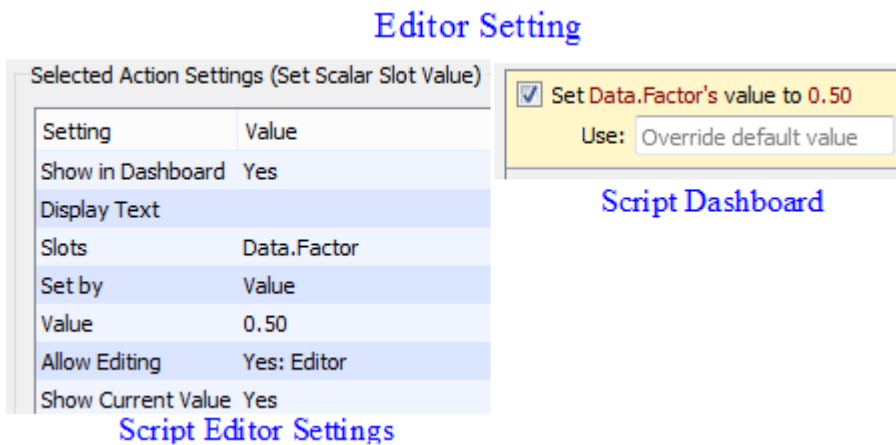
**Figure 1.17**



### Editor

When the Allow Editing setting is Yes: Editor, an editable field is shown on the Script Manager. The default value will be used unless the user enters a different value in the field. [Figure 1.18](#) shows the settings to get this control.

Figure 1.18

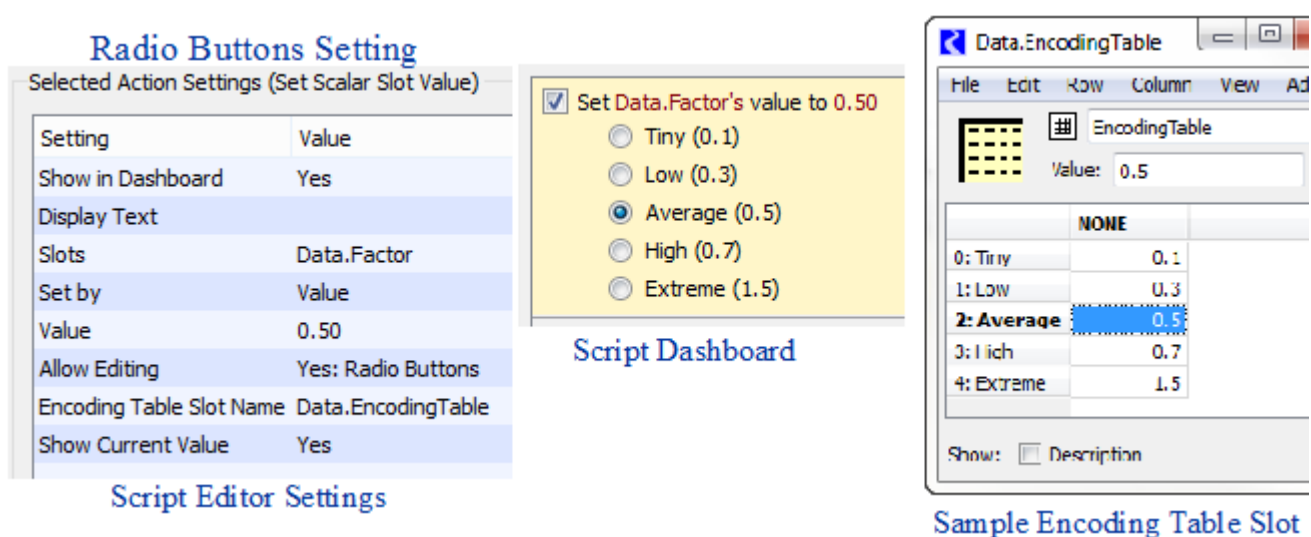


**Radio Buttons**

When the Allow Editing setting is Yes: Radio Buttons and an Encoding Table Slot is specified, radio buttons are shown on the Script Manager. The default value will be used unless the user selects one of the radio buttons to use that value.

The Encoding Table Slot Name setting specifies a Table Slot whose row labels and values establish a textual encoding of values. Figure 1.19 shows a sample encoding table slot. The row labels are used as the labels on the radio buttons. The values in the table are displayed on the Script Manager and will be used by the action for that button.

Figure 1.19

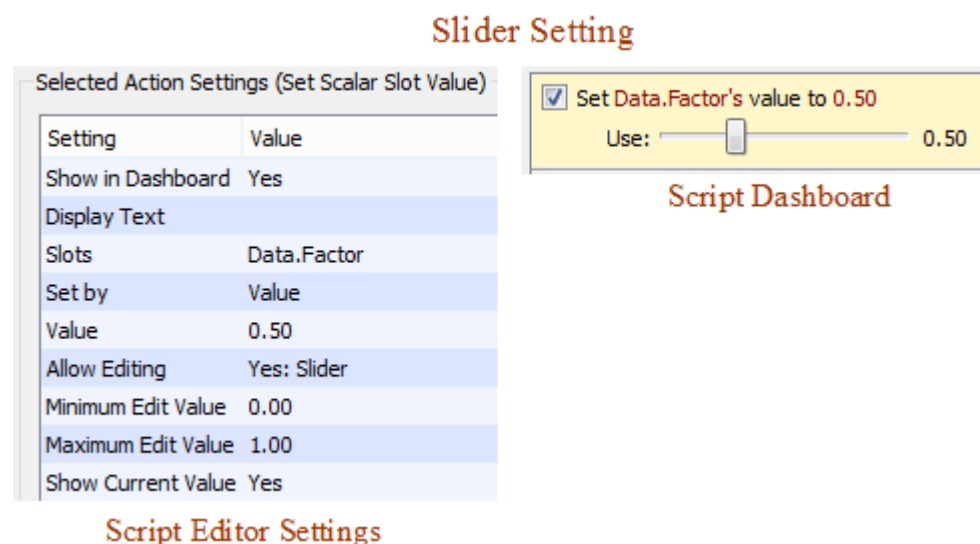


**Sliders**

When the Allow Editing setting is Yes: Slider and Minimum and Maximum Edit Values are defined, a slider will be shown on the Script Manager. The default value will be used unless the user moves the slider to a different

value. [Figure 1.20](#) shows the settings to configure this control. The available slider positions are computed by dividing the interval between the minimum and maximum by 100.

**Figure 1.20**



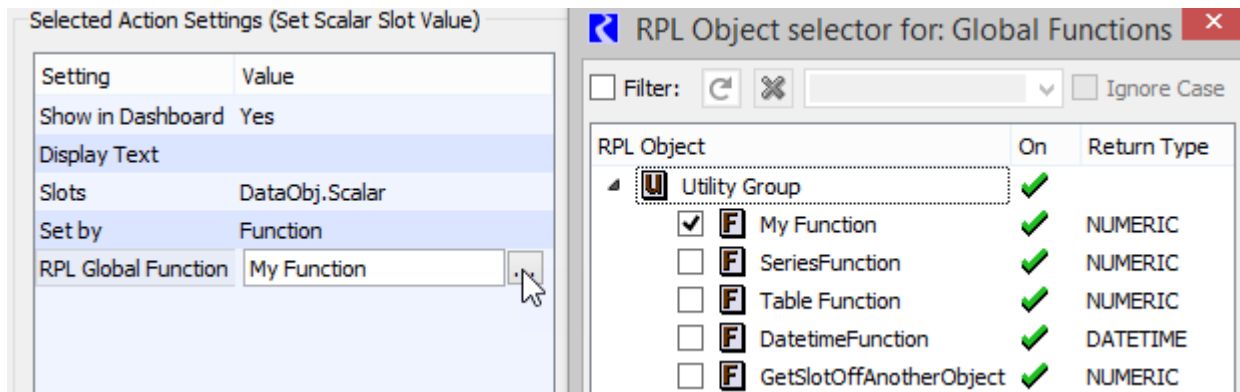
## Setting Slot Values Using a Global Function

The three script actions allow slot values to be set to values as described above. They also allow the slot value to be set to numeric or datetime values using a RPL Global Function.

- “[Set Scalar Slot Value](#)” on page 74
- “[Set Series Slot Values](#)” on page 77
- “[Set Series Slot With Periodic Inputs](#)” on page 76
- “[Set Table Slot Value](#)” on page 78
- “[Set Periodic Slot Value](#)” on page 71

To use this functionality, in any of these actions, in the “Set by” setting, choose “Function”. Then choose the desired RPL Global Functions Set and the specific RPL Global Function to use as shown in [Figure 1.21](#).

**Figure 1.21** Screenshot of selecting a Global Function



**Note:** The Global Function must return a NUMERIC or DATETIME value. Setting Text Series Slots is not supported by this mechanism.

The global function can contain various combinations of arguments based on the action. You can use zero, one or more arguments. The arguments passed for each action are:

- **“Set Scalar Slot Value”**: SLOT slotToSet. This action loops over each slot and calls the function passing in the slot as the argument.
- **“Set Series Slot Values”**: SLOT slotToSet, DATETIME date: This action loops over each slot and datetime and calls the function passing in the slot and datetime as the arguments.
- **“Set Series Slot With Periodic Inputs”**: SLOT slotToSet, NUMERIC row. This action loops over each slot and datetime and calls the function passing in the slot and row as the arguments. This action sets the periodic portion of the slot. Use the **“Set Series Slot Values”** to set the series input if desired. Use the **“Set Series Slot Input Mode”** action to switch between the modes.
- **“Set Table Slot Value”**: SLOT slotToSet, NUMERIC row, NUMERIC col. This action loops over each slot and calls the function passing in the slot, row, and column as the arguments. This action sets the slot at the specified rows and columns. Use the **“Rectangular Region”** setting to loop over a set of contiguous rows and columns.
- **“Set Periodic Slot Value”**: SLOT slotToSet, NUMERIC row, NUMERIC col. This action loops over each slot and calls the function passing in the slot, row, and column as the arguments. This action sets the slot at the specified rows and columns. Use the **“Rectangular Region”** setting to loop over a set of contiguous rows and columns.

**Example 1.1** Using a RPL Global Function to set multiple series slots.

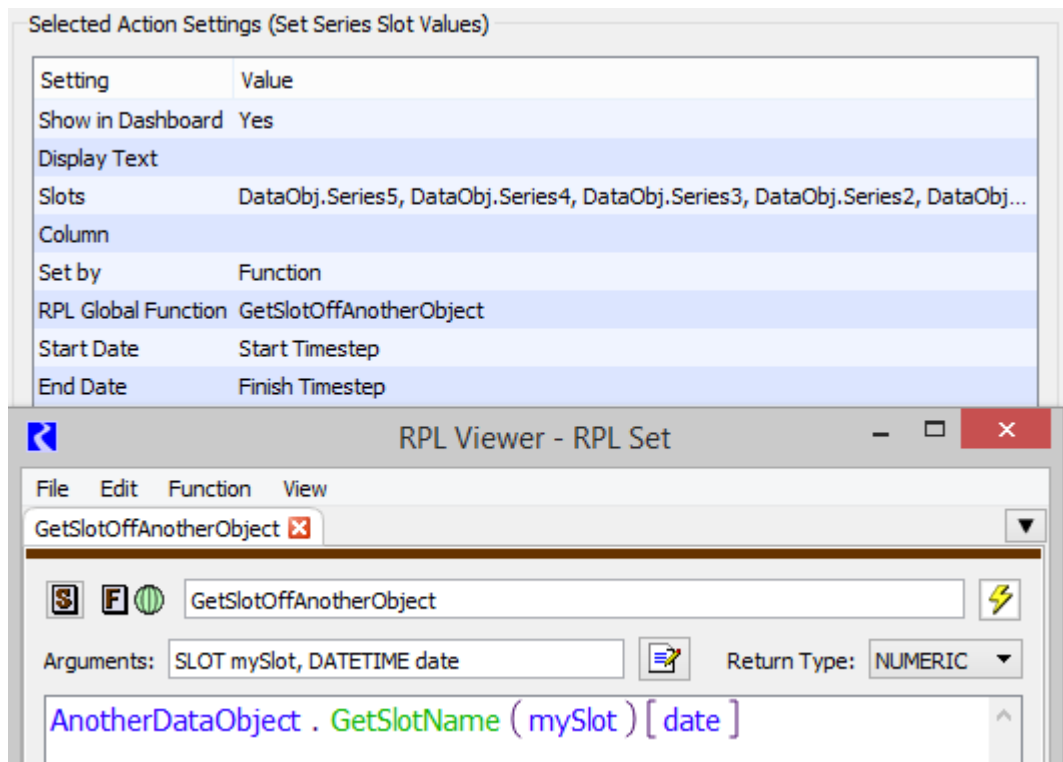
**This is best explained with an example. Let’s assume we want to set five series slots to a value on a data object at each timestep in the run.**

Set up the script as follows:

1. In the Set Series Slot Values action, “Slots” setting, choose the 5 slots to set.
2. Choose the Global Functions Set and the desired Global Function
3. Specify the Start and End Date

In our example, the settings and RPL function is shown in [Figure 1.22](#).

**Figure 1.22** Screenshot of a Set Series Slot Values that calls a numeric RPL function



When this action is executed, it will loop over the 5 slots and all timesteps in the run. For each slot/timestep, it will call the GetSlotOffAnotherObject function and pass the slot and datetime into the function. The function computes a numeric, in this case looking up the value on another object. The action then sets that value.

## Script Variable Values and Dependencies

Whether an action executes or not can depend on the value of a variable set by another action, often controlled by the script dashboard user or operator. This allows scripts that can be executed with different settings or in different “modes” without requiring scripts to be duplicated many times for each potential variation.

To set this up, following these three steps as described in this section.

1. In the Script Editor, define a script variable with the **Specify Script Variable Value** action.
2. In the Script Editor, for each dependent action, define the dependencies, that is, the conditions of the variables in which the action should execute.
3. In the dashboard, specify the script variable value (or leave as the default) before running the script.

These three steps are described below:

### Define a Specify Script Variable Value Action

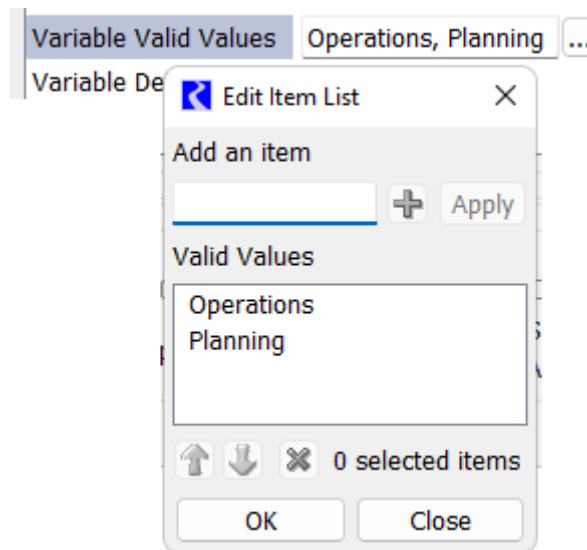
In the Script Editor, create Specify Script Variable Value action. This action uses three settings to define a variable:

- **Variable Name** - The name of the script variable.

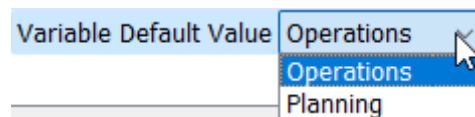
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- **Variable Valid Values** - The valid or possible values the script variable can have. Specify the list of text items in an **Edit Item List** dialog that is opened by clicking the “...” button. A sample is shown in the screenshot. Two items have been added.

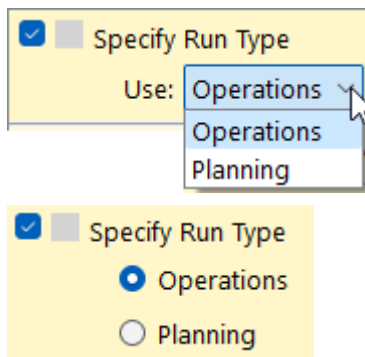


- **Variable Default Value** - The default value of the variable on the Script Dashboard. Select the default value from the list of values specified above. In the screenshot, Operations will be the default value.

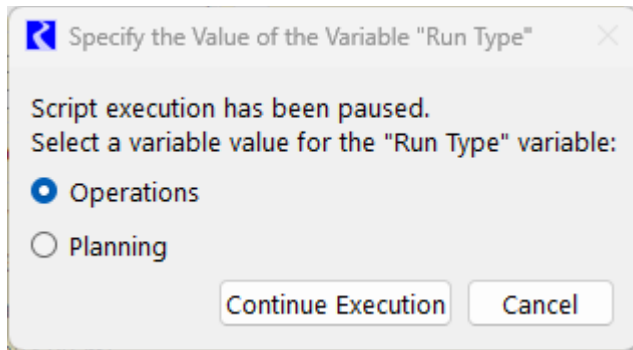


Then, optionally select how you want the selection to appear on the dashboard.

- **Variable Display Type** - Specify whether to use a menu or radio buttons.



- **Pause Execution and Prompt User** - Instead of specifying it on the dashboard, you can pause the script and ask the user for a value at the time of script execution. Then user will get a dialog that looks like this, depending on the Variable Display Type



- **Allow Sub-actions to Override Variable Values** - Specify whether actions in referenced subscripts (i.e. sub-actions) are allowed to override or change the script variable values. The original approach was to not allow any redefinition of variables (default “No” setting). See [“Not Allowing Variable Overrides”](#) on page 35 for more information. Specify “Yes” to allow sub-actions to modify the variable value. See [“Allowing Variable Overrides”](#) on page 36.

For more information on this specific action, see [“Specify Script Variable Value”](#) on page 79.

Once configured, the dashboard user is able to select the value of the variable. The dashboard will reflect this selection by disabling actions that will not execute due to a dependency on the variable. Actions disabled this way have a special icon and cannot be manually re-enabled until the variable selection changes. See [“Specify the Script Variable Values on the Dashboard”](#) on page 34

## Specify Dependency Settings for other Actions

Three settings are used to control an action’s execution based on the value of a script variable. These settings are available on any action:

- **Dependency Type** - Is the action dependent on a variable or not? There are two options for this setting:
  - **None:** the default. This is the standard behavior of script execution; the action is not dependent on a variable value.
  - **Script Variable Has Value:** the action is dependent on a variable. The next two settings will be shown and you must specify the name and value.
- **Dependency Variable Name** - The variable name on which the action depends. Enter the text string name of the variable. The text must be entered exactly the same as the definition in the Specify Script Variable Value or the action won’t do anything.
- **Dependency Variable Value** - The variable value on which the action depends. Enter the text string value of the variable. The text must be entered exactly the same as the definition in the Specify Script Variable Value or the action won’t do anything

**Tip:** Variable names and values must match exactly, including case, and must be entered for each action. Be careful when typing the items. To minimize needing to enter variable values for many

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items, create a subscript and add the dependencies on the Execute Script action that calls the subscript.

Thus, an action will execute when:

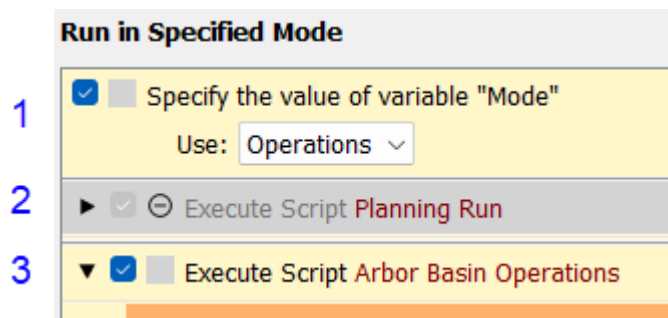
- its Dependency Type is set to **None**, or
- its Dependency Type is set to **Script Variable Has Value**, the variable specified by their Dependency Variable Name exists, and that variable has the value specified by the action's Dependency Variable Value.

## Specify the Script Variable Values on the Dashboard

The script dashboard user selects the value of the variable from the dashboard. The dashboard will reflect this selection by turning off actions that will not execute due to a dependency on the variable. Actions disabled this way

have a special icon  and cannot be manually re-enabled until the variable selection changes.

**Figure 1.23** Screenshot of dashboard with conditional items



1. This action defines the variable “Mode” with two values Planning and Operations. The default is Operations.
2. This action executes a script, but is dependent on the variable Mode having the value Planning. It will not be executed and is displayed with the icon and grey background.
3. This action executes a script and is dependent on the variable Mode having the value Operations. It will execute based on the setting in item 1.

If the script operator switches the variable in action/row 1 to Planning, the action/row 2 would be turn to yellow to be executed and action/row 3 would be disabled.

## Additional Information on Conditional Script Action Execution

The following section describes some additional information on using the Script Variables.

The action **Specify Script Variable Value** creates the variable when they are executed. That is, a variable cannot be referenced before the Specify Script Variable Value action defines it. Script execution will abort and an error will be posted to the Diagnostics Output if the variable on which an action depends has not been defined. Thus, it is generally a good idea to insert these actions early in a script to ensure that their variables are present when they are needed.

**Note:** The setting of the top most script (the first time **Specify Script Variable Value** sets the script variable) controls all subscript actions. The remainder of the script either allows or doesn't allow overrides.

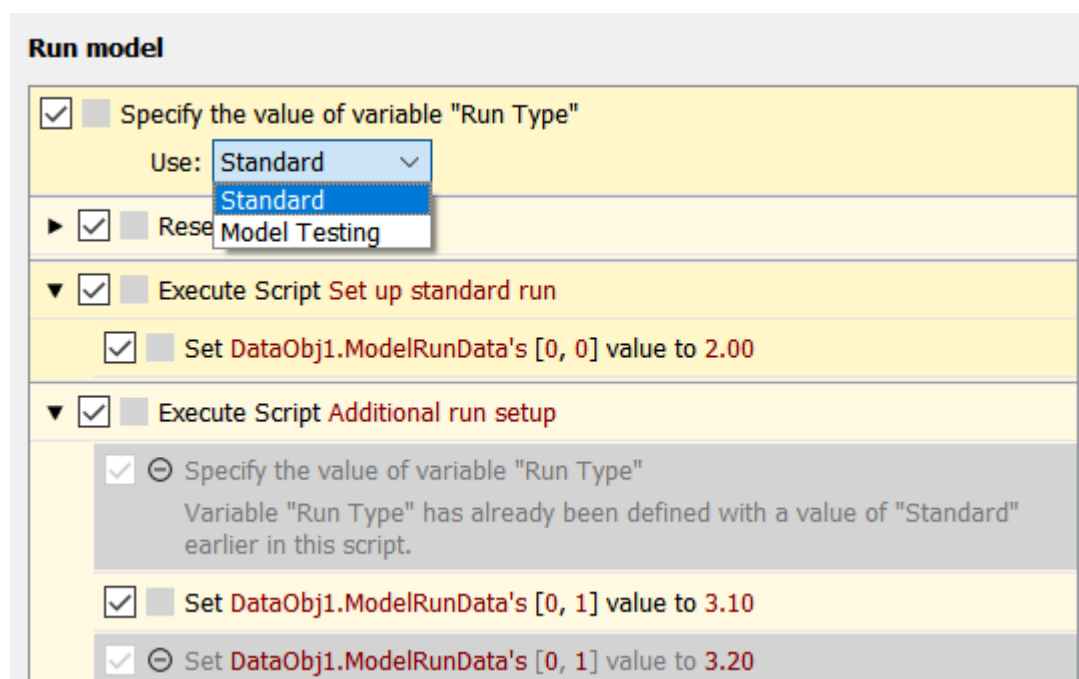


**Tip:** Variable names and values must match exactly, including case. Be careful when typing the items.

## Not Allowing Variable Overrides

The rest of this section applies when the **Allow Sub-actions to Override Variable Values** is set to “No”. In that case, a script variable can only be *defined* once within any given script and any subscripts it executes via the Execute Script action. If a Specify Script Variable Value action attempts to redefine a variable that has already been defined (including in parent scripts), then any definition other than the first will be ignored. This can be seen in [Figure 1.24](#); the “Additional run setup” script defines the “Run Type” variable as its first action.

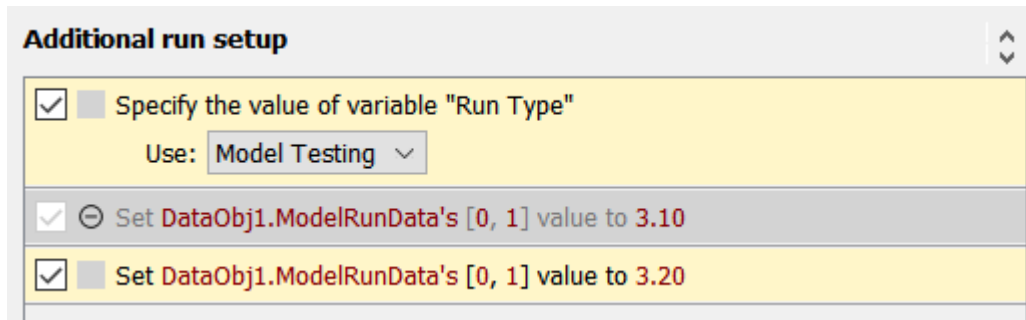
**Figure 1.24** Screenshot of dashboard with script variable redefinition of the Run Type variable.



However, because the “Additional run setup” script is being executed as a subscript of the “Run model” script, only the variable definition in the “Run model” script takes effect. Furthermore, the first of the “Set DataObj1.ModelRunData” actions is enabled because it executes when “Run Type” has a value of “Standard” while the second is disabled because it executes when the variable has a value of “Model Testing”.

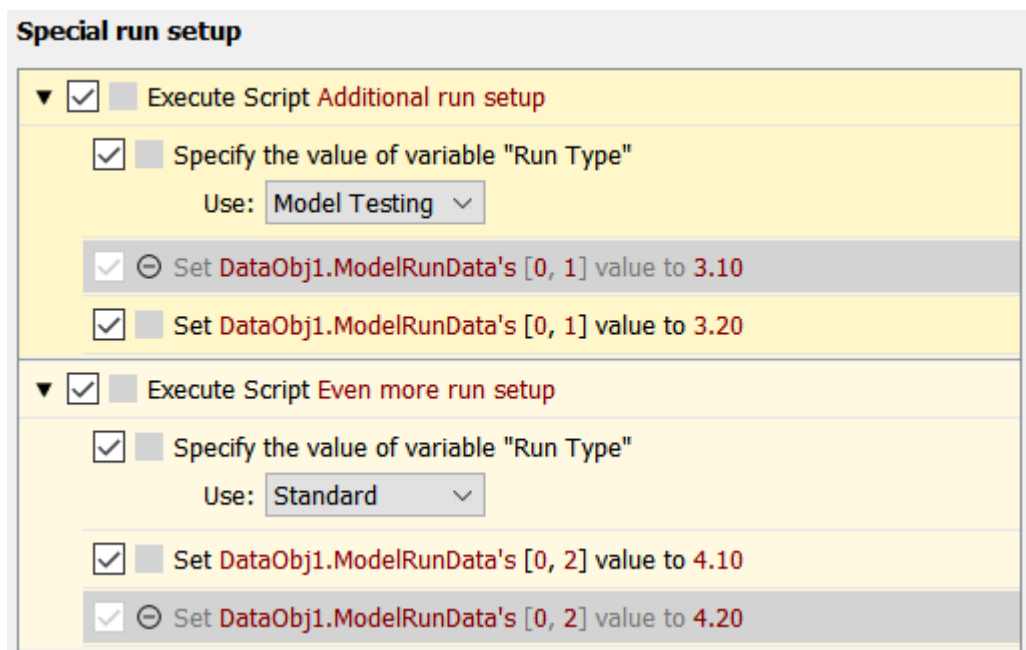
[Figure 1.25](#) shows that if the “Additional run setup” script is executed on its own, that its definition of the variable is used. As before, the first of the “Set DataObj1.ModelRunData” actions executes when “Run Type” has a value of “Standard” while the second executes when the variable has a value of “Model Testing”.

**Figure 1.25** Screenshot of dashboard with a script variable for its own use



Script variables can be defined multiple times, although later or subscript definitions will be ignored. [Figure 1.26](#) shows an example of the “Run Type” variable being defined in both the “Additional run setup” script as well as the “Even more run setup” script. However, they do not interfere with each other because the “Run Type” variable is not also defined in the “Special run setup” parent script.

**Figure 1.26** Screenshot of dashboard with script variable redefinition



## Allowing Variable Overrides

When **Allow Sub-actions to Override Variable Value** is set to “Yes” on the first **Specify Script Variable Value** action encountered, then all subsequent **Specify Script Variable Value** actions can redefine or override the variable value. This allows you to either modify the value or prompt the user for a new value after first defining.

# Action Type Settings Reference

This section provides details about settings specific to each type of action.

## • Activate Unit Scheme

Activate the specified Unit Scheme as defined in the Unit Scheme Manager. For more information on Unit Schemes, see [“Unit Schemes” in User Interface](#).

| Setting     | Type      | Notes                                       |
|-------------|-----------|---------------------------------------------|
| Unit Scheme | Selection | Select the desired Unit Scheme to activate. |

## • Analyze Ensemble Data Set

Compute the specified statistic for slots in an ensemble Data Set. For more information on ensemble Data Sets, see [“About Data Set Objects” in Output Utilities and Data Visualization](#). Statistics are computed for each selected slot at each timestep across traces. The outcome are one or more time series of results for each selected slot.

See [“Ensemble Data Set Analysis” in Output Utilities and Data Visualization](#) for more information on interactive use of this action in the Data Analysis Tool.

For more information on a similar action to visualize the ensemble data set, see [“Visualize Ensemble Data Set”](#) on page 82.

| Setting                   | Type           | Notes                                                                                                                                                 |
|---------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Which Data Set To Analyze | Selection      | Indicates how the ensemble data set will be specified. Select either “Most Recently Opened” or “Specified by Name”.                                   |
| Analysis Data Set Name    | Text           | If the above selection is 'Specified by Name', then this setting is used to specify the name of the data set.                                         |
| How to Filter Slots       | Selection      | How to filter the analysis data by slot. Options in include All Slots and Include Only Specific Slots                                                 |
| Slots To Include          | Slot Selection | When “Include Specific Slots” is selected, this setting defines the slots.                                                                            |
| How to Filter Dates       | Selection      | How to filter the analysis data by date. Options include All Dates, Include Dates In A Specific Range                                                 |
| Start Date                | Datetime       | When “Dates in a Specific Range” is selected, this setting defines the start date. See <a href="#">“Datetime Information”</a> on page 22 for details. |
| End Date                  | Datetime       | When “Dates in a Specific Range” is selected, this setting defines the end date. See <a href="#">“Datetime Information”</a> on page 22 for details.   |
| Allow Start Date Editing  | Yes/No         | Yes indicates that users should be allowed to override the start date in the Dashboard.                                                               |
| Allow End Date Editing    | Yes/No         | Yes indicates that users should be allowed to override the end date in the Dashboard.                                                                 |
| How To Filter Traces      | Selection      | How to filter the analysis data by traces. Options in include All Traces, Trace Range, and Include Specific Traces.                                   |
| First Trace Index         | Integer        | When “Trace Range” is selected, this setting defines the first trace number of the trace range to be included in the analysis.                        |
| Last Trace Index          | Integer        | When “Trace Range” is selected, this setting defines the last trace number of the trace range to be included in the analysis.                         |

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| Setting                             | Type                  | Notes                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trace Selection                     | Integer List          | When “Include Specific Traces” is selected, this setting defines which traces are included in the analysis. Click in the setting field and open up a trace selector dialog to select which trace numbers should be included.                                                                                                                                                           |
| How To Handle NaNs                  | Selection             | Options include Ignore NaNs, Error If Any NaNs and Error if All NaNs.                                                                                                                                                                                                                                                                                                                  |
| Analysis Type                       | Selection             | Specify the statistic to compute for the selected ensemble data. Options are: Exceedance, RPL Function, Max, Mean, Median, Min, Multiple Statistics, Percentile, or Sum.                                                                                                                                                                                                               |
| Percentage (0% - 100%)              | Number (as a percent) | When Exceedance or Percentile is the Analysis Type, this setting is the percentage to use. For example for Exceedance analysis and a value of 90%, then the computation will return the values for which 90% of traces have larger values larger (that is, a low number). For Percentile analysis, a percentage of 90% will give the 90th percentile values (that is, a large number). |
| RPL Global Function                 | Selection             | When RPL Function is selected for the Analysis Type, select a global RPL function that specifies the computation for the statistic. See below for more information.                                                                                                                                                                                                                    |
| Write Results to Simulation Objects | Yes/No                | Yes indicates that the results will get written to the same object as the original source slot. The default is Yes in the script editor.                                                                                                                                                                                                                                               |
| Write Results to Data Set           | Yes/No                | Yes indicates that the results will get written to a results ensemble data set object. The default is No in the script editor.                                                                                                                                                                                                                                                         |
| Results Data Set Name               | Text                  | The name of the results data set object to which the results will get written. If the object does not exist, it will get created.                                                                                                                                                                                                                                                      |
| Save Results Data Set With Model    | Yes/No                | If Yes, the object specified in the Results Data Set Name setting will get saved in with the model when the model is saved. If No, the object and its slots and data will not get saved and will no longer be present when the model is reloaded. The default is Yes. This setting could change the object’s current configuration if it already exists.                               |
| Results Data Set Slot Group         | Text                  | Specify the name of the slot group on the results data set object in which the results slots will get added. If the slot group does not already exist, it will get created.                                                                                                                                                                                                            |
| Results Slot Tag                    | Text                  | This text string is added to the results slot name, see below. This is not applicable for the Multiple Statistics analysis type.                                                                                                                                                                                                                                                       |
| Results Slot Column                 | Integer               | If the results slot is an Agg Series slot and the results slot column value is other than zero, the results are written to that column of the results slot.                                                                                                                                                                                                                            |

**Warning:** Ensemble data analysis is not supported for aggregate series slots. Including aggregate series slots in ensemble data analysis will result in an error.

### Results

Results are written to a set of series slots or agg series slots, one for each slot included in the analysis. Results can be written to the same object as the source slot, a results data set object, or both. When written to the same object as the source slot, the results slot’s name is created using the template: “<slot’s object>.<slot name> <results slot tag>”. For example, if the slot “Res A.Inflow” participates in the analysis and the Results Slot Tag is “Median”, then results associated with that slot will be written to the slot “Res A.Inflow Median”.

When written to a results data set object, the results slot name is created using the template: “<results data set name>.<slot’s object> <slot name><results slot tag>”. For the same example of the slot “Res A.Inflow” and the Results Slot Tag of “Median” and a Results Data Set Name “Analysis Data Set,” the results will be written to the slot “Analysis Data Set.Res A Inflow Median”.

If the **Write Results to Simulation Objects** is “yes”, the action will create the data set series slot if it does not already exist and, if it does exist, resize it as necessary to the minimum required dimensions. All values are given the input (I) flag. If the results slot is an agg series slot and the results slot column value is other than zero, the results are written to that column of the results slot.

If the **Write Results to Data Set** is “yes”, the action will create the data set series slots on a new or existing Ensemble Data Object with the given tag in the slot name. All values are given the output (O) flag but are read only.

## Analysis Type

In the **Analysis Type** setting, specify the statistic to compute for each timestep across traces. Options are:

- **Percentile:** The percentile is computed as follows: The values (for a particular timestep for all traces) are sorted from smallest to largest. The rank index is then computed as the specified percentile multiplied by the number of values, rounded up to the nearest integer. Then the result is selected from the sorted list based on the rank index. For Percentile analysis, a percentage of 90% will give the 90th percentile values (that is, a large number).
- **Exceedance:** The exceedance is the inverse of percentile:  $\text{Exceedance}(P) = \text{Percentile}(100\% - P)$  For example for Exceedance analysis and a value of 90%, then the computation will return the values for which 90% of traces have larger values larger (that is, a small number).
- **Max:** This statistic finds the largest value at each timestep.
- **Mean:** This statistic finds the mean of all values at each timestep.
- **Median:** This is the equivalent of selecting Percentile or Exceedance and specifying 50% for the percentage.
- **Min:** This statistic finds the lowest value at each timestep.
- **Multiple Statistics:** This type computes a slot for the following seven percentiles: Min (0th), 10th, 25th, Mean (50th), 75th, 90th, and Max (100th). Currently the percentiles are hard-wired and cannot be changed.
- **Sum:** This statistic sums all values at each timestep. It may be meaningless for certain values, like flows.
- **RPL Function:** If none of the above statistics meet your needs, you can create your own Global RPL Function. In the script, specify the function in the RPL Global Function setting.

The function must be in a Global Functions Set. It should perform the computation for the set of values associated with all traces in a data set for a single slot and date.

The following arguments are supported. That is, your function should define these arguments and can use them as needed:

- A LIST of NUMERIC values representing the data set values for the slot[datetime] for all traces. For example, it would be a list containing the Jan 1 slot value in trace1, trace 2, trace 3 and so on.
- A SLOT, representing the slot associated with the values.
- An OBJECT representing the object which contains the slot associated with the values.
- A DATETIME associated with the value.

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- A LIST of NUMERIC values representing the trace number.

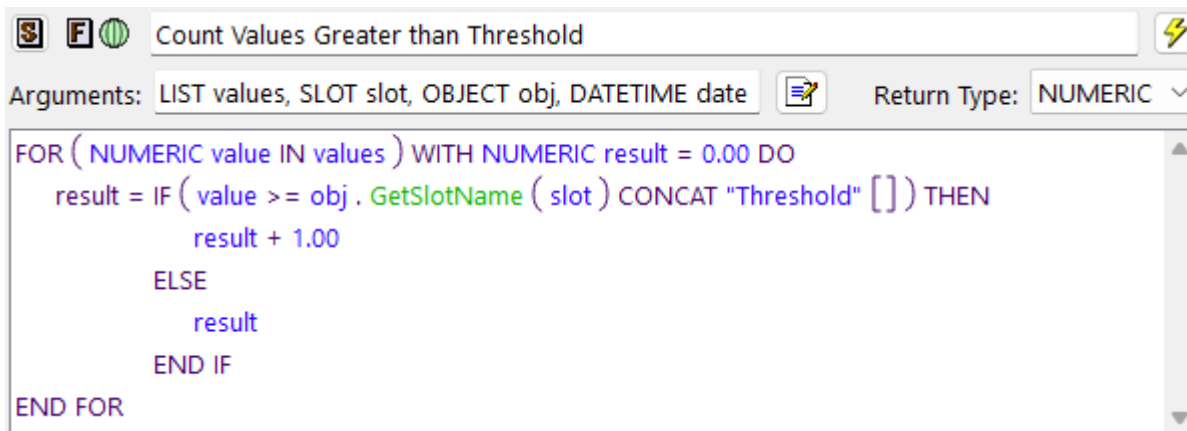
Although all five arguments are supported, the function actually matches up the type of the argument with the appropriate value. So you could specify it as:

1. A single LIST argument:  
GlobalFunction(<LIST>)
2. All five arguments: LIST, SLOT, OBJECT, DATETIME, LIST:  
GlobalFunction(<LIST>, <SLOT>, <OBJECT>, <DATETIME>, <LIST>)
3. GlobalFunction(<LIST>, <OBJECT>)

**Note:** If you plan to use the second list of trace numbers, you must also include the first list of numeric values or it will not find the correct list.

The called function should return a numeric value representing the desired statistic computed on the set of numeric values. For example, the function below return a count of the number of traces that have a value of the slot at the given timestep that is greater than a threshold value, stored in a scalar slot on the original object:

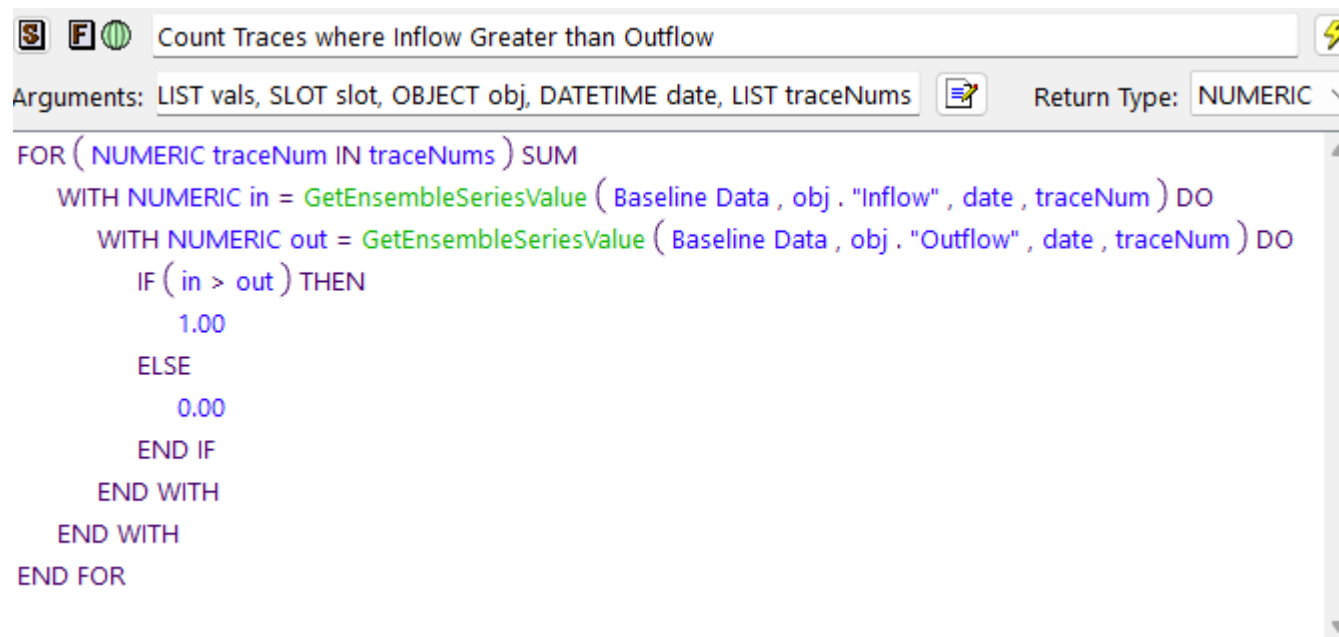
**Figure 1.27** Screenshot of a global RPL function that does per timestep across trace analysis:



For the above function, note that three arguments are used, values, slot, and obj. The date is defined but not used in the body.

The predefined function `GetEnsembleSeriesValue` is also very useful as you can get the value of a slot for a trace and a datetime. See [“GetEnsembleSeriesValue” in RiverWare Policy Language \(RPL\)](#). With this predefined function, you can create a function that counts the number of traces where a trace slot is greater than another trace slot.

**Figure 1.28** Screenshot of a global RPL function that counts number of traces that meet a condition



**Note:** This function has all five arguments but does not use the LIST of vals or the SLOT slot. It is configured to always look at the Inflow and Outflow on the OBJECT obj.

## • Apply Window Layout

Apply a window layout created with the Window Manager. See [“Window Manager” in User Interface](#).

| Setting     | Type | Notes                                                                                                                |
|-------------|------|----------------------------------------------------------------------------------------------------------------------|
| Layout Name | Text | Enter the name of the Window Layout. An error in script execution will occur if the specified layout does not exist. |

## • Clear DMI Values

Clear the values set by a DMI (by using the DMI invocation manager functionality).

| Setting  | Type | Notes                                                |
|----------|------|------------------------------------------------------|
| DMI Name | Text | Enter the DMI name or choose the DMI using the menu. |

## • Clear Scalar Slot Values

Clear the values on a set of Scalar Slots (i.e. set them to NaN).

| Setting | Type           | Notes                                                                            |
|---------|----------------|----------------------------------------------------------------------------------|
| Slots   | Slot Selection | Enter or choose one or more scalar slots for which you wish to clear the values. |

## • Clear Table Slot Values

Clear one or more values on a set of Table slots (i.e. set them to NaN).

| Setting              | Type           | Notes                                                                                                                                                                                                                                                                               |
|----------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slots                | Slot Selection | Enter or choose one or more table slots. The values in these slots will be cleared.                                                                                                                                                                                                 |
| How To Specify Cells | Selection      | Select from one of the following: <ul style="list-style-type: none"><li>• Row and Column (Single Cell): clear only one value (this is the default)</li><li>• Rectangular Region: clear a contiguous range of cells</li><li>• All Cells: clear all cells in the table slot</li></ul> |
| Row                  | Text           | When clearing only a Single Cell, specify the row as either the row number (zero based) or the row heading. For Periodic slots, use either the row's zero-based integer index or the label for its interval (e.g., "Feb").                                                          |
| Column               | Text           | When clearing only a Single Cell, specify the column as either the column number (zero based) or column heading                                                                                                                                                                     |
| Begin Row            | Value          | When clearing a Rectangular Region, specify the starting row as either row index (zero based) or the row heading.                                                                                                                                                                   |
| End Row              | Value          | When clearing a Rectangular Region, specify the ending row as either row index (zero based) or the row heading.                                                                                                                                                                     |
| Begin Column         | Value          | When clearing a Rectangular Region, specify the starting column as either column index (zero based) or the column heading.                                                                                                                                                          |
| End Column           | Value          | When clearing a Rectangular Region, specify the ending column as either column index (zero based) or the column heading.                                                                                                                                                            |

## • Close RPL Set

Close a selected RPL set from the model.

**Note:** Previously, this action was called Remove RPL Set.

When executed, an RBS Ruleset or Optimization Goal Set is unloaded and closed, a Global Function Set is closed, or the Initialization Rules Set is cleared.

**Note:** If you have any of these sets saved in the model file and you remove them using this action, any changes will be lost! This action should only be used if you have the set saved to a file and wish to remove it from the model and reopen it from the file. Use with caution!

| Setting         | Type             | Notes                                                                          |
|-----------------|------------------|--------------------------------------------------------------------------------|
| Set             | Single Selection | Choose the desired RPL set.                                                    |
| Confirm Removal | Yes/No           | Yes indicates that a dialog should be presented to confirm removal of the set. |



## • Compute Duration Curve

Compute duration curves for a set of series slots across multiple traces and timesteps from an ensemble data set. (For more information on ensemble Data Sets, see [“Ensemble Data Set Analysis”](#) in *Output Utilities and Data Visualization*.) The results are written to table slots (one results table slot for each input slot) with a format similar to a statistical table slot using the Duration Curve function (see [“Statistical Table Slots”](#) in *User Interface*) with all of the slot values sorted (ranked) from largest to smallest. The results table has five columns: Percent of observations equaled or exceeded, the slot values, Rank, Date, and Trace.

| Setting                             | Type           | Notes                                                                                                                                                                                                                        |
|-------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Which Data Set TO Analyze           | Selection      | Indicates how the ensemble data set will be specified. Select either “Most Recently Opened” or “Specified by Name”.                                                                                                          |
| Analysis Data Set Name              | Text           | If the above selection is 'Specified by Name', then this setting is used to specify the name of the data set.                                                                                                                |
| How to Filter Slots                 | Selection      | How to filter the analysis data by slot. Options in include All Slots and Include Only Specific Slots                                                                                                                        |
| Slots To Include                    | Slot Selection | When “Include Only Specific Slots” is selected, this setting defines the slots.                                                                                                                                              |
| How to Filter Dates                 | Selection      | How to filter the analysis data by date. Options include All Dates, Include Only Dates In A Specific Range                                                                                                                   |
| Start Date                          | Datetime       | When “Dates in a Specific Range” is selected, this setting defines the start date. See <a href="#">“Datetime Information”</a> on page 22 for details.                                                                        |
| End Date                            | Datetime       | When “Dates in a Specific Range” is selected, this setting defines the end date. See <a href="#">“Datetime Information”</a> on page 22 for details.                                                                          |
| Allow Start Date Editing            | Yes/No         | Yes indicates that users should be allowed to override the start date in the Dashboard.                                                                                                                                      |
| Allow End Date Editing              | Yes/No         | Yes indicates that users should be allowed to override the end date in the Dashboard.                                                                                                                                        |
| How To Filter Traces                | Selection      | How to filter the analysis data by traces. Options in include All Traces, Trace Range, and Include Specific Traces.                                                                                                          |
| First Trace Index                   | Integer        | When “Trace Range” is selected, this setting defines the first trace number of the trace range to be included in the analysis.                                                                                               |
| Last Trace Index                    | Integer        | When “Trace Range” is selected, this setting defines the last trace number of the trace range to be included in the analysis.                                                                                                |
| Trace Selection                     | Integer List   | When “Include Specific Traces” is selected, this setting defines which traces are included in the analysis. Click in the setting field and open up a trace selector dialog to select which trace numbers should be included. |
| How to Handle NaNs                  | Selection      | Options include Ignore NaNs, Error If Any NaNs, Error If All NaNs.                                                                                                                                                           |
| Write Results to Simulation Objects | Yes/No         | Yes indicates that the results will get written to the same object as the original source slot. The default is Yes.                                                                                                          |
| Write Results to Data Set           | Yes/No         | Yes indicates that the results will get written to a results data set object. The default is No.                                                                                                                             |
| Results Data Set Name               | Text           | The name of the results data set object to which the results will get written. If the object does not exist, it will get created.                                                                                            |

| Setting                          | Type   | Notes                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Save Results Data Set With Model | Yes/No | If Yes, the object specified in the Results Data Set Name setting will get saved in with the model when the model is saved. If No, the object and its slots and data will not get saved and will no longer be present when the model is reloaded. The default is Yes. This setting could change the object's current configuration if it already exists. |
| Results Data Set Slot Group      | Text   | Specify the name of the slot group on the results data set object in which the results slots will get added. If the slot group does not already exist, it will get created.                                                                                                                                                                              |
| Results Slot Tag                 | Text   | This text string is added to the results slot name, see below.                                                                                                                                                                                                                                                                                           |

**Warning:** The Compute Duration Curve action does not support aggregate series slots and accounting slots. Including aggregate series slots or accounting slots will result in an error.

Results are written to a set of table slots, one for each slot included in the analysis. Results can be written to the same object as the source slot, a results data set object, or both. When written to the same object as the source slot, the results slot's name is created using the template: "<slot's object>.<slot name> <results slot tag>". For example, if the slot "Res A.Inflow" participates in the analysis and the Results Slot Tag is "Duration Curve", then results associated with that slot will be written to the slot "Res A.Inflow Duration Curve".

When written to a results data set object, the results slot name is created using the template: "<results data set name>.<slot's object> <slot name><results slot tag>". For the same example of the slot "Res A.Inflow" and the Results Slot Tag of "Duration Curve" and a Results Data Set Name "Analysis Data Set," the results will be written to the slot "Analysis Data Set.Res A Inflow Duration Curve".

The action will create the results table slot if it does not already exist and, if it does exist, resize it and change column units and column labels as necessary.

## • Compute Scalar Regression

Perform a multiple linear regression analysis on an ensemble data set. (For more information on Ensemble Data Sets, see ["Data Analysis Tool \(DAT\)" in Output Utilities and Data Visualization](#).) The dependent variable and independent variables must be scalar slots. The primary outputs, the regression coefficients, are written to one table slot. Four other table slots are written out with statistical information about the solution quality. Details are provided below.

**Note:** This action was previously called "Compute Regression".

| Setting                              | Type           | Notes                                                                                                               |
|--------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------|
| Which Data Set To Analyze            | Selection      | Indicates how the ensemble data set will be specified. Select either "Most Recently Opened" or "Specified by Name". |
| Analysis Data Set Name               | Text           | If the above selection is 'Specified by Name', then this setting is used to specify the name of the data set.       |
| Dependent Variable Slot              | Slot Selection | Specify the dependent variable for the regression. It must be a scalar slot in the data set.                        |
| How To Specify Independent Variables | Selection      | Options are List of Slot Names and RPL Function                                                                     |

| Setting                             | Type                      | Notes                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Independent Variable Slots          | Slot Selection            | When "List of Slot Names" is selected for How to Specify Independent Variables, this setting becomes available to specify one or more independent variables for the regression. The independent variables must be scalar slots in the data set.                                                                                                          |
| Independent Variables RPL Function  | Single Function Selection | If the "RPL Function" option is selected in the How to Specify Independent Variables setting, use the RPL function chooser to select the function. The function must be in an open global functions set at the time of execution, must have no arguments, and must evaluate to a list of scalar slots that are in the data set.                          |
| How To Filter Traces                | Selection                 | How to filter the analysis data by traces. Options include All Traces, Trace Range, and Include Specific Traces. The resulting number of traces must be at least one more than the number of independent variables.                                                                                                                                      |
| First Trace Index                   | Integer                   | When "Trace Range" is selected, this setting defines the first trace number of the trace range to be included in the analysis.                                                                                                                                                                                                                           |
| Last Trace Index                    | Integer                   | When "Trace Range" is selected, this setting defines the last trace number of the trace range to be included in the analysis.                                                                                                                                                                                                                            |
| Trace Selection                     | Integer List              | When "Include Specific Traces" is selected, this setting defines which traces are included in the analysis. Click in the setting field and open up a trace selector dialog to select which trace numbers should be included.                                                                                                                             |
| How to Handle NaNs                  | Selection                 | Options include Ignore NaNs, Error If Any NaNs, Error If All NaNs. If Ignore NaNs is selected, if a NaN is encountered in any variable for a trace, that trace is omitted from the regression. An error will still be issued if the resulting number of traces is not at least one more than the number of independent variables.                        |
| Force Zero Intercept                | Yes/No                    | Specify whether to force the intercept (constant) in the regression to zero. The default is No. When yes is selected, output table slots will still contain a column for the intercept, but the values will be NaN.                                                                                                                                      |
| Write Results to Simulation Objects | Yes/No                    | Yes indicates that the results will get written to an existing non-data set object on the workspace. The default is Yes.                                                                                                                                                                                                                                 |
| Results Simulation Object Name      | Single Object Selection   | Select the object to which results will get written                                                                                                                                                                                                                                                                                                      |
| Write Results to Data Set           | Yes/No                    | Yes indicates that the results will get written to a results data set object. The default is No.                                                                                                                                                                                                                                                         |
| Results Data Set Name               | Text                      | The name of the results data set object to which the results will get written. If the object does not exist, it will get created.                                                                                                                                                                                                                        |
| Save Results Data Set With Model    | Yes/No                    | If Yes, the object specified in the Results Data Set Name setting will get saved in with the model when the model is saved. If No, the object and its slots and data will not get saved and will no longer be present when the model is reloaded. The default is Yes. This setting could change the object's current configuration if it already exists. |
| Results Data Set Slot Group         | Text                      | Specify the name of the slot group on the results data set object in which the results slots will get added. If the slot group does not already exist, it will get created.                                                                                                                                                                              |

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| Setting                             | Type | Notes                                                                                                                                                                                                 |
|-------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Coefficient Estimates Slot Name     | Text | Specify the name of the results slot for the primary result of the regression analysis, the regression coefficient estimates. The default name is "Coefficient Estimates."                            |
| Coefficient Variability Slot Name   | Text | Specify the name of the results slot that reports the variability of the regression coefficients: standard error, and lower and upper 95th percentile. The default name is "Coefficient Variability." |
| Coefficient Probabilities Slot Name | Text | Specify the name of the results slot that reports the significance of the regression coefficients: the T-statistic and P-value. The default name is "Coefficient Probabilities."                      |
| Summary Statistics Slot Name        | Text | Specify the name of the results slot that reports general statistics on the regression quality. The default name is "Summary Statistics."                                                             |
| Residuals Slot Name                 | Text | Specify the name of the results slot that reports the residuals from the regression analysis, the difference between the predicted and observed values. The default name is "Residuals."              |

The Compute Scalar Regression action uses the ordinary least squares approach to compute the regression coefficients for a linear equation of the following form.

$$DepVar = C_0 + C_1IndVar_1 + C_2IndVar_2 + \dots + C_{N-1}IndVar_{N-1} + C_NIndVar_N$$

The dependent variable and independent variables are scalar slot outputs from an ensemble that are part of an existing ensemble data set.

**Note:** The number of traces included in the regression must be at least one more than the number of independent variables. Otherwise an error will be issued.

### Regression Results Slots

Results can be written to an existing object on the workspace (not an ensemble data set object), a results data set object, or both.

The five results table slots from the Compute Scalar Regression action must be specified in the associated five settings. Leaving one of the result slot name settings empty will result in an error when the action is executed. If the results slot does not exist, it will be created when the script action executes. If the results slot does exist, it will be resized as necessary to fit the regression results. The units on each of the columns of the results slots will be modified as necessary to have the appropriate units from the regression analysis. For results slots on standard objects (non-data set objects) row and column labels can be edited. Any existing column and row labels will be maintained. Results slots written to results data set objects will always have default column and row labels that cannot be edited. The five results slots are described below.

- Coefficient Estimates Slot

This is the primary result from the regression. The table slot has a single row, a column for the computed intercept ( $C_0$ ), and a column for the computed coefficient for each of the independent variables ( $C_1$  to  $C_N$ ). The units of the intercept column are the units of the dependent variable. If the "Force Intercept to Zero" setting is "Yes," the intercept value will be NaN. The units of the independent variable columns are the units of the dependent variable divided by the unit of that independent variable. If the resulting unit type for one of the coefficients is not an existing unit type in RiverWare, the script action will fail with an error message.

- Coefficient Variability Slot

This result table slot has the same columns and units as the Coefficient Estimates slot. It contains three rows. The first row contains the standard error for each coefficient. The second and third rows contain the lower and upper bounds on the 95% confidence interval for each coefficient. If the “Force Intercept to Zero” setting is “Yes,” all of the values in the intercept column will be NaN.

- Coefficient Probabilities Slot

This result table slot has a column for the intercept and a column for each independent variable. The unit type for all columns is NONE. The table has two rows:

- T-statistic: The first row contains the Student’s t-statistic for each coefficient, which is computed as the estimated coefficient value divided by the standard error of the coefficient.

$$tStat_i = \frac{C_i}{StdError_i}$$

- P-Value: The second row contains the P-value for each coefficient, which is the probability of generating the t-statistic with a null hypothesis that the real coefficient is zero. A low P-value indicates that you can reject the null hypothesis. In other words, a low P-value indicates that the relationship between the independent variable and dependent variable is statistically significant, and thus, the independent variable is a meaningful predictor for your regression model. 0.05 is a commonly used threshold to determine if an estimated coefficient value is significant.

**Tip:** P-values are often on the order of  $10^{-4}$  or smaller. For this reason, it is recommend that the display unit settings for all columns in the Coefficient Probabilities Slot use the scientific format. This can be configured either in the individual slot configuration (select **View** then **Configure Columns Identically...** in the table slot dialog) or by adding an exception to the Unit Scheme (see “Unit Schemes” in *User Interface*).

- Summary Statistics Slot

This result table slot has a single row with a column for each of the following statistics.

- Multiple R: The multiple correlation coefficient between three or more variables, the square root of  $R^2$
- R Squared: The coefficient of determination, a measure of the proportion of the variance of the dependent variable that is explained by the independent variables.

$$R^2 = 1 - \frac{\sum (y_i - f_i)^2}{\sum (y_i - \bar{y})^2}$$

where  $y$  is the vector of observed values for the dependent variable,  $f$  is the vector of predicted values for the dependent variable from the regression model, and  $\bar{y}$  is the mean of the observed values for the dependent variable.

- Adjusted R Squared: Corrects for the effect of improving  $R^2$  as additional independent variables are added  
When a non-zero intercept (constant) is used, Adjusted R Squared is computed as

$$\overline{R^2} = 1 - (1 - R^2) \frac{n - 1}{n - k - 1}$$

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When the intercept is forced to zero (no constant term), Adjusted R Squared is computed as

$$\overline{R^2} = 1 - (1 - R^2) \frac{n}{n - k}$$

where  $n$  is the number of observations, and  $k$  is the number of independent variables.

- Standard Error: The average distance that observed values are from the predicted value
- Observations: The number of trials (traces) included in the regression analysis

The Standard Error column has the units of the dependent variable. All other columns have a unit type of NONE.

- Residuals Slot

This table slot reports on the difference between the observed value of the dependent variable and the predicted value for each observation given the coefficient estimates. It has one row for each observation (trace) included in the regression analysis and the following columns.

- Trace: The trace number of the observation (unit type of NONE)
- Observation: The observed value for the dependent variable for each trace, which is the dependent variable slot value in the ensemble data set for each trace (units of the dependent variable)
- Predicted: The predicted value for the dependent variable using the computed regression coefficients and the observed values for the independent variables (units of the dependent variable)
- Residual: The Predicted value minus the Observed value (units of the dependent variable)
- Standard Residual: The Residual divided by the standard deviation of the residuals (units of NONE)
- Independent Variables: One column for each independent variable with the observed value for that variable, which is the slot value in the ensemble data set for each trace (units of the independent variable)

### • Compute Series Regression

Perform a single or multiple linear regression analysis on series slot data. The dependent variable and independent variables must be series slots. The primary outputs, the regression coefficients, are written to a table slot. Another table slot is written with statistical information about the solution quality. A fitted series slot is also provided.

Results can be written to slots on a simulation object or a results data set object. These can be automatically plotted and managed in the Data Analysis Tool. For more information on Ensemble Data Sets, see [“Data Analysis Tool \(DAT\)” in Output Utilities and Data Visualization](#).

Details are provided below.

| Setting                              | Type                      | Notes                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dependent Variable Slot              | Slot Selection            | Specify the dependent variable for the regression. This is often called the “Y” variable or the variable you are trying to predict. It must be a series slot in the model.                                                                                                                                                                                                                                                          |
| How To Specify Independent Variables | Selection                 | Options are <b>List of Slot Names</b> or <b>RPL Function</b> .                                                                                                                                                                                                                                                                                                                                                                      |
| Independent Variable Slots           | Slot Selection            | When “List of Slot Names” is selected for How to Specify Independent Variables, this setting becomes available to specify one or more independent variables for the regression. These are often called the “X” variables or the observed values. The independent variables must be series slots in the model.                                                                                                                       |
| Independent Variables RPL Function   | Single Function Selection | If the “RPL Function” option is selected in the How to Specify Independent Variables setting, use the RPL function chooser to select the function. The function must be in an open global functions set at the time of execution, must have no arguments, and must evaluate to a list of series slots.                                                                                                                              |
| How to Handle NaNs                   | Selection                 | Options include <b>Ignore NaNs</b> , <b>Warn if Any NaNs</b> , <b>Error if Any NaNs</b> . If Ignore NaNs is selected, if a NaN is encountered in any time series value, that timestep’s values is omitted from the regression.<br><br>For all options, an error will be issued if the resulting number of timesteps is not at least one more than the number of independent variables. This can happen if a time series is all NaNs |
| Force Zero Intercept                 | Yes/No                    | Specify whether to force the intercept (constant) in the regression to zero. The default is No. When yes is selected, output table slots will still contain a column for the intercept, but the values will be NaN.                                                                                                                                                                                                                 |
| Write Results to Simulation Objects  | Yes/No                    | Yes indicates that the results will get written to an existing non-data set object on the workspace. The default is Yes.                                                                                                                                                                                                                                                                                                            |
| Results Simulation Object Name       | Single Object Selection   | Select the object to which results will get written                                                                                                                                                                                                                                                                                                                                                                                 |
| Write Results to Data Set            | Yes/No                    | Yes indicates that the results will get written to a results Data Set object. The default is No.                                                                                                                                                                                                                                                                                                                                    |
| Results Data Set Name                | Text                      | The name of the results Data Set object to which the results will get written. If the object does not exist, it will get created.                                                                                                                                                                                                                                                                                                   |
| Save Results Data Set With Model     | Yes/No                    | If Yes, the object specified in the Results Data Set Name setting will get saved in with the model when the model is saved. If No, the object and its slots and data will not get saved and will no longer be present when the model is reloaded. The default is Yes. This setting could change the object’s current configuration if it already exists.                                                                            |
| Results Data Set Slot Group          | Text                      | Specify the name of the slot group on the results data set object in which the results slots will get added. If the slot group does not already exist, it will get created.                                                                                                                                                                                                                                                         |
| Regression Statistics Slot Name      | Text                      | Specify the name of the results table slot for the primary result of the regression analysis, the regression coefficient estimates. The default name is “Regression Statistics.”                                                                                                                                                                                                                                                    |
| Summary Statistics Slot Name         | Text                      | Specify the name of the results table slot that reports general statistics on the regression quality. The default name is “Summary Statistics.”                                                                                                                                                                                                                                                                                     |

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| Setting                      | Type | Notes                                                                                                                                                                                                                            |
|------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fitted Data Slot Name        | Text | Specify the name of the results series slot that gives the fitted data. This will contain the predicted value for each timestep. The default name is "Fitted Data."                                                              |
| Residuals Slot Name          | Text | Specify the name of the results series slot that reports the residuals from the regression analysis. This is the difference between the predicted and observed values. The default name is "Residuals."                          |
| Standard Residuals Slot Name | Text | Specify the name of the results series slot that reports the standard residuals from the regression analysis. This is the residual divided by the standard deviation of the residuals. The default name is "Standard Residuals." |

The Compute Series Regression action uses the ordinary least squares approach to compute the regression coefficients for a linear equation of the following form.

$$DepVar = C_0 + C_1IndVar_1 + C_2IndVar_2 + \dots + C_{N-1}IndVar_{N-1} + C_NIndVar_N$$

The dependent variable and independent variables are the time series slot values.

**Note:** The number of time series values included in the regression must be at least one more than the number of independent variables. Otherwise an error will be issued.

### Input Slots

When the Compute Series Regression action is performed, the independent and dependent slots are copied to the destination object and placed in a Slot Group named Input Slots. These are read-only copies of the slots that are used for visualization by the DAT.

### Regression Results Slots

Results can be written to an existing object on the workspace (not a Data Set object), a results Data Set object, or both.

The five slots from the Compute Series Regression action must be specified in the associated five settings. Leaving one of the result slot name settings empty will result in an error when the action is executed. If the results slot does not exist, it will be created when the script action executes. If the results slot does exist, it will be resized as necessary to fit the regression results. The units on each of the columns of the table slots will be modified as necessary to have the appropriate units from the regression analysis. For results slots on standard objects (non-data set objects) row and column labels can be edited. For Table Slots, any existing column and row labels will be maintained. Results slots written to results data set objects will always have default column and row labels that cannot be edited. The five results slots are described below.

- Regression Summary

This is the primary result from the regression. The table slot has a single row, a column for the computed intercept ( $C_0$ ), and a column for the computed coefficient for each of the independent variables ( $C_1$  to  $C_N$ ). The units of the intercept column are the units of the dependent variable. If the "Force Intercept to Zero" setting is "Yes," the intercept value will be NaN. The units of the independent variable columns are the units of the dependent variable divided by the unit of that independent variable. If the resulting unit type for one of the coefficients is not an existing unit type in RiverWare, the script action will fail with an error message.

- Summary Statistics Slot

This result table slot has a single row with a column for each of the following statistics.



- Multiple R: The multiple correlation coefficient between three or more variables, the square root of  $R^2$
- R Squared: The coefficient of determination, a measure of the proportion of the variance of the dependent variable that is explained by the independent variables.

$$R^2 = 1 - \frac{\sum (y_i - f_i)^2}{\sum (y_i - \bar{y})^2}$$

where  $y$  is the vector of observed values for the dependent variable,  $f$  is the vector of predicted values for the dependent variable from the regression model, and  $\bar{y}$  is the mean of the observed values for the dependent variable.

- Adjusted R Squared: Corrects for the effect of improving  $R^2$  as additional independent variables are added  
When a non-zero intercept (constant) is used, Adjusted R Squared is computed as

$$\bar{R}^2 = 1 - (1 - R^2) \frac{n - 1}{n - k - 1}$$

When the intercept is forced to zero (no constant term), Adjusted R Squared is computed as

$$\bar{R}^2 = 1 - (1 - R^2) \frac{n}{n - k}$$

where  $n$  is the number of observations, and  $k$  is the number of independent variables.

- Standard Error: The average distance that observed values are from the predicted value
- Observations: The number of trials (traces) included in the regression analysis

The Standard Error column has the units of the dependent variable. All other columns have a unit type of NONE.

- Fitted Data

This series slot provides time series of the predicted values computed using the regression and the observed values. Plotting the fitted data and the dependent variable shows a graphical view of how well the regression performed. The residuals listed below gives the difference between the two slots.

- Residuals Slot

This series slot reports the difference between the fitted data and the dependent variable. It has one value for each timestep.

- Standard Residuals Slot

This series slot reports the standardized residual, which is each residual value divided by the standard deviation of the residuals.

## • Configure Excel Dataset

For the specified Database DMI Excel Dataset, configure various aspects including the workbook, run name type, run name, and run suffix.

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**Note:** This action includes functionality provided by Set Excel Dataset Run Name, but is more general. See [Set Excel Dataset Run Name](#), page 68 for more information.

| Setting                   | Type         | Notes                                                                                                                                                                                                                                                                                                                                         |
|---------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Excel Dataset             | Text         | Enter or choose the name of an Excel Dataset defined in the Dataset Manager. See <a href="#">“Excel Datasets” in Data Management Interface (DMI)</a> for details on Excel Datasets.                                                                                                                                                           |
| Set the Workbook          | Yes/No       | Select yes to set the workbook file path; the Workbook setting below is then shown. If no, the workbook in the dataset will not be changed.                                                                                                                                                                                                   |
| Workbook                  | File Chooser | When Set the Workbook is yes, enter the file path to the workbook. See <a href="#">“Filename Information”</a> on page 24 for details.                                                                                                                                                                                                         |
| Set the Run Name Type     | Yes/No       | Select Yes to set the Run Name Type; the Run Name Type setting below is then shown. If no, the Run Name Type will not be changed.                                                                                                                                                                                                             |
| Run Name Type             | Selection    | When Set the Run Name Type is “yes”, choose one of the available Run Name Types either: <ul style="list-style-type: none"><li>• Single Run Name</li><li>• Specify the Single Run Name when the DMI is invoked.</li><li>• Run with MRM Number (i.e. Run0, Run1, ...)</li><li>• Trace with MRM Trace Number (i.e. Trace1, Trace2,...)</li></ul> |
| Set the Single Run Name   | Yes/No       | Select Yes to set the Single Run Name; the Single Run Name setting below is then shown. If no, the Single Run Name will not be changed.                                                                                                                                                                                                       |
| Single Run Name           | Text         | This setting is shown when the above setting is set to yes. Enter a new Single Run Name or choose one from the list; see <a href="#">“Run Name Type” in Data Management Interface (DMI)</a> for details. In the Script Dashboard you can enter a new Single Run Name or choose one from the same list.                                        |
| Set the Single Run Suffix | Yes/No       | Select Yes to change the Single Run Suffix; the Single Run Suffix setting below is then shown. If no, the Single Run Suffix will not be changed.                                                                                                                                                                                              |
| Single Run Suffix         | Integer      | Specify the integer to use for the Single Run Suffix. This is appended to the Single Run Name. It is automatically incremented when using the Data Extractor Tool. See <a href="#">“Run Name Type” in Data Management Interface (DMI)</a> for more information.                                                                               |

#### • Configure MRM Index Sequential

For the specified MRM configuration, modify the configuration in terms of Index Sequential. This is shown on the Input tab in the MRM configuration. See [“Index Sequential” in Solution Approaches](#) for details.

| Setting           | Type    | Notes                                                                              |
|-------------------|---------|------------------------------------------------------------------------------------|
| MRM Configuration | Text    | Enter or choose the name of an MRM configuration defined in the MRM Run Controller |
| Number of Runs    | Integer | Specify the Number of Runs                                                         |
| Initial Offset    | Integer | Specify the Initial Offset                                                         |
| Interval          | Integer | Specify the Interval                                                               |

## • Configure MRM Input DMI

For the specified MRM configuration, modify the configuration in terms of Input DMIs, particularly the repeat count. This is shown on the Input tab in the MRM configuration. See [“Input Tab” in Solution Approaches](#) for details.

| Setting                    | Type    | Notes                                                                                        |
|----------------------------|---------|----------------------------------------------------------------------------------------------|
| MRM Configuration          | Text    | Enter or choose the name of an MRM configuration defined in the MRM Run Controller.          |
| DMI Name                   | Text    | Choose the Input DMI to configure. This should already be selected in the MRM configuration. |
| Repeat Count               | Integer | Specify the Repeat Count.                                                                    |
| Allow Repeat Count Editing | Yes/No  | Yes indicates that users can override the Repeat Count from the dashboard.                   |

## • Configure MRM Output

For the specified MRM configuration, modify the configuration in terms of the Output. This is shown on the Output tab in the MRM configuration. See [“Output Tab” in Solution Approaches](#) for details.

| Setting                    | Type         | Notes                                                                                   |
|----------------------------|--------------|-----------------------------------------------------------------------------------------|
| MRM Configuration          | Text         | Enter or choose the name of an MRM configuration defined in the MRM Run Controller.     |
| Control File               | File Chooser | Enter the file path. See <a href="#">“Filename Information”</a> on page 24 for details. |
| Generate CSV Files         | Yes/No       | Specify whether or not to generate CSV output                                           |
| Generate NetCDF Files      | Yes/No       | Specify whether or not to generate NetCDF output                                        |
| Allow Control File Editing | Yes/No       | Specify if you would like the control file to be editable on the dashboard.             |

## • Configure MRM Traces

For the specified MRM configuration, modify the first trace and number of traces.

| Setting                        | Type    | Notes                                                                               |
|--------------------------------|---------|-------------------------------------------------------------------------------------|
| MRM Configuration              | Text    | Enter or choose the name of an MRM configuration defined in the MRM Run Controller. |
| First Trace                    | Integer | Specify the first trace of the range of traces to be simulated.                     |
| Number of Traces               | Integer | Specify the number of traces to simulate.                                           |
| Allow Number of Traces Editing | Yes/No  | Yes indicates that users can override the Number of Traces from the dashboard.      |

## • Configure Trace Directory DMI

For the specified Trace Directory DMI, set the Trace Number to use when the DMI is executed outside of MRM. See “[Trace Directory DMI](#)” in *Data Management Interface (DMI)* for details.

| Setting                    | Type    | Notes                                                                      |
|----------------------------|---------|----------------------------------------------------------------------------|
| DMI Name                   | Text    | Choose the Input DMI to configure. It must be a Trace Directory DMI.       |
| Trace Number               | Integer | Specify the Trace Number to use if the DMI is executed outside of MRM.     |
| Allow Trace Number Editing | Yes/No  | Yes indicates that users can override the Trace Number from the dashboard. |

## • Create Ensemble Data Set

Create an ensemble data set object from an RDF file. This action will create an ensemble data set object on the workspace and will add the data set to the Data Analysis Tool. For more information on Ensemble Data Sets, see “[Ensemble Data Set Analysis](#)” in *Output Utilities and Data Visualization*.

| Setting                | Type   | Notes                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RDF File Name          | Text   | Enter the file path. See “ <a href="#">Filename Information</a> ” on page 24 for details.                                                                                                                                                                                                                                                                                        |
| Ensemble Data Set Name | Text   | The name to give to the data set object on the workspace when it is created                                                                                                                                                                                                                                                                                                      |
| Save With Model        | Yes/No | If Yes, the ensemble data set object created on the workspace by this action will get saved in the model when the model is saved. If No, the object and its slots and data will not get saved and will no longer be present when the model is reloaded. The default is Yes. Ensemble data sets can be very large, so it may not always be desirable to save them with the model. |

If an ensemble data set or any object with the same name already exists, an error will be issued. A [Delete Ensemble Data Set](#) action can be included before this action to always guarantee that any existing ensemble data set with this name will get deleted before trying to create the new ensemble data set.

## • Create Slot Cache

Creates a slot cache that contains the current values of all series slots on the workspace, in the range of the current controller. See “[Slot Cache](#)” in *User Interface* for details on the cache.

**Note:** The slot cache is under development. Contact [riverware-support@colorado.edu](mailto:riverware-support@colorado.edu) for details and the current status of this feature.

| Setting | Type | Notes |
|---------|------|-------|
| None    |      |       |

## • Create Snapshot

Create a Snapshot of the specified slots.

| Setting                    | Type           | Notes                                                                                                                                                                                                                    |
|----------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slots                      | Slot Selection | Enter or choose one or more slots for which you wish to take a snapshot.                                                                                                                                                 |
| Snapshot Base Name         | Text           | Specify the name of the snapshot object to create.                                                                                                                                                                       |
| Allow Base Name Editing    | Yes/No         | Yes indicates that the users can override the default Base Name from the dashboard.                                                                                                                                      |
| Append Timestamp?          | Yes/No         | Choose whether to have a timestamp appended to the Base Name for the object's name. If Yes, the timestamp, reflecting the time of snapshot creation based on system time, will be appended in the format YYYYMMDDHHMMSS. |
| Replace Existing Snapshot? | Yes/No         | Choose to replace a snapshot of the same name. If No, then a unique name for the snapshot will be created by appending an integer to the Base Name.                                                                      |

## • Delete Custom Slots

Delete the selected custom slots.

| Setting          | Type           | Notes                                                                                                                                                                                                                                                                                                                                                                             |
|------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slots            | Slot Selection | The custom slots to delete. Only custom slots can be deleted. If simulation slots are selected, they will not be deleted.<br><br><b>Note:</b> The slot selection is stored in the script as a list of strings. This preserves the list of slots even after it has deleted them allowing you to import objects of the same name and the Delete Custom Slots will continue to work. |
| Confirm Deletion | Yes/No         | Yes indicates that a dialog should be presented to confirm removal of the slots. No indicates the slots will be deleted without a confirmation.                                                                                                                                                                                                                                   |

## • Delete Ensemble Data Set

Remove an ensemble data set. This will delete the associated object and slots from the workspace. For more information on Ensemble Data Sets, see [“Organizing and Deleting Data Sets” in Output Utilities and Data Visualization](#).

| Setting                           | Type      | Notes                                                                                                                                                                                                                                    |
|-----------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Which Ensemble Data Set to Delete | Selection | Indicates how the ensemble data set will be specified. Select either “Most Recently Opened” or “Specified by Name”.                                                                                                                      |
| Ensemble Data Set Name            | Text      | If the above selection is 'Specified by Name', then this setting is used to specify the name of the data set.                                                                                                                            |
| Data Set Must Exist               | Yes/No    | Choose whether the Ensemble Data Set must exist or not. If “No” (the default), the set does not need to exist and the action will succeed even if the set doesn't exist. If “Yes”, an error will be issued if there is no set to delete. |

## • Delete Link

Delete a link between the two specified slots. For more information on links, see [“Linking Slots on Objects” in Building a Model](#). It is not an error if the two slots are not linked. See also the complimentary action to [“Link Slots”](#) on page 63.

**Note:** Interactively, when you delete a link and it is the only link between those two objects, and there are underlying accounting supplies, you are presented with a dialog to choose to **Delete**, **Delete and Show Supply Manager**, or **Cancel** the delete. When using the Delete Link action, the script will delete the link with only a simple confirmation (if configured that way) but won’t delete the accounting supplies.

| Setting          | Type           | Notes                                                                                                                                         |
|------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Slot 1           | Slot Selection | The slot at one end of the link.                                                                                                              |
| Slot 2           | Slot Selection | The slot at the other end of the link.                                                                                                        |
| Confirm Deletion | Yes/No         | Yes indicates that a dialog should be presented to confirm removal of the link.No indicates the links will be deleted without a confirmation. |

## • Delete Objects

Delete the selected objects.

| Setting          | Type             | Notes                                                                                                                                                                                                                                                                  |
|------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objects          | Object Selection | The objects to delete.<br><br><b>Note:</b> The objects are stored in the script as a list of strings. This preserves the list of objects even after it has deleted them, allowing you to import objects of the same name and the Delete Objects will continue to work. |
| Confirm Deletion | Yes/No           | Yes indicates that a dialog should be presented to confirm removal of the objects. No indicates the objects will be deleted without a confirmation.                                                                                                                    |

## • Delete Optimization Restore Points

Delete all of the optimization restore points currently in existence. For more information on optimization restore points, see [“Restore Point” in Optimization](#).

| Setting | Type | Notes |
|---------|------|-------|
| None    |      |       |

## • Delete Snapshots

Delete any snapshot objects older than a specified number of days.

| Setting                     | Type    | Notes                                                                                                                                                                                                                                                                                     |
|-----------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Delete Snapshots Older Than | Numeric | Specify the number of days. Any snapshots older than this number of days, based on the current system time and the timestamp of the snapshot creation, will be deleted. Any decimal fraction of days may be entered. For example, 0.5 days will delete all snapshots older than 12 hours. |
| Confirm Deletion            | Yes/No  | Yes indicates that a dialog should be presented to confirm removal of the snapshots. No indicates the snapshots will be deleted without a confirmation.                                                                                                                                   |

## • Divider

This item adds a divider to the Script Editor and Script Dashboard. It does not do anything in script execution; it is strictly for to provided a visual separation between actions.

- The Display Text can be used to add text to the divider.
- The Background Color can be used to select the background color for the divider. (This was previously called the “Divider Color”)

## • Enable Dispatching

For the specified objects, enable or disable the dispatch method execution. When disabled, the object icons on the workspace and in other dialogs appear dimmed.

| Setting            | Type                                     | Notes                                                                                                        |
|--------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Objects            | Object Selection                         | Choose the desired objects using the object selector.                                                        |
| Enable or Disable? | Enable Dispatching / Disable Dispatching | Specify <b>Enable</b> to enable dispatching. Specify <b>Disable</b> to disable dispatching of those objects. |

## • Enable MRM Distributed Runs

For the specified MRM configuration, specify whether to distribute concurrent runs to multiple processors on the same machine. See “[Distributed Concurrent Runs](#)” in *Solution Approaches* for details on distributed MRM.

| Setting            | Type             | Notes                                                                                                                      |
|--------------------|------------------|----------------------------------------------------------------------------------------------------------------------------|
| MRM Configuration  | Text             | Enter or choose the name of an MRM configuration defined in the MRM Run Controller.                                        |
| Enable or Disable? | Enable / Disable | Specify <b>Enable</b> to run in distributed mode. Specify <b>Disable</b> to run the concurrent runs on a single processor. |

## • Enable RPL Item

Enable or disable a RPL item (policy group, rule, method, or goal).

## Chapter 1

### Script Management

| Setting            | Type             | Notes                                                      |
|--------------------|------------------|------------------------------------------------------------|
| Set                | Single Selection | Choose the set to which the RPL item belongs.              |
| Item               | Single Selection | Use the RPL item chooser to select the item.               |
| Enable or Disable? | Yes/No           | Specify Yes to Enable or No to disable the specified item. |

#### • Evaluate Expression Slots

Evaluate a set of expression slots.

| Setting | Type           | Notes                                |
|---------|----------------|--------------------------------------|
| Slots   | Slot Selection | Choose the slots using the selector. |

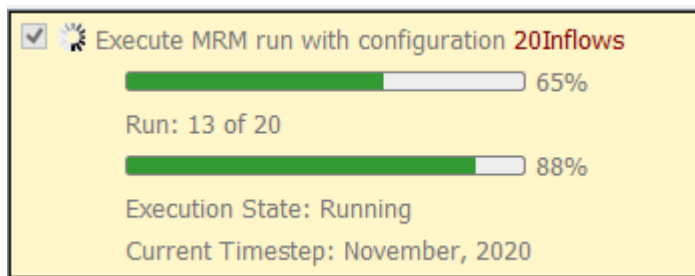
#### • Execute DMI

Invoke a DMI or DMI group.

| Setting  | Type | Notes                                                |
|----------|------|------------------------------------------------------|
| DMI Name | Text | Enter the DMI name or choose the DMI using the menu. |

#### • Execute MRM Run

Execute the runs for the specified MRM configuration. The script dashboard displays the overall and/or individual run progress and status. For example, for a concurrent MRM, the dashboard shows the following:



For distributed MRM, the run status will be shown in the separate Distributed MRM Controller, not in the script dashboard.



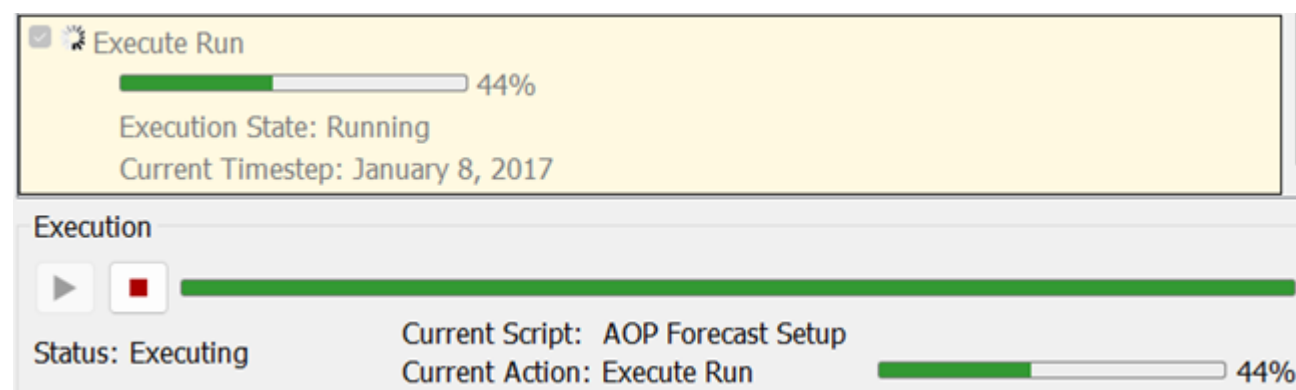
| Setting                             | Type      | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MRM Configuration                   | Text      | Enter the name of an MRM configuration you wish to run.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Allow Distributed MRM               | Yes/No    | If "Yes" and the MRM configuration is configured to use distributed MRM, the multiple run will be executed via distributed MRM. (See <a href="#">"Distributed Concurrent Runs" in Solution Approaches</a> ) If "Yes" and the MRM configuration is not configured to use distributed MRM, this setting does nothing. If "No" and the MRM configuration is configured to use distributed MRM, an error will be issued.                                                                                                                                                                                                                                                                                                                                                                                      |
| Show the distributed MRM interface  | Yes/No    | This setting is available when "Yes" is selected for Allow Distributed MRM setting. This setting indicates whether to open the interface for the Distributed MRM Controller. The default is Yes. Typically this is the desired behavior in interactive mode, but in batch, it may not be necessary or desired to see the user interface open.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Close the distributed MRM interface | Selection | This setting is available when "Yes" is selected for Show the distributed MRM interface. The options are Always, Never, and After a successful run. When "Always" is selected, the Distributed MRM Controller will automatically close after the multiple run completes, and the script execution will continue. When "Never" is selected, the Distributed MRM Controller must be closed manually after the multiple run completes before the script execution will advance. When "After a successful run" is selected, when the multiple run completes successfully, it behaves as for the "Always" selection. If the multiple run fails, the Distributed MRM Controller remains open until closed manually. This provides an interface to view the diagnostic messages from the failed distributed run. |

## • Execute Run

Run the model using the currently selected controller.

| Setting | Type | Notes |
|---------|------|-------|
| None    |      |       |

The script dashboard displays the run progress and status. The following image shows the progress bar on the action row and a separate progress bar in the overall script status area. This is particularly useful in the Execute Run is part of a subscript or set of collapsed actions as it will always show the progress of the run.



## • Execute Script

Execute a script from within this script.

**Note:** The script called by this action can have additional Execute Script actions, up to 5 levels deep. Rephrased, a script can call a script that calls another script, but only 5 Execute Script levels in total.

See [“Display of Execute Script and Sub-actions”](#) on page 90 for more information on how sub-actions are displayed.

| Setting                        | Type   | Notes                                                                                                                                                                                                                                  |
|--------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Script Name                    | Text   | Type the script name or choose the desired single script from the script selector.                                                                                                                                                     |
| Expand in Dashboard by default | Yes/No | If 'Yes', then the sub-actions will be expanded by default on the Script Dashboard. If 'No', the sub-actions on the dashboard will be collapsed. Regardless of the setting, the actions can be collapsed or expanded on the dashboard. |

## • Export DMIs

Export Data Management Interfaces (DMIs) to a file and optionally include name maps and datasets. See [“Export” in \*Data Management Interface \(DMI\)\*](#) for more information. Also, see [“Import DMIs”](#) on page 62 for information on the script action to import the DMI.

| Setting                        | Type   | Notes                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File Name                      | Text   | Enter the file path to create or overwrite. See <a href="#">“Filename Information”</a> on page 24 for details.                                                                                                                                                                                                                                        |
| DMI Names                      | Text   | Select one or more DMIs using the DMI Selector.                                                                                                                                                                                                                                                                                                       |
| Include Datasets and Name Maps | Yes/No | Select Yes to also export the Datasets and Name Maps referenced by Database DMI. Select No to export only the Database DMI and not the referenced Datasets and Name Maps. See <a href="#">“Export” in <i>Data Management Interface (DMI)</i></a> for more information. This setting has no effect on Control File-Executable or Trace Directory DMIs. |

## • Export Objects

Export Objects to a file and optionally include links, object attributes, slot notes and accounting items. See [“Export” in \*User Interface\*](#) for more information on exporting objects. The options described in that section correspond to settings in this script action. Also, see [“Import Objects”](#) on page 62 for information on the script action to import Objects from a file.

| Setting            | Type             | Notes                                                                                                            |
|--------------------|------------------|------------------------------------------------------------------------------------------------------------------|
| File Name          | Text             | Enter the file path to create or overwrite. See <a href="#">“Filename Information”</a> on page 24 for details.   |
| Objects            | Object Selection | Choose the desired objects using the object selector. Note, exporting clusters is not supported within a script. |
| Include Links      | Yes/No           | Yes indicates workspace links between slots will also be exported.                                               |
| Include Attributes | Yes/No           | Yes indicates object attributes will be exported.                                                                |
| Include Text Notes | Yes/No           | Yes indicates series slot text notes will be exported.                                                           |

| Setting                        | Type   | Notes                                                                              |
|--------------------------------|--------|------------------------------------------------------------------------------------|
| Include Storage Accounts       | Yes/No | Yes indicates storage accounts on specified reservoir objects will be exported.    |
| Include Diversion Accounts     | Yes/No | Yes indicates diversion accounts on the specified objects will be exported.        |
| Include Instream Flow Accounts | Yes/No | Yes indicates instream flow accounts on specified control points will be exported. |
| Include Passthrough Accounts   | Yes/No | Yes indicates passthrough accounts on specified objects will be exported.          |
| Include Supplies               | Yes/No | Yes indicates supplies (accounting links) between accounts will be exported.       |

### • Export Optimization Restore Point

Export the specified optimization restore point to a file. For more information on optimization restore points, see [“Restore Point” in Optimization](#) and see [“Export a Restore Point” in Optimization](#) for details on exporting a restore point. Also see [“Import Optimization Restore Point”](#) on page 63 for details to import a restore point from a file. The named restore point does not need to exist when the script action is configured, but it must exist when the script action is executed. Otherwise an error will be issued.

| Setting            | Type   | Notes                                                                                                                                                                                 |
|--------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File Name          | Text   | Enter the file path to create or overwrite. See <a href="#">“Filename Information”</a> on page 24 for details. Optimization restore point files typically use the .orp file extension |
| Restore Point Name | Text   | Enter the name of the restore point to be exported. The restore point does not need to exist when the setting is configured, but it must exist when the script action is executed.    |
| Confirm Overwrite? | Yes/No | Yes requires interactive confirmation when exporting the restore point to a preexisting file.                                                                                         |

### • Export Table Slot as CSV

Export the selected table slot to a comma-separated values (CSV) file. The data will always be exported to the CSV file with display precision. The first column of the CSV file will be the row headers of the table slot. The first row of the CSV file will be the column headers of the table slot with units included in parentheses. Periodic table slots and statistical table slots can be exported to CSV files as well as standard table slots.

| Setting    | Type           | Notes                                                                                                                           |
|------------|----------------|---------------------------------------------------------------------------------------------------------------------------------|
| File Name  | Text           | Enter the file path for the CSV file to create or overwrite. See <a href="#">“Filename Information”</a> on page 24 for details. |
| Table Slot | Slot Selection | Choose the table slot to export using the selector.                                                                             |

## • Generate Output Device

Generate the specified Output Device. This could be a file based device like a Model Report or a graphical device like a Plot Page.

| Setting     | Type | Notes                                                                   |
|-------------|------|-------------------------------------------------------------------------|
| Device Name | Text | Enter or choose the name of a output device (plot, file, report, chart) |

## • Global Time Scroll

Scroll time and series displays to the specified symbolic date/time.

| Setting            | Type      | Notes                                                                                   |
|--------------------|-----------|-----------------------------------------------------------------------------------------|
| Scroll to DateTime | Datetime  | See <a href="#">“Datetime Information”</a> on page 22 for details.                      |
| Allow Editing      | Selection | Yes indicates that users should be allowed to override the start date in the Dashboard. |

## • Import DMIs

Import Data Management Interfaces (DMIs) from a file and specify how to handle conflicts. See [“Import” in Data Management Interface \(DMI\)](#) for more information on the feature. See also [“Export DMIs”](#) on page 60 for information on the script action to export the Database DMI.

| Setting           | Type      | Notes                                                                                                                                                                                                                                                                        |
|-------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File Name         | Text      | Enter the file path to the file containing the exported DMIs. See <a href="#">“Filename Information”</a> on page 24 for details.                                                                                                                                             |
| Resolve Conflicts | Selection | Select either “Import the DMI items, replacing those in the model” or “Import the DMI items with unique names”. These choices correspond to the <b>Import conflicts with a unique name</b> option described in <a href="#">“Import” in Data Management Interface (DMI)</a> . |

## • Import Objects

Import Objects from a file and specify how to import object attributes, slot notes and object clusters, and how to locate the imported objects. See [“Import” in User Interface](#) for more information on importing objects. The options described in that section correspond to the settings on this action. See also [“Export Objects”](#) on page 60 for information on the script action to export Objects.

| Setting           | Type   | Notes                                                                                                                               |
|-------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------|
| File Name         | Text   | Enter the file path to the file containing the exported objects. See <a href="#">“Filename Information”</a> on page 24 for details. |
| Import Attributes | Yes/No | Yes indicates object attributes will be imported.                                                                                   |
| Import Text Notes | Yes/No | Yes indicates series slot text notes will be imported.                                                                              |

| Setting         | Type             | Notes                                                                                                                                                               |
|-----------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Object Clusters | Selection        | Specify how exported object clusters are imported. There are two choices “Import member objects and cluster configuration” and “Import cluster configuration only”. |
| Import Location | Selection        | Choose either “Use exported coordinates” or “Near object”.                                                                                                          |
| Near Object     | Object Selection | When the Import Location setting above is “Near object”, select an object. The imported objects will be imported just to the right of the specified object.         |

## • Import Optimization Restore Point

Import an optimization restore point from the specified file. For more information on optimization restore points, see [“Restore Point” in Optimization](#) and see [“Import a Restore Point” in Optimization](#) for details on importing a restore point. Also see [“Export Optimization Restore Point”](#) on page 61 for details to export a restore point to a file.

| Setting                  | Type      | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File Name                | Text      | Enter the file path to the file containing the restore point. See <a href="#">“Filename Information”</a> on page 24 for details.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Rename New Restore Point | Yes/No    | No indicates that the restore point will be imported with the name that it has in the file, the name it had when exported.<br><br>Yes allows you to give the restore point a different name on import than it has in the file.                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| New Restore Point Name   | Text      | This setting is available when “Yes” is selected for “Rename New Restore Point.” Enter the desired name to give the restore point on import.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Name Conflict Strategy   | Selection | Choose from three options for how to handle cases that the restore point being imported has the same name as an existing restore point. <ul style="list-style-type: none"> <li>• Replace the Existing Restore Point - overwrites an the existing restore point with the restore point being imported</li> <li>• Rename the Existing Restore Point - the existing restore point will be given a new name to make it unique, and the restore point being imported will maintain the name from the file (or the name specified in the New Restore Point Name setting)</li> <li>• Treat as an error - the script execution will terminate with an error</li> </ul> |

## • Link Slots

Link the two specified slots. The slots must be physical slots (non-accounting) on two different objects. For more information on links, see [“Linking Slots on Objects” in Building a Model](#). It is not an error if the two slots are already linked. See also the complimentary action to [“Delete Link”](#) on page 56.

| Setting | Type           | Notes                                                                               |
|---------|----------------|-------------------------------------------------------------------------------------|
| Slot 1  | Slot Selection | The first slot to link. Only physical (non-accounting) series slots can be linked.  |
| Slot 2  | Slot Selection | The second slot to link. Only physical (non-accounting) series slots can be linked. |

## • Load Goal Set

Open and load a goal set (for Optimization runs).

| Setting             | Type   | Notes                                                                                                                                          |
|---------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------|
| File name           | Text   | Enter the file path. See <a href="#">"Filename Information"</a> on page 24 for details.                                                        |
| Confirm Replacement | Yes/No | Yes indicates that if the specified set is already open, a dialog should be presented to confirm replacement of the existing set.              |
| Show Dialog         | Yes/No | When this setting is Yes, the RPL set dialog is shown when the action is executed. Otherwise, the set is loaded into the model, but not shown. |

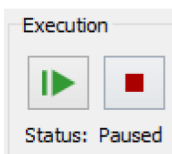
## • Load Ruleset

Open and load a ruleset (for Rulebased Simulation runs).

| Setting             | Type   | Notes                                                                                                                                          |
|---------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------|
| File name           | Text   | Enter the file path. See <a href="#">"Filename Information"</a> on page 24 for details.                                                        |
| Confirm Replacement | Yes/No | Yes indicates that if the specified set is already open, a dialog should be presented to confirm replacement of the existing set.              |
| Show Dialog         | Yes/No | When this setting is Yes, the RPL set dialog is shown when the action is executed. Otherwise, the set is loaded into the model, but not shown. |

## • Memo

Pause script execution and optionally open a confirmation dialog with your user specified message. While script execution is paused, you can interact with RiverWare or other applications in ways not directly supported by scripts. Select the script **Continue** button to resume the script execution.

| Setting                           | Type            | Notes                                                                                                                                                                                                                             |
|-----------------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Explanation                       | Multi-line text | Enter the desired text. This will optionally be displayed in the memo dialog and or the Dashboard dialog along with the Display Text.                                                                                             |
| Pause Execution                   | Yes/No          | Yes indicates that script execution will pause when the memo is reached. Select the <b>Continue</b> button to resume the Script execution.<br> |
| Show Memo Dialog                  | Yes/No          | Yes indicates that memo execution will open a dialog which presents the memo display and explanation texts.                                                                                                                       |
| Show Explanation in the Dashboard | Yes/No          | Yes indicates that the explanation text will be displayed with the display text in the Dashboard.                                                                                                                                 |

## • Move Slots to Object

Move custom slots from one or more objects to another object. For more information on moving slots, see [“Move Slots to Object” in User Interface](#).

| Setting              | Type                      | Notes                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| How to Specify Slots | Selection                 | There are two options: <ul style="list-style-type: none"> <li>All Slots on an Object: Use settings below to select the desired object. All slots on that object will be moved to the Destination Object.</li> <li>Select Slots: Use a slot selection to define the list of slots to move. The slots can be on one or more objects. The slots will be moved to the Destination Object.</li> </ul> |
| Source Object        | Object Selection (single) | This setting is shown when “All Slots on Object” is selected above. Select the object that contains the slots to move.                                                                                                                                                                                                                                                                           |
| Slots                | Slot Selection (multiple) | This setting is shown when “Select Slots” is selected above. Select the list of slots on one or more object to move.                                                                                                                                                                                                                                                                             |
| Destination Object   | Object Selection (single) | Specify the destination object; that is, where you want to move the slots.                                                                                                                                                                                                                                                                                                                       |
| Slot Groups Too      | Yes/No                    | If yes, if the slots are in slot groups, the groups will be copied too. They will not be deleted from the source slot. Also, if the group already exists on the destination object, the slots will be moved to that group.<br>If no, slot groups will not be copied to the destination object.                                                                                                   |

## • Open Data Analysis Tool

Open the Data Analysis Tool and bring it to the top of the windows. For more information on this tool, see [“Accessing the Data Analysis Tool” in Output Utilities and Data Visualization](#).

**Note:** This action was previously called “Open Ensemble Data Tool”. That action will automatically update to the new name.

| Setting | Type | Notes |
|---------|------|-------|
| None    |      |       |

## • Open File

Open the specified external file using the windows file association. If you would like to pause script execution after opening the file, add a Memo action, described at [“Memo”](#) on page 64, after the Open File action.

| Setting   | Type | Notes                                                                                                                                                                                                                                     |
|-----------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File name | Text | Enter the file path. See <a href="#">“Filename Information”</a> on page 24 for details. The file will be opened using the application associated with the file's extension in Windows. For example, temp.xlsx will be opened using Excel. |

## • Open Global Functions Set

Open a global function set.

| Setting             | Type   | Notes                                                                                                                                          |
|---------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------|
| File name           | Text   | Enter the file path. See <a href="#">“Filename Information”</a> on page 24 for details.                                                        |
| Confirm Replacement | Yes/No | Yes indicates that if the specified set is already open, a dialog should be presented to confirm replacement of the existing set.              |
| Show Dialog         | Yes/No | When this setting is Yes, the RPL set dialog is shown when the action is executed. Otherwise, the set is opened into the model, but not shown. |

## • Open Objects

Open the specified objects’ dialogs.

| Setting | Type             | Notes                       |
|---------|------------------|-----------------------------|
| Objects | Object Selection | Choose the objects to open. |

## • Open SCT

Open a System Control Table (SCT) with the specified filename.

| Setting   | Type | Notes                                                                                   |
|-----------|------|-----------------------------------------------------------------------------------------|
| File Name | Text | Enter the file path. See <a href="#">“Filename Information”</a> on page 24 for details. |

## • Open Slots

Open the specified slots.

| Setting | Type           | Notes                                |
|---------|----------------|--------------------------------------|
| Slots   | Slot Selection | Choose the slots using the selector. |

## • Reorder RPL Set

Reorder the items (rules, goals, statements) in a RPL set according to priority or index defined on the specified table slot.

| Setting    | Type                  | Notes                                                                                                                                |
|------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Set        | Single Selection      | Choose the desired RPL set.                                                                                                          |
| Order Slot | Single Slot Selection | Choose the table slot which specifies the new RPL set ordering. See <a href="#">“RPL Reorder Set Action”</a> on page 24 for details. |



## • Replace Initialization Rules Set From File

Replace the existing Initialization Rules Set from a file. Any changes made to the Initialization Rules in the model will be lost. Use with caution!

| Setting             | Type   | Notes                                                                                                                                          |
|---------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------|
| File name           | Text   | Enter the file path. See <a href="#">“Filename Information”</a> on page 24 for details.                                                        |
| Confirm Replacement | Yes/No | Yes indicates that a dialog should be presented to confirm replacement of the existing Initialization Rules Set.                               |
| Show Dialog         | Yes/No | When this setting is Yes, the RPL set dialog is shown when the action is executed. Otherwise, the set is replaced in the model, but not shown. |

## • Save Model

Save the model to a file.

| Setting                 | Type    | Notes                                                                                   |
|-------------------------|---------|-----------------------------------------------------------------------------------------|
| File Name               | Text    | Enter the file path. See <a href="#">“Filename Information”</a> on page 24 for details. |
| Include Output Values   | Yes/No  | Yes indicates output values will be saved.                                              |
| Output Values Precision | Integer | Indicates the precision to use for output values.                                       |
| Confirm Overwrite       | Yes/No  | Yes requires interactive confirmation when saving the model to a preexisting file.      |

## • Set Account Method

Set the selected method in one of an account’s method categories.

| Setting      | Type | Notes                                 |
|--------------|------|---------------------------------------|
| Account Name | Text | Choose the account using the selector |
| Category     | Text | Choose the category on the account    |
| Method       | Text | Choose the desired method             |

## • Set Accounting Period

Set the time range of the accounting system (i.e., the Begin Accounting Period and End Accounting Period dates).

| Setting    | Type     | Notes                                                              |
|------------|----------|--------------------------------------------------------------------|
| Start Date | Datetime | See <a href="#">“Datetime Information”</a> on page 22 for details. |
| End Date   | Datetime | See <a href="#">“Datetime Information”</a> on page 22 for details. |

## • Set Controller

Set the current controller.

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| Setting                 | Type      | Notes                                                                                                                                                                                                                                       |
|-------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Controller              | Selection | Choose the controller from the list                                                                                                                                                                                                         |
| Run From Restore Point? | Yes/No    | Available only when the Optimization controller is selected. Yes indicates that the optimization run will use a Restore Point to give the optimization an advanced start (see <a href="#">“Restore Point” in Optimization</a> for details). |
| Restore Point Name      | Text      | Available only when the Optimization controller is selected. Enter or choose the name of the optimization restore point to use. If the “Run From Restore Point?” setting is “No,” this setting will be ignored.                             |

#### • Set Dataset File

Set the DSS file for a specified DSS Dataset or the Excel workbook for a specified Excel Dataset.

| Setting            | Type      | Notes                                                                                                                    |
|--------------------|-----------|--------------------------------------------------------------------------------------------------------------------------|
| Dataset            | Text      | Enter or choose the name of a DSS Dataset or an Excel Dataset defined in the Dataset Manager.                            |
| File Name          | Text      | Enter or choose the DSS file or Excel workbook.                                                                          |
| Allow File Editing | Selection | Yes means the file can be entered or chosen in the Dashboard. No means that the file cannot be changed in the Dashboard. |

#### • Set Evaluation Time for Expression Slots

Set the evaluation time for the selected expression slots. For more information on evaluation time, see [“Evaluation Time” in User Interface](#).

| Setting         | Type           | Notes                                                                                                                                                                                                                                                                                       |
|-----------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slots           | Slot Selection | Choose the expression slots (series or scalar) using the selector. Non expression slots are silently skipped. The slot selection can include wildcards.                                                                                                                                     |
| Evaluation Time | Selection      | Select from the six possible evaluation times: <ul style="list-style-type: none"><li>• Never</li><li>• Only Interactively</li><li>• Beginning of Run</li><li>• End of Run</li><li>• Beginning of Timestep, current timestep only</li><li>• End of Timestep, current timestep only</li></ul> |

#### • Set Excel Dataset Run Name

For the specified Database DMI Excel Dataset, set the run name type and/or specify a new single run name.

**Note:** The Configure Excel Dataset action provides more general functionality to set many aspects of an Excel Dataset. See [Configure Excel Dataset](#), page 51 for more information.

| Setting         | Type      | Notes                                                                                                                                                                                                                                                                                                              |
|-----------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Excel Dataset   | Text      | Enter or choose the name of an Excel Dataset defined in the Dataset Manager. See <a href="#">“Excel Datasets” in Data Management Interface (DMI)</a> for details on Excel Datasets.                                                                                                                                |
| Run Name Type   | Selection | Choose one of the available Run Name Types either: <ul style="list-style-type: none"> <li>• Single Run Name</li> <li>• Specify the Single Run Name when the DMI is invoked.</li> <li>• Run with MRM Number (i.e. Run0, Run1, ...)</li> <li>• Trace with MRM Trace Number (i.e. Trace1, Trace2,...)</li> </ul>      |
| Single Run Name | Text      | This setting is shown when the above setting is set to Single Run Name. Enter a new Single Run Name or choose one from the list; see <a href="#">“Run Name Type” in Data Management Interface (DMI)</a> for details. In the Script Dashboard you can enter a new Single Run Name or choose one from the same list. |

### • Set Init. Rules Exec. Flag

Set the Execute Initialization Rules run control flag. This flag controls whether or not the initialization rules are executed as part of the run. The flag is independent for each major controller, Simulation, Rulebased Simulation and Optimization. For more information, see [“Can I disable the Initialization Rules?” in RiverWare Policy Language \(RPL\)](#).

| Setting      | Type   | Notes                                                                                                                                       |
|--------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Sim and RBS  | Yes/No | Yes = execute initialization rules before each Simulation or Rulebased Simulation run, including those controllers that include Accounting. |
| Optimization | Yes/No | Yes = execute initialization rules before an Optimization run.                                                                              |

### • Set Method

Set the selected method in the specified category for a set of objects.

| Setting     | Type | Notes                                                                                                        |
|-------------|------|--------------------------------------------------------------------------------------------------------------|
| Object Name | Text | Choose one or more the objects using the selector. The object selection can include wildcards.               |
| Category    | Text | Choose the category on the object. The list shown is for the first object.                                   |
| Method      | Text | Choose the desired method. If the category or method is not possible on the object, an error will be issued. |

### • Set MRM Descriptor

In a given MRM configuration, set the specified keyword, value pair MRM Descriptor. The descriptors are shown on the **Description** tab of the MRM configuration. See [“Description Tab” in Solution Approaches](#) for details.

| Setting           | Type | Notes                                                                                          |
|-------------------|------|------------------------------------------------------------------------------------------------|
| MRM Configuration | Text | Enter or choose the name of an MRM configuration defined in the MRM Run Controller             |
| Keyword           | Text | Specify the Keyword. If the Keyword does not exist in the configuration, a new one is created. |
| Value             | Text | Specify the desired Value to use with the Keyword.                                             |

### • Set MRM Ruleset

In a given MRM configuration, set a new file path to a ruleset. The ruleset is shown on the **Policy** tab of the MRM configuration. See [“Policy Tab” in Solution Approaches](#) for details.

| Setting           | Type    | Notes                                                                                   |
|-------------------|---------|-----------------------------------------------------------------------------------------|
| MRM Configuration | Text    | Enter or choose the name of an MRM configuration defined in the MRM Run Controller      |
| Ruleset           | Integer | Specify which of the rulesets you wish to change. The default is the first ruleset, 1.  |
| File Name         | Text    | Enter the file path. See <a href="#">“Filename Information”</a> on page 24 for details. |

### • Set MRM Run Range

Set the time range of the run (i.e. the Run Start and End dates) for the specified MRM configuration. The run range is shown on the **Run Parameters** tab of the MRM configuration. See [“Concurrent Runs” in Solution Approaches](#) for details.

| Setting                  | Type     | Notes                                                                                   |
|--------------------------|----------|-----------------------------------------------------------------------------------------|
| MRM Configuration        | Text     | Enter or choose the name of an MRM configuration defined in the MRM Run Controller      |
| Start Date               | Datetime | See <a href="#">“Datetime Information”</a> on page 22 for details.                      |
| End Date                 | Datetime | See <a href="#">“Datetime Information”</a> on page 22 for details.                      |
| Allow Start Date Editing | Yes/No   | Yes indicates that users should be allowed to override the start date in the Dashboard. |
| Allow End Date Editing   | Yes/No   | Yes indicates that users should be allowed to override the end date in the Dashboard.   |

### • Set Optimization Parameter

Set the value of any optimization parameter that is either an integer or a real number.

| Setting             | Type      | Notes                                                                                                                                                                                                                                                              |
|---------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameter Name      | Text      | Specify the name of the parameter to set. The name is not case sensitive. If the specified parameter does not match the parameter type, an error will be issued upon execution.                                                                                    |
| Parameter Type      | Selection | The available options are Integer, Real Number, and Enumeration. The default is Integer.                                                                                                                                                                           |
| Value               | Integer   | Specify the value to set. When the Parameter Type is Real Number, scientific notation can be used, for example, 1e-4. When the Parameter Type is Enumeration, the value must be a text string that matches one of the options for that parameter (case sensitive). |
| Allow Value Editing | Yes/No    | When the Parameter Type is Integer, the is setting is available. Yes allows users to override the value in the Dashboard. Dashboard overrides are not available when the Parameter Type is Real Number.                                                            |

When the Parameter Type is Enumeration, the text string specified for the value only needs to uniquely match the initial portion of one option. For example the LPMETHOD parameter has the following options

- Automatic: let CPLEX choose
- Primal simplex
- Dual simplex
- Network simplex
- Barrier
- Sifting
- Concurrent

The Barrier method can be selected by typing the full name, **Barrier**, or by typing a shortened version such as **Bar**.

## • Set Periodic Slot Value

Set the value at the specified rows and columns on a set of periodic slots.

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| Setting              | Type                    | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slots                | Slot Selection          | Choose the periodic slot using the selector.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| How To Specify Cells | Selection               | <p>Select from one of the following:</p> <ul style="list-style-type: none"> <li>Row and Column (Single Cell): set only one value (this is the default)</li> <li>Rectangular Region: set a contiguous range of cells</li> <li>All Cells: set all cells in the periodic slot.</li> </ul> <p><b>Caution:</b> When using “Rectangular Region” or “All Cells” and setting a single value (not via a function), all columns must have the same unit type or an error will be issued. Unit types must also be consistent between the value and the column when set by a function.</p> |
| Row                  | Text                    | When setting only a Single Cell, specify the row as either the row number (zero based) or the row heading. For Periodic slots, use either the row's zero-based integer index or the label for its interval (e.g., “Feb”).                                                                                                                                                                                                                                                                                                                                                      |
| Column               | Text                    | When setting only a Single Cell, specify the column as either the column number (zero based) or column heading                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Begin Row            | Value                   | When setting a Rectangular Region, specify the starting row as either row index (zero based) or the row heading.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| End Row              | Value                   | When setting a Rectangular Region, specify the ending row as either row index (zero based) or the row heading.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Begin Column         | Value                   | When setting a Rectangular Region, specify the starting column as either column index (zero based) or the column heading.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| End Column           | Value                   | When setting a Rectangular Region, specify the ending column as either column index (zero based) or the column heading.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Set by               | Value/Function          | <p>This setting indicates how you wish to set the slot, either by specifying values or by a global RPL function. When set to “Value”, the “Value” setting and editing setting, below, are shown. When set to “Function”, the RPL Global Function setting is shown.</p> <p><b>Note:</b> Only numeric and datetime slots can be set via the “Function” option.</p>                                                                                                                                                                                                               |
| Value                | Value                   | If the “Value” option is selected in the “Set by” setting, specify the value to set at each timestep specified. This value can remain or be set to NaN.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Allow Editing        | Selection               | No indicates that the value can not be changed in the Dashboard, Yes: Editor indicates that it can be changed using a line editor; Yes: Radio Buttons indicate that a set of radio buttons should be presented in the Dashboard to override the value; and Yes: Slider indicates that a slider should be presented to allow changes to the value.                                                                                                                                                                                                                              |
| Minimum Edit Value   | Numeric<br>(slot value) | Specify the minimum value that is allowed in the Dashboard (applies to Slider only).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Setting                  | Type                    | Notes                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maximum Edit Value       | Numeric<br>(slot value) | Specify the maximum value that is allowed in the Dashboard (applies to Slider only)                                                                                                                                                                                                                                                                                 |
| Encoding Table Slot Name | Slot Selection          | The table slot row labels and values establish a textual encoding of values. Applies to Radio Buttons only. See <a href="#">“Configuring User Control on the Script Dashboard”</a> on page 27 for details.                                                                                                                                                          |
| RPL Global Function      | RPL function selector   | When using the “Function” mode, specify the Global Function to use. Currently, this function must return a Numeric or a Datetime depending on the slot’s unit type. The function is passed the SLOT, row NUMERIC, and column NUMERIC, as arguments. Use these as desired. See <a href="#">“Setting Slot Values Using a Global Function”</a> on page 29 for details. |

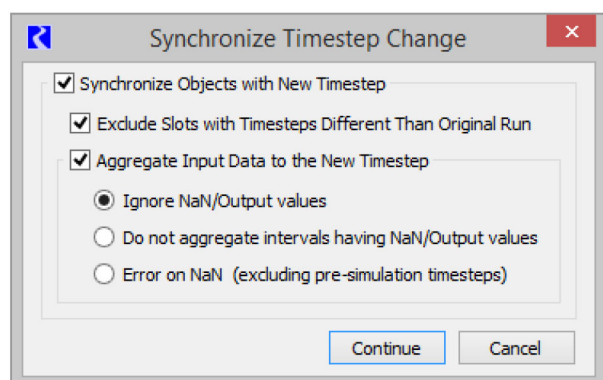
## • Set Run Range

Set the time range of the run (i.e., the Run Start and End dates).

| Setting                  | Type     | Notes                                                                                   |
|--------------------------|----------|-----------------------------------------------------------------------------------------|
| Start Date               | Datetime | See <a href="#">“Datetime Information”</a> on page 22 for details.                      |
| End Date                 | Datetime | See <a href="#">“Datetime Information”</a> on page 22 for details.                      |
| Allow Start Date Editing | Yes/No   | Yes indicates that users should be allowed to override the start date in the Dashboard. |
| Allow End Date Editing   | Yes/No   | Yes indicates that users should be allowed to override the end date in the Dashboard.   |

## • Set Run Timestep

Set the run timestep size. This action is analogous to changing the run control timestep size. The settings in the action are analogous to the options presented in the Synchronize Timestep Change dialog; see [“Synchronize From Run Control”](#) in *User Interface* for details.



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| Setting                                                  | Type      | Notes                                                                                                                                                                        |
|----------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Timestep                                                 | Selection | Choose from 5 Minute, 15 Minute, 1 Hour, 6 Hour, 12 Hour, 1 Day, 1 Month, or 1 Year.                                                                                         |
| Synchronize Objects with New Timestep                    | Yes/No    | Should objects be synchronized to the new timestep?                                                                                                                          |
| Exclude Slots with Timesteps Different Than Original Run | Yes/No    | Should slots with a timestep that is different than the original run be synchronized with the new timestep?                                                                  |
| Aggregate Input Data to the New Timestep                 | Yes/No    | When changing to a larger timestep (1 Month or 1 Year), should Input values be aggregated?                                                                                   |
| Aggregate NaN Handling                                   | Selection | If aggregating, how should NaN and Output values in input slots be handled? See <a href="#">"Synchronize From Run Control" in User Interface</a> for details on aggregation. |

#### • Set Scalar Slot Value

Set one or more Scalar Slot values.

| Setting             | Type                    | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slots               | Slot Selection          | Choose the scalar slots using the selector                                                                                                                                                                                                                                                                                                                                                                                                       |
| Set by              | Value/Function          | This setting indicates how you wish to set the slot, either by specifying values or by a global RPL function. When set to "Value", the "Value" setting and Allow editing setting, below, are shown. When set to "Function", the RPL Global Function setting is shown.<br><br><b>Note:</b> Only numeric or datetime slots can be set via the "Function" option.                                                                                   |
| Value               | Value                   | If the "Value" option is selected in the "Set by" setting, specify the value to set at each timestep specified. This value can remain or be set to NaN.                                                                                                                                                                                                                                                                                          |
| RPL Global Function | Single Selection        | If the "Function" option is selected in the "Set by" setting, use the RPL function chooser to select the function.                                                                                                                                                                                                                                                                                                                               |
| Allow Editing       | Selection               | No indicates that the value can not be changed in the Dashboard, Yes: Editor indicates that it can be changed using a line editor; Yes: Radio Buttons indicate that a set of radio buttons should be presented in the Dashboard to override the value; and Yes: Slider indicates that a slider should be presented to allow changes to the value. See <a href="#">"Configuring User Control on the Script Dashboard"</a> on page 27 for details. |
| Minimum Edit Value  | Numeric<br>(slot value) | Specify the minimum value that is allowed in the Dashboard (applies to Slider only).                                                                                                                                                                                                                                                                                                                                                             |
| Maximum Edit Value  | Numeric<br>(slot value) | Specify the maximum value that is allowed in the Dashboard (applies to Slider only)                                                                                                                                                                                                                                                                                                                                                              |



| Setting                  | Type                  | Notes                                                                                                                                                                                                                                                                                                             |
|--------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Encoding Table Slot Name | Slot Selection        | The table slot row labels and values establish a textual encoding of values. Applies only to Radio Button display. See <a href="#">“Configuring User Control on the Script Dashboard”</a> on page 27 for details.                                                                                                 |
| Show Current Value       | Yes/No                | When Allow Editing is Yes, then this setting allows you to show the current value for the scalar slot on the Dashboard. This only works when one scalar slot is specified.                                                                                                                                        |
| RPL Global Function      | RPL function selector | When using the “Set by:” “Function” mode, specify the Global Function to use. Currently, this function must return a Numeric value or a Datetime. The function is passed the SLOT as an argument. Use these as desired. See <a href="#">“Setting Slot Values Using a Global Function”</a> on page 29 for details. |

## • Set Series Slot Flags

Set the specified flag (input or output) on a set of series slots within the specified time range.

| Setting                  | Type           | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slots                    | Slot Selection | Enter or choose one or more series slots. The flags in these slots will be set to the specified value.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Column                   | Value or Text  | For multi-column series slots, specify the column as either the column number (zero based) or column heading. If not specified, the first column is used.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Start Date               | Datetime       | The first timestep for which you wish to set the flag. See <a href="#">“Datetime Information”</a> on page 22 for details.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| End Date                 | Datetime       | The last timestep for which you wish to set the flag. See <a href="#">“Datetime Information”</a> on page 22 for details.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Allow Start Date Editing | Yes/No         | Yes indicates that users should be allowed to override the start date in the Dashboard.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Allow End Date Editing   | Yes/No         | Yes indicates that users should be allowed to override the end date in the Dashboard.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Flag                     | Selection      | The flag to set on each value in the range. The two options are <b>Input</b> or <b>Output</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Behavior if No Value     | Selection      | If <b>Input</b> is specified for the Flag, this setting controls the behavior if there is no value (i.e. NaN) on the slot at that timestep. The three options are: <ul style="list-style-type: none"> <li>Ignore - since there is no value, leave the value as NaN and don't post any message.</li> <li>Issue warning - leave the value as NaN but post a warning message to the diagnostics output.</li> <li>Stop with an error - stop the script and post an error to diagnostics output. Use this option if you do not expect there to be a missing or NaN value.</li> </ul> |

## • Set Series Slot Input Mode

Set the input mode (series or periodic) for one or more Series Slots With Periodic Input. More information on these slots can be found in [“Series Slots With Periodic Input”](#) in *User Interface*.

| Setting                  | Type            | Notes                                                                                                                                                                                                                                                                                                                                      |
|--------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slots                    | Slot Selection  | Enter or choose one or more series slots with periodic input. Any other type of slot will issue an error when the script is run.                                                                                                                                                                                                           |
| Input Mode               | Series/Periodic | Choose either series or periodic mode.                                                                                                                                                                                                                                                                                                     |
| Confirm Input Overwrite? | Yes/No          | When switching from series mode to periodic mode, it is possible that the input values do not match the periodic values and the series would be overwritten by the periodic values. If this setting is Yes, a confirmation is posted (in interactive mode) and you can choose to overwrite or skip. If No, the values will be overwritten. |

## • Set Series Slot With Periodic Inputs

For one or more Series Slots with Periodic Input, set a **periodic** value at a given row and column. If you’d like to instead set the series value, use [“Set Series Slot Values”](#). More information on these slots can be found in [“Series Slots With Periodic Input”](#) in *User Interface*.

**Note:** This actions does not check if the slots are currently in the Periodic Input Mode; if one of the slot is not in periodic input mode (i.e., it is in Series Input Mode), then executing the action will have no immediate or visible impact. In this case, to see the new periodic input value and for that value to have any impact on the series values, either open the slot dialog for that slot and switch the mode to Periodic or use the Set Series Slot Input Mode. This is described in [“Set Series Slot Input Mode”](#) on page 76.

| Setting             | Type                  | Notes                                                                                                                                                                                                                                                                                                                                             |
|---------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slots               | Slot Selection        | The series slots with periodic input whose periodic input value will be set.                                                                                                                                                                                                                                                                      |
| Row                 | Value or Text         | The row of the table of periodic input whose value is to be set. Use either the row's zero-based integer index or the label for its interval (e.g., “Feb”).                                                                                                                                                                                       |
| Set by              | Value/Function        | This setting indicates how you wish to set the slot, either by specifying values or by a global RPL function. When set to “Value”, the “Value” setting and allow editing setting, below, are shown. When set to “Function”, the RPL Global Function setting is shown.                                                                             |
| Value               | Value                 | If the “Value” option is selected in the “Set by” setting, specify the value to set. This value can remain or be set to NaN.                                                                                                                                                                                                                      |
| RPL Global Function | RPL function selector | When using the “Function” mode, specify the Global Function to use. Currently, this function must return a Numeric or a Datetime depending on the slot's unit type. The function is passed the SLOT and row NUMERIC as arguments. Use these as desired. See <a href="#">“Setting Slot Values Using a Global Function”</a> on page 29 for details. |
| Allow Editing       | Selection             | No indicates that the value can not be changed in the Dashboard, Yes: Editor indicates that it can be changed using a line editor; Yes: Radio Buttons indicate that a set of radio buttons should be presented in the Dashboard to override the value; and Yes: Slider indicates that a slider should be presented to allow changes to the value. |

| Setting                  | Type                 | Notes                                                                                                                                                                                                      |
|--------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Minimum Edit Value       | Numeric (slot value) | Specify the minimum value that is allowed in the Dashboard (applies to Slider only).                                                                                                                       |
| Maximum Edit Value       | Numeric (slot value) | Specify the maximum value that is allowed in the Dashboard (applies to Slider only).                                                                                                                       |
| Encoding Table Slot Name | Slot Selection       | The table slot row labels and values establish a textual encoding of values. Applies to Radio Buttons only. See <a href="#">“Configuring User Control on the Script Dashboard”</a> on page 27 for details. |

## • Set Series Slot Values

Set Series Slot values within a time range.

**Note:** This action does not support the setting of Text Series Slots.

| Setting                  | Type                 | Notes                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slots                    | Slot Selection       | Choose the series slot using the selector                                                                                                                                                                                                                                                                                                                |
| Set by                   | Value/Function       | This setting indicates how you wish to set the slot, either by specifying values or by a global RPL function. When set to “Value”, the “Value” setting and editing setting, below, are shown. When set to “Function”, the RPL Global Function setting is shown.<br><br><b>Note:</b> Only numeric or datetime slots can be set via the “Function” option. |
| Column                   | Value or Text        | For multi-column series slots, specify the column as either the column number (zero based) or column heading. If not specified, the first column is used.                                                                                                                                                                                                |
| Value                    | Value                | If the “Value” option is selected in the “Set by” setting, specify the value to set at each timestep specified. This value can remain or be set to NaN.                                                                                                                                                                                                  |
| Start Date               | Datetime             | See <a href="#">“Datetime Information”</a> on page 22 for details.                                                                                                                                                                                                                                                                                       |
| End Date                 | Datetime             | See <a href="#">“Datetime Information”</a> on page 22 for details.                                                                                                                                                                                                                                                                                       |
| Allow Editing            | Selection            | No indicates that the value can not be changed in the Dashboard, Yes: Editor indicates that it can be changed using a line editor; Yes: Radio Buttons indicate that a set of radio buttons should be presented in the Dashboard to override the value; and Yes: Slider indicates that a slider should be presented to allow changes to the value.        |
| Allow Start Date Editing | Yes/No               | Yes indicates that users should be allowed to override the start date in the Dashboard.                                                                                                                                                                                                                                                                  |
| Allow End Date Editing   | Yes/No               | Yes indicates that users should be allowed to override the end date in the Dashboard.                                                                                                                                                                                                                                                                    |
| Minimum Edit Value       | Numeric (slot value) | Specify the minimum value that is allowed in the Dashboard (applies to Slider only).                                                                                                                                                                                                                                                                     |

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| Setting                  | Type                  | Notes                                                                                                                                                                                                                                                                                                                                          |
|--------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maximum Edit Value       | Numeric (slot value)  | Specify the maximum value that is allowed in the Dashboard (applies to Slider only).                                                                                                                                                                                                                                                           |
| Encoding Table Slot Name | Slot Selection        | The table slot row labels and values establish a textual encoding of values. Applies to Radio Buttons only. See <a href="#">"Configuring User Control on the Script Dashboard"</a> on page 27 for details.                                                                                                                                     |
| RPL Global Function      | RPL function selector | When using the "Function" mode, specify the Global Function to use. Currently, this function must return a Numeric or a Datetime depending on the slot's unit type. The function is passed the SLOT and DATETIME as arguments. Use these as desired. See <a href="#">"Setting Slot Values Using a Global Function"</a> on page 29 for details. |

#### • Set Table Slot Value

Set a value on a set of Table Slots at the specified rows and columns.

| Setting              | Type           | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slots                | Slot Selection | Choose the table slot or periodic slot using the selector. Although this action will set periodic slots, see also <a href="#">"Set Periodic Slot Value"</a> on page 71.                                                                                                                                                                                                                                                                                                                                                                                                     |
| How To Specify Cells | Selection      | <p>Select from one of the following:</p> <ul style="list-style-type: none"> <li>Row and Column (Single Cell): set only one value (this is the default)</li> <li>Rectangular Region: set a contiguous range of cells</li> <li>All Cells: set all cells in the table slot.</li> </ul> <p><b>Caution:</b> When using "Rectangular Region" or "All Cells" and setting a single value (not via a function), all columns must have the same unit type or an error will be issued. Unit types must also be consistent between the value and the column when set by a function.</p> |
| Row                  | Text           | When setting only a Single Cell, specify the row as either the row number (zero based) or the row heading. For Periodic slots, use either the row's zero-based integer index or the label for its interval (e.g., "Feb").                                                                                                                                                                                                                                                                                                                                                   |
| Column               | Text           | When setting only a Single Cell, specify the column as either the column number (zero based) or column heading                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Begin Row            | Value          | When setting a Rectangular Region, specify the starting row as either row index (zero based) or the row heading.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| End Row              | Value          | When setting a Rectangular Region, specify the ending row as either row index (zero based) or the row heading.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Begin Column         | Value          | When setting a Rectangular Region, specify the starting column as either column index (zero based) or the column heading.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| End Column           | Value          | When setting a Rectangular Region, specify the ending column as either column index (zero based) or the column heading.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Setting                  | Type                    | Notes                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Set by                   | Value/Function          | This setting indicates how you wish to set the slot, either by specifying values or by a global RPL function. When set to "Value", the "Value" setting and editing setting, below, are shown. When set to "Function", the RPL Global Function setting is shown.<br><br><b>Note:</b> Only numeric and datetime slots can be set via the "Function" option.           |
| Value                    | Value                   | If the "Value" option is selected in the "Set by" setting, specify the value to set at each timestep specified. This value can remain or be set to NaN.                                                                                                                                                                                                             |
| Allow Editing            | Selection               | No indicates that the value can not be changed in the Dashboard, Yes: Editor indicates that it can be changed using a line editor; Yes: Radio Buttons indicate that a set of radio buttons should be presented in the Dashboard to override the value; and Yes: Slider indicates that a slider should be presented to allow changes to the value.                   |
| Minimum Edit Value       | Numeric<br>(slot value) | Specify the minimum value that is allowed in the Dashboard (applies to Slider only).                                                                                                                                                                                                                                                                                |
| Maximum Edit Value       | Numeric<br>(slot value) | Specify the maximum value that is allowed in the Dashboard (applies to Slider only)                                                                                                                                                                                                                                                                                 |
| Encoding Table Slot Name | Slot Selection          | The table slot row labels and values establish a textual encoding of values. Applies to Radio Buttons only. See <a href="#">"Configuring User Control on the Script Dashboard"</a> on page 27 for details.                                                                                                                                                          |
| RPL Global Function      | RPL function selector   | When using the "Function" mode, specify the Global Function to use. Currently, this function must return a Numeric or a Datetime depending on the slot's unit type. The function is passed the SLOT, row NUMERIC, and column NUMERIC, as arguments. Use these as desired. See <a href="#">"Setting Slot Values Using a Global Function"</a> on page 29 for details. |

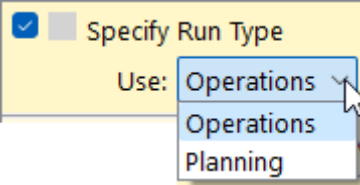
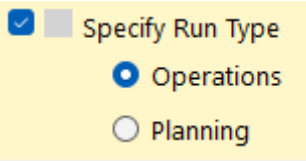
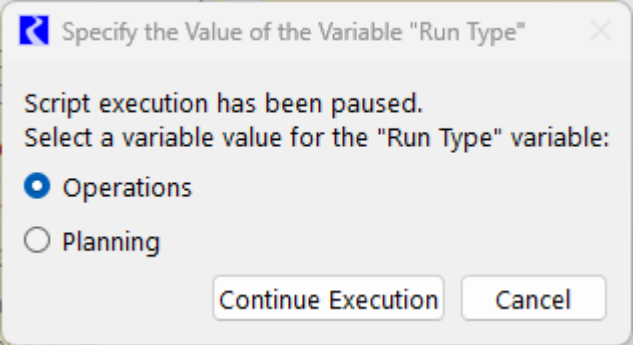
## • Specify Script Variable Value

Specify a script variable and the default value; other actions can be dependent on the value of the variable. In the dashboard, the script operator can select the value from the valid values; dependent actions will then be executed or not based on their settings. For more information on using Script Variable Values for conditional execution of actions, see ["Script Variable Values and Dependencies"](#) on page 31.

| Setting                | Type                | Notes                                                                                                                                                        |
|------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Variable Name          | Text                | Specify the variable name by typing in a string.                                                                                                             |
| Variable Valid Values  | List of text values | Specify the list of possible valid values of the variable. Use the "..." button to open a dialog to create and delete items and organize the list of values. |
| Variable Default Value | Text selection      | Specify the default value of the variable using the menu.                                                                                                    |

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| Setting                                       | Type      | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Variable Display Type                         | Selection | <p>Specify whether to use a Menu or Radio Buttons on the dashboard for the variable values. The two options are shown below.</p> <p><b>Menu:</b></p>  <p><b>Radio Buttons:</b></p>                                                                                                                                                                                               |
| Pause Execution and Prompt User               | Yes/No    | <p>Specify whether a dialog opens to prompt the user to specify the script variable value. The Variable Default Value and Variable Display Type will be shown in the dialog.</p>                                                                                                                                                                                                                                                                                |
| Allow Sub-actions to Override Variable Values | Yes/No    | <p>This setting indicates whether actions in referenced sub-scripts (i.e. sub-actions) are allowed to override or change the script variable values. The original approach was to not allow any redefinition of variables. This corresponds to the default “No” setting. See <a href="#">“Not Allowing Variable Overrides”</a> on page 35 for more information.</p> <p>Use the “Yes” setting to allow one or more sub-actions to modify the variable value. See <a href="#">“Allowing Variable Overrides”</a> on page 36 for more information.</p> |

## • Synchronize Objects

Set the time range for the specified objects.

| Setting                             | Type             | Notes                                                                                                                                                                                                                                                             |
|-------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objects to Synchronize              | Object Selection | Use the selector to choose the desired objects. All series slots on the selected objects will be synchronized, except custom slots configured to Prevent Time Synchronization as described in <a href="#">“Time Series Range—Series Slot” in User Interface</a> . |
| Exclude Slots with Non-run Timestep | Yes/No           | Yes indicates that slots with different timesteps will be excluded from the synchronize action.                                                                                                                                                                   |
| Start Date                          | Datetime         | See <a href="#">“Datetime Information”</a> on page 22 for details.                                                                                                                                                                                                |
| End Date                            | Datetime         | See <a href="#">“Datetime Information”</a> on page 22 for details.                                                                                                                                                                                                |
| Allow Start Date Editing            | Yes/No           | Yes indicates that users should be allowed to override the start date in the Dashboard.                                                                                                                                                                           |
| Allow End Date Editing              | Yes/No           | Yes indicates that users should be allowed to override the end date in the Dashboard.                                                                                                                                                                             |

## • Synchronize Slots

Set the time range and timestep size for the specified slots.

| Setting                  | Type           | Notes                                                                                                                                                                                                                                        |
|--------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slots to Synchronize     | Slot Selection | Use the selector to choose the desired slots. All slots selected will be synchronized, including custom slots configured to Prevent Time Synchronization as described in <a href="#">“Time Series Range—Series Slot” in User Interface</a> . |
| Start Date               | Datetime       | See <a href="#">“Datetime Information”</a> on page 22 for details.                                                                                                                                                                           |
| End Date                 | Datetime       | See <a href="#">“Datetime Information”</a> on page 22 for details.                                                                                                                                                                           |
| Use Run Timestep         | Yes/No         | Yes indicates that the slot's timestep size will match the timestep on the run control at the time of script execution.                                                                                                                      |
| Timestep                 | Selection      | When Use Run Timestep is “No” this setting specifies the timestep size that the slot will have.                                                                                                                                              |
| Allow Start Date Editing | Yes/No         | Yes indicates that users should be allowed to override the start date in the Dashboard.                                                                                                                                                      |
| Allow End Date Editing   | Yes/No         | Yes indicates that users should be allowed to override the end date in the Dashboard.                                                                                                                                                        |

When the timestep specification is something different than the Run Timestep, symbolic datetime math for the Start Date and End Date settings (for example, Start Timestep -  $N$  Timesteps) is computed as follows. Start Timestep and Finish Timestep are the times for the Run Control Start Timestep and Finish Timestep at the time of the script action execution. The  $N$  Timesteps are the timestep size for the slot from the Timestep setting. Then the resulting datetime is rounded up to the nearest timestep based on the Timestep setting.

## • Visualize Ensemble Data Set

Visualize data for an ensemble data set. Currently, this action will create a plot of line curves:

- For an RDF data set, it will include one curve per trace specified.
- For an Analysis data set, it will include one curve for each slot representing the computed statistic.

The plot can be shown in the Data Analysis Tool (DAT) and/or in a Plot Page. See [“Visualizing Data Sets” in Output Utilities and Data Visualization](#) for more information on interactive use of this action in the DAT.

For more information on Ensemble Data Sets, see [“Data Analysis Tool \(DAT\)” in Output Utilities and Data Visualization](#).

For more information on a similar action to analyze the ensemble data set, see [“Analyze Ensemble Data Set”](#) on page 37.

| Setting                   | Type                       | Notes                                                                                                                                                                                                                                                            |
|---------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Which Data Set To Analyze | Selection                  | Indicates how the ensemble data set will be specified. Select either “Most Recently Opened” or “Specified by Name”.                                                                                                                                              |
| Analysis Data Set Name    | Text                       | If the above selection is 'Specified by Name', then this setting is used to specify the name of the data set.                                                                                                                                                    |
| Data Source               | Selection: RDF or Analysis | Specify the type of data set to visualize. The selection then leads to different options below on how the data set can be visualized                                                                                                                             |
| How to Filter Slots       | Selection                  | How to filter the data by slot. Options in include All Slots and Include Only Specific Slots                                                                                                                                                                     |
| Slots To Include          | Slot Selection             | When “Include Specific Slots” is selected, this setting defines the slots.                                                                                                                                                                                       |
| How to Filter Dates       | Selection                  | How to filter the analysis data by date. Options include All Dates, Include Dates In A Specific Range                                                                                                                                                            |
| Start Date                | Datetime                   | When “Include Dates In A Specific Range” is selected, this setting defines the start date. This start date is used as the Time Range Start on the X Axis on the plot. See <a href="#">“Datetime Information”</a> on page 22 for details on specifying datetimes. |
| End Date                  | Datetime                   | When “Include Dates In A Specific Range” is selected, this setting defines the end date. This end date is used as the Time Range End on the X Axis on the plot. See <a href="#">“Datetime Information”</a> on page 22 for details on specifying datetimes.       |
| Allow Start Date Editing  | Yes/No                     | Yes indicates that users should be allowed to override the start date in the Dashboard.                                                                                                                                                                          |
| Allow End Date Editing    | Yes/No                     | Yes indicates that users should be allowed to override the end date in the Dashboard.                                                                                                                                                                            |
| How To Filter Traces      | Selection                  | When using an RDF data set as the Data Source setting, this allows you to filter the analysis data by traces. Options in include All Traces, Trace Range, and Include Specific Traces.                                                                           |
| First Trace Index         | Integer                    | When “Trace Range” is selected, this setting defines the first trace number of the trace range to be included in the analysis.                                                                                                                                   |
| Last Trace Index          | Integer                    | When “Trace Range” is selected, this setting defines the last trace number of the trace range to be included in the analysis.                                                                                                                                    |



| Setting                       | Type         | Notes                                                                                                                                                                                                                                                                  |
|-------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trace Selection               | Integer List | When “Include Specific Traces” is selected, this setting defines which traces are included in the analysis. Click in the setting field and open up a trace selector dialog to select which trace numbers should be included.                                           |
| Create Plot Pages             | Yes/No       | Yes indicates that a Plot Page should be created in the Output Manager for each plot. No, indicates no Plot Pages will be created. If No, the Show in Data Analysis Tool setting should be set to Yes.                                                                 |
| Confirm Plot Page Replacement | Yes/No       | When creating Plot Pages with the above setting, Yes indicates that a confirmation dialog will be shown if the Plot Page already exists asking if you want to replace it with the new Plot Page. No, indicates the Plot Page will be overwritten with no confirmation. |
| Open Plot Pages               | Yes/No       | When creating Plot Pages with the above setting, Yes indicates that the plot pages should be opened and shown. No indicates they are not opened but are available in the Output Manager.                                                                               |
| Show in Data Analysis Tool    | Yes/No       | When Yes, the plots are shown as tabs in the Data Analysis Tool. Also, open the DAT if it is not already open. If No, the Create Plot Page setting should be set to yes or no plots will be created.                                                                   |

## • Visualize Series Regression Data Set

Visualize the results of a Series Regression results data set. Currently, this action will create a plot of line curves for an Analysis data set. The plot will show one curve for each of the following slots:

- Dependent slot
- Each Independent Slot (as possible given the units)
- Fitted Data

The plot can be shown in the Data Analysis Tool (DAT) and/or in a Plot Page. See [“Visualizing Data Sets” in Output Utilities and Data Visualization](#) for more information on interactive use of this action in the DAT.

For more information on Ensemble Data Sets, see [“Data Analysis Tool \(DAT\)” in Output Utilities and Data Visualization](#).

For more information on a similar action to analyze the ensemble data set, see [“Analyze Ensemble Data Set”](#) on page 37.

| Setting                | Type   | Notes                                                                                                                                                                                                  |
|------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Analysis Data Set Name | Text   | Specify the name of the data set.                                                                                                                                                                      |
| Create Plot Pages      | Yes/No | Yes indicates that a Plot Page should be created in the Output Manager for each plot. No, indicates no Plot Pages will be created. If No, the Show in Data Analysis Tool setting should be set to Yes. |

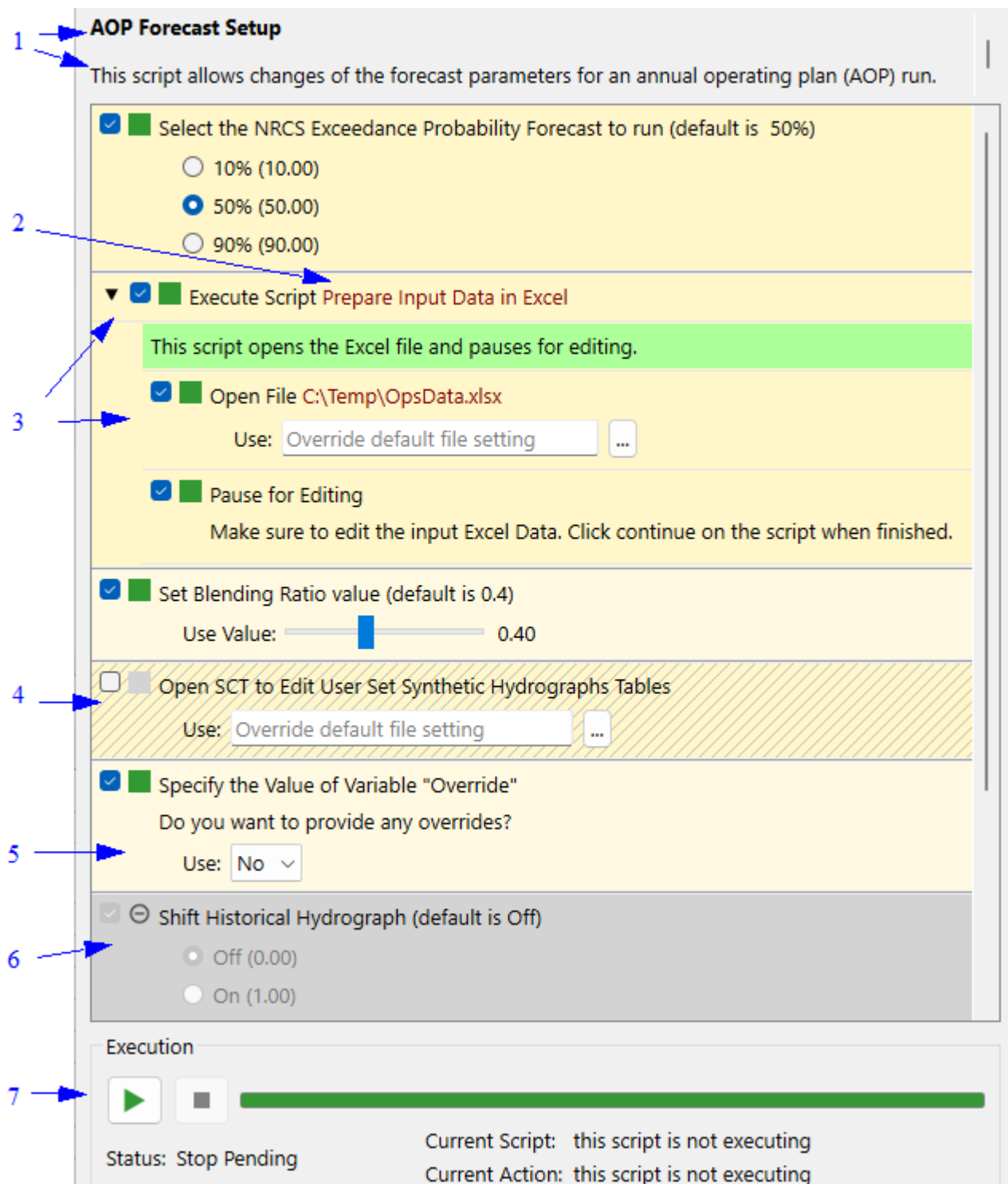
| Setting                       | Type   | Notes                                                                                                                                                                                                                                                                  |
|-------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Confirm Plot Page Replacement | Yes/No | When creating Plot Pages with the above setting, Yes indicates that a confirmation dialog will be shown if the Plot Page already exists asking if you want to replace it with the new Plot Page. No, indicates the Plot Page will be overwritten with no confirmation. |
| Open Plot Pages               | Yes/No | When creating Plot Pages with the above setting, Yes indicates that the plot pages should be opened and shown. No indicates they are not opened but are available in the Output Manager.                                                                               |
| Show in Data Analysis Tool    | Yes/No | When Yes, the plots are shown as tabs in the Data Analysis Tool. Also, open the DAT if it is not already open. If No, the Create Plot Page setting should be set to yes or no plots will be created.                                                                   |

## Script Dashboard

The Script Dashboard is the main interface for executing the script. This dialog box presents the actions and controls for the script, although not all actions are necessarily shown. See [“Editing a Script”](#) on page 10 for details about configuring the settings for each action. Each row in the dashboard represents an action. Within each action, the action display text, optional action description, and any user controls are presented.

[Figure 1.29](#) identifies additional features of this dialog.

**Figure 1.29** Annotated Script Dashboard



1. The script name and description are shown at the top.

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2. The script variable values are shown in red. Editable values are shown as either text edit fields, buttons, or sliders.
3. Actions which are part of an Execute Script action (i.e. a subscript) are indented. See [“Display of Execute Script and Sub-actions”](#) on page 90 for more information. The arrows can be used to expand or collapse those sub-actions. Each Execute Script action has a setting that indicates whether the sub-actions should be expanded (yes is the default) but the dashboard user can expand or collapse as desired. See **Expand All** and **Collapse All** in [“Operations for Actions on the Dashboard”](#) on page 87.
4. Disable an action by selecting the check box. This makes the row cross hatched. Use **Edit**, then **Enable All Action** to enable all actions shown in the dashboard at once; similarly **Edit**, then **Disable All Action** disables them all. Use the **Edit**, then **Reset Values** option to revert all values in the Script Dashboard back to the default value as defined in the Script Editor.
5. Specify script variable values using the menu or radio buttons. See [“Script Variable Values and Dependencies”](#) on page 31 for more information.
6. Actions that are dependent on Script Variables are shown in gray when they won’t be executed. See [“Script Variable Values and Dependencies”](#) on page 31 for more information.
7. To run the entire script, specify any desired override values (where configured) in the Script Dashboard and then select the **Run** button. 

## Executing Actions From the Script Dashboard

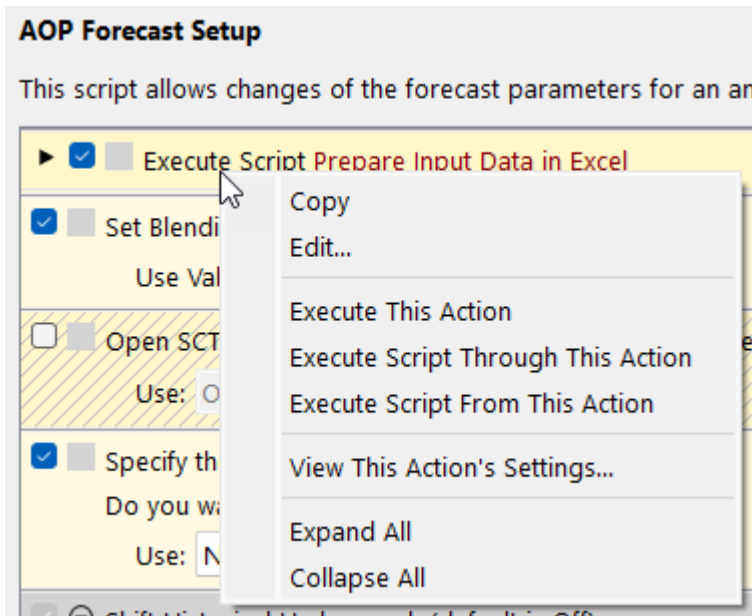
Use the following options to execute the script:

- Use the **Start** button to run the entire script.
- Use the **Execute Action** context menus to run a single action.
- Use the **Execute Script Through This Action** context menu to run the script from the start up to and including this action.
- Use the **Execute Script from this Action** context menu to run that action and the remainder of the script.

[Figure 1.30](#) Shows the context menu options.

**Note:** The execute options are available for both actions in the main script and in sub scripts.

**Figure 1.30** Screenshot of dashboard context menu operations

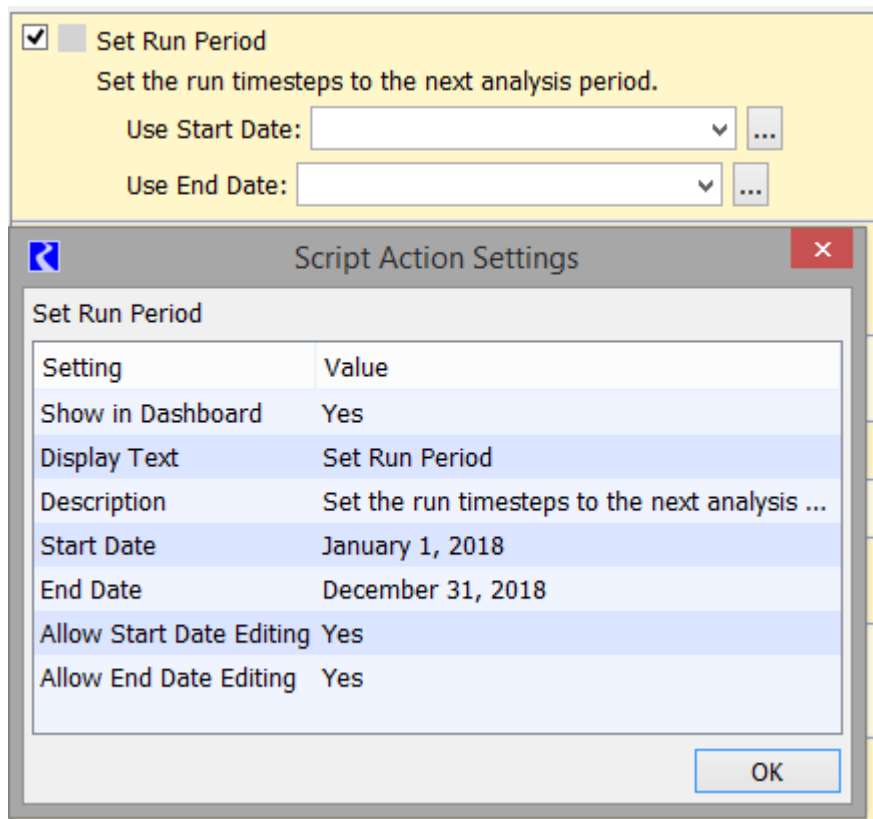


## Operations for Actions on the Dashboard

[Figure 1.30](#) shows three additional operations from the context menu on the dashboard:

- **Copy:** Use this menu to copy the action and its settings and then paste it into a Script Editor.
- **Edit:** Use this menu to open the Script Editor with this action selected. Then edit the settings in the Script Editor.
- **Open Slots:** For actions that involve slots (Set Series Slot, Open Slots, and so on), you can open those slots directly from the context menu.
- **View This Action's Settings:** Open a read-only dialog box that shows all of the settings for that action. This allows you to see the settings and configuration for this action without opening the Script Editor. [Figure 1.31](#) shows a sample.
- **Expand All:** Show all sub-actions called by Execute Script actions.
- **Collapse All:** Hide all sub-actions called by Execute Script Actions.

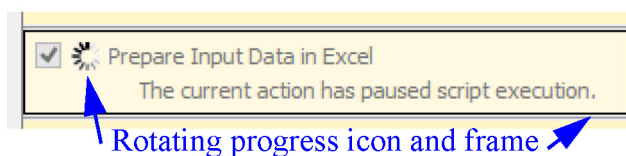
**Figure 1.31** Screenshot of the Dashboard and corresponding Script Action Settings dialog box



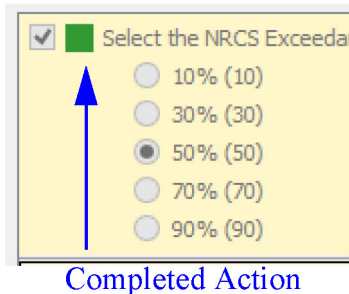
## Progress and Status on the Dashboard

The Script Dashboard provides a progress bar at the bottom of the dialog showing the current action and the overall progress (based on the number of actions). As script actions execute, the Script Dashboard scrolls to keep the currently executing action in view.

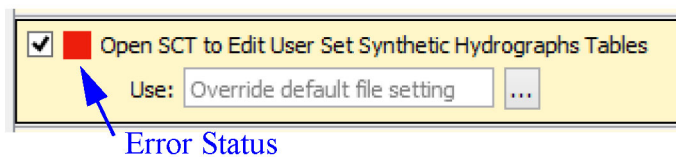
A rotating icon to the right of the **Enable/Disable** checkbox and darkened frame progress through the script to indicate the currently executing action. For Execute Script actions, both the Execute Script action and any sub-actions shown the same.



Actions that have completed successfully are indicated by a small green square to the right of the **Enable/Disable** checkbox.



Actions resulting in an error, show a red square indicator.

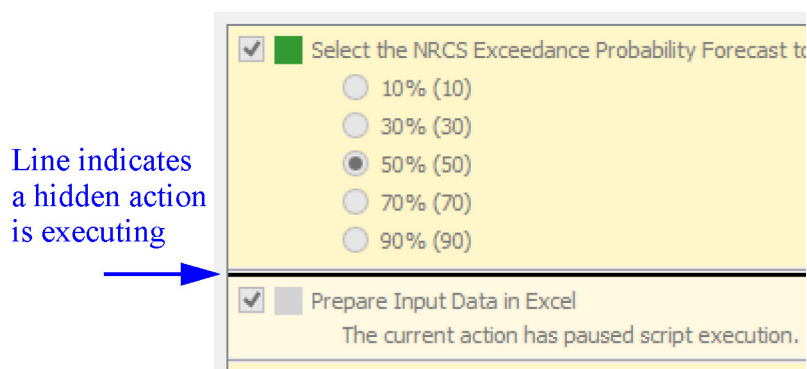


A partial green square indicates that some but not all actions in a subscript were executed.

#### ▼ Execute Script Arbor Basin Operations

A subset of actions in a subscript can be executed using the “Execute This Action” or “Execute Through/From This Action” operations on the context menu as described in [“Executing Actions From the Script Dashboard”](#) on page 86. If only some of the actions are executed, any upper level, calling scripts will have this partial green square as the status.

During script execution, a dark horizontal line between actions indicates a hidden action (one that is not shown in the dashboard as configured in the editor) is executing.

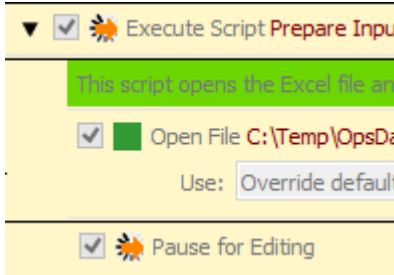


If the script fails or a Memo with confirmation is encountered, a dialog will open alerting you.

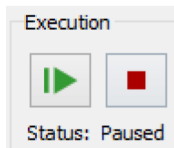
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When paused in a Memo action, the Dashboard shows an orange arrow on both the Memo action and the calling Execute Script action, if necessary.



For the Memo, you must select the **Continue** button to resume execution.



**Note:** The script developer can specify which actions are shown on the Script Dashboard. Therefore, the dashboard may not contain all the actions executed by the script.

## Display of Execute Script and Sub-actions

A script can call another script using the Execute Script actions. Up to 5 levels (5 Execute Script actions) are supported. Actions called as part of an Execute Script are termed sub-actions. This section explains some of the behavior supported for sub-actions in the dashboard.

**Note:** Sub-actions are not shown in the script editor.

Figure 1.32 shows a script with two levels of sub-actions. Set Run Info calls Set Run Range (light blue background) Notice that all sub-actions are indented including the dividers and disabled status background. Use the override controls to modify the sub-actions as desired.



**Figure 1.32** Screenshot of a script dashboard with multiple levels of Execute Script actions

☒ ☐ Select the NRCS Exceedance Probability Forecast to run (default is 50%)

☐ 10% (10.00)

☒ 50% (50.00)

☐ 90% (90.00)

▼ ☒ ☐ Execute Script **Set Run Info**

☒ ☐ Set Blending Ratio value (default is 0.4)

Use Value:  0.40

▼ ☒ ☐ Execute Script **Set Run Range**

This script sets the run range, synchronizes objects (if necessary), and sets the controller.

☒ ☐ Set the Run Range to: Jan 1, Start Year -- Start Timestep + 15 Timesteps

Use Start Date:  ...

Use End Date:  ...

☐ ☐ Synchronize Objects to: Start Timestep - 1 -- End Timestep

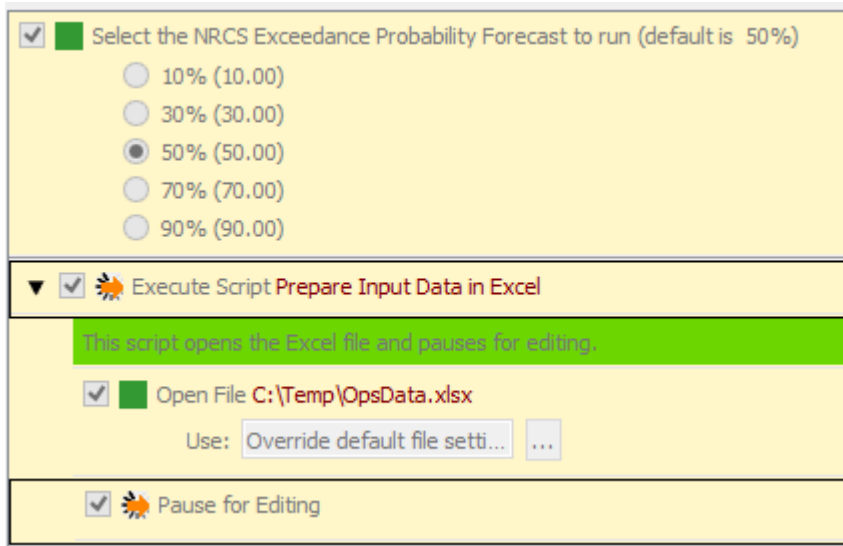
☒ ☐ Set the Controller to **Inline Simulation and Accounting**

When the script is paused in a sub-action, the dashboard shows that information as shown in [Figure 1.33](#). Notice the orange arrow that indicates the top level script(s) and sub-action at which execution is paused due to a Memo action.

## Chapter 1

### Script Management

**Figure 1.33** Screenshot of a script dashboard paused in a sub-action



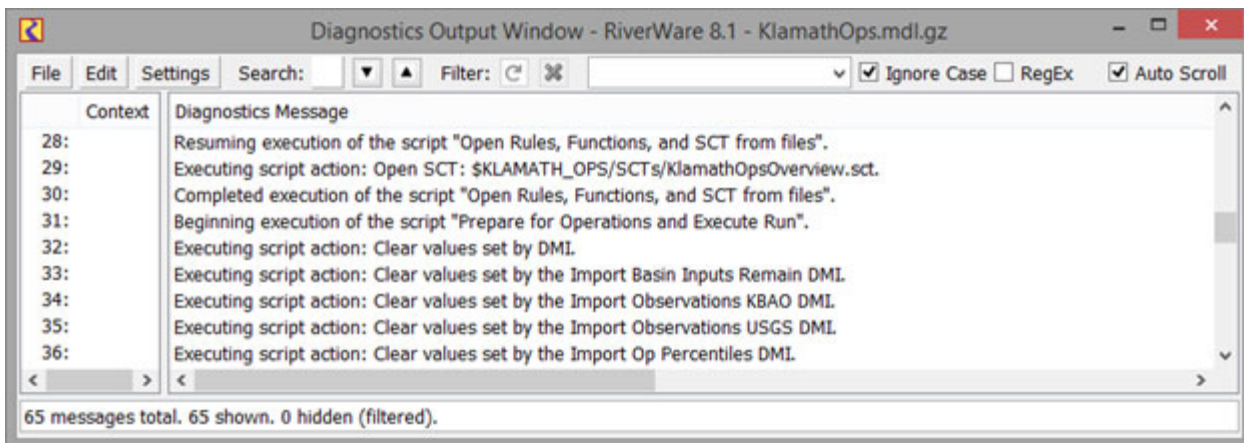
If an error occurs during the execution of a sub-action, then all Execute Script actions that it is nested under will also be marked as having an error to aid in locating the source of an error in a large script.

The script execution control panel also displays the currently executing script in addition to the currently executing action to aid in identifying the specific action that is executing. This is especially useful in scripts in which the same action is executed multiple times under different Execute Script actions.

## Script Diagnostics

Diagnostics are available for script execution. In the Diagnostics Manager, use the **Workspace** settings. Then select the Script Execution diagnostics group. A sample of diagnostic messages are shown in [Figure 1.34](#).

**Figure 1.34** Screenshot of the diagnostic output window showing script execution diagnostics.



# Chapter 2

## Batch Mode and RiverWare Command Language

---

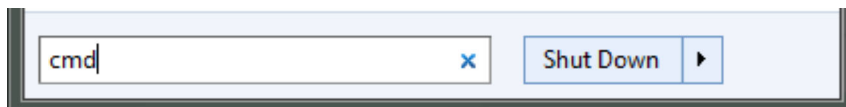
This section describes methodology to run RiverWare from the command line and optionally execute a RiverWare Command Language (RCL) script. This Batch Mode allows you to run RiverWare without seeing the user interface and allows for automation of many processes including advancing the run dates, running DMIs to bring in new data, running the model, generating output, and saving the model.

### Topics

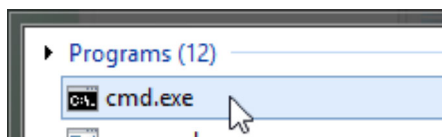
- [Running RiverWare From the Command Line, page 93](#)
- [Running RiverWare in Batch Mode, page 94](#)
- [About RiverWare Command Language, page 95](#)
- [RCL Commands Reference, page 96](#)

## Running RiverWare From the Command Line

Running RiverWare from the command line provides an alternative to selecting the RiverWare icon from the Windows system. In addition, RiverWare supports certain commands directly from the command line like loading models and rulesets and certain other operations. To access the command line, in the Windows Start menu Search field, type `cmd`.

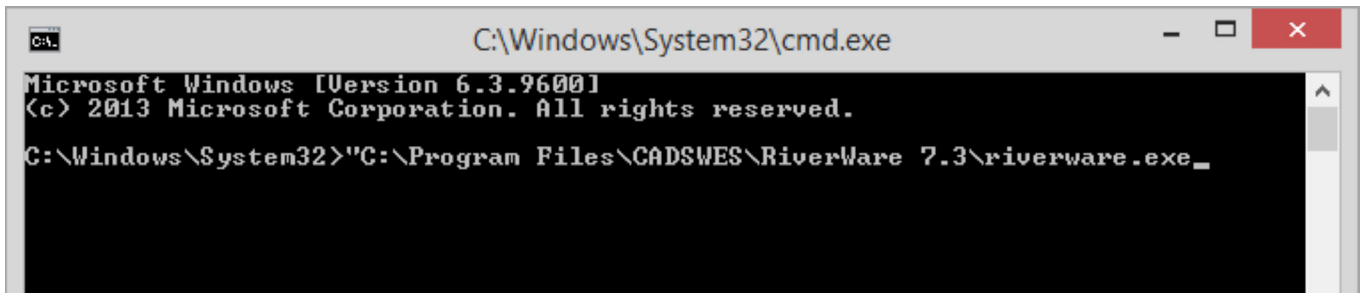


Then select the option for **cmd.exe**.



A window similar to that shown in [Figure 2.1](#) will open. To start RiverWare from this window, enter the full RiverWare path (depending on your PATH configuration).

**Figure 2.1** Starting RiverWare from the Windows Command Line



A complete list of command line options can be viewed by invoking RiverWare with the `--help` command line argument. On the Windows command line, type:

```
riverware.exe --help
```

When RiverWare is executed from a command line window, it is run as an independent process from the command shell, so exiting the first command window will not automatically terminate RiverWare. The Windows task manager can be used to terminate batch runs of RiverWare.

Sometimes, the you may wish to run one or more models as automatic tasks (as Windows “Scheduled Tasks”). Setting up the runs using batch mode and the command line options can be useful in this approach. For example, a user may wish to run a series of models overnight when the machine is more available. One command line option that is useful is the `--log <file>` switch. This option sends any diagnostic output to the specified file.

Instead of logging to a file as described above, you may like to send the messages to a console window. Then, use the `--log console` option and a new command window will open and show the messages.

## Running RiverWare in Batch Mode

The RiverWare Command Language (RCL) facility allows models to be run in batch mode. While in this mode, RiverWare does not open a user interface, but instead runs in background mode.

In batch mode, the user provides RiverWare with a script file, which contains the commands RiverWare executes. Commands are entered through the RCL file, rather than through the user interface. The commands are described in detail in the following document.

The syntax to invoke RiverWare in batch mode from a command prompt is:

```
riverware --batch <script file>
```

The following example will invoke the batch script file called `BasinOperationsScript.rcl` and will write the diagnostics output to file called `BasinOperationsDiagnostics.txt`.

```
riverware --batch "C:\MyRiverWareFiles\BasinOperationsScript.rcl" --log "C:\MyOut-puts\BasinOperationsDiagnostics.txt"
```

The `<script file>` defines how the batch run should proceed. In the file, you specify commands to open a model, load a ruleset if necessary, start a run, and save the results. In the simplest case, you will open a model, run a simulation, save the model, and exit batch mode. The following RCL commands support these actions:

**# Open the model**

```
Openworkspace <Model Name>
```

```
# Run the simulation
startController
# Save the model
saveWorkspace <Model Name>.saved.mdl
# Close the opened model and exit RiverWare
closeWorkspace
```

This is a very simple example. The full complement of key words and commands is presented in the following section.

## About RiverWare Command Language

This section provides general information about the RiverWare Command Language (RCL) syntax and variables.

### Syntax Key

- Angled brackets (<>) indicate a required parameter.
- Brackets ([]) indicate an optional parameter.
- A vertical bar “|” within angled brackets or brackets indicates a choice of parameters. For example, “<#RunInfo|#MRM>” indicates that either “#RunInfo” or “#MRM” must be provided as a required parameter.
- Parameters starting with “!” are tokens, which should appear exactly as is.
- Parameters starting with “#” are RiverWare names, which should appear exactly as is.
- A “#” sign at the beginning of the line represents a comment. Anything occurring after the “#” will be ignored.
- Parameters which contain embedded white space characters must be enclosed in curly braces {}, or quotes “ ”.
- Anywhere there is a file referenced below, the user can specify the path using either a full reference or using environment variables (see next section). On Windows, path names should use “/” or “\”. Also, the file may reference a shared network: RCL accepts “\” but it has to be escaped with “\\”, so the RCL syntax is as follows:  
\\\\\\ComputerName\\ShareName\\...

### Environment Variables

Environment variables are *system variables* that allow portability of models across machines and platforms.

### Setting Environment Variables

If you wish to set environment variables and then reference them in the RCL script, they can be set using the TCL command “set env” as follows:

```
set env(variable) value
```

## Chapter 2

### Batch Mode and RiverWare Command Language

Then the variable can be accessed within the RCL script using the `$env(variable)` syntax. These variables are only available within the script, RiverWare has already inherited the environment before this is set and are then not available. To set variables to be available to RiverWare, use the `SetEnv` command; see “[SetEnv](#)” on page 107.

This command has the following syntax:

```
SetEnv <variable> <value>
```

These variables are then available in RiverWare but not available within the script. Most likely, you will want to set both variables within the script. Consider the following example:

```
# for use in the script:
```

```
set env(DMI_DIR) C:/DMI/data
```

```
# for use by RiverWare:
```

```
SetEnv DMI_DIR $env(DMI_DIR)
```

```
# open the model using the TCL environment variable syntax:
```

```
openworkspace $env(DMI_DIR)/model.mdl.gz
```

Then within RiverWare, you can reference `DMI_DIR` using the `$DMI_DIR` syntax in DMI’s, MRM or other places.

## Using Environment Variables

Environment variables set outside of the RCL script can be referenced using the form:

- `$VARIABLE`
- `$(VARIABLE)`
- `${VARIABLE}`

`VARIABLE` is a letter followed by zero or more letters, digits or underscores. If the variable is followed by a letter, digit or underscore it must be “quoted” with `()` or `{}`.

## Special Parameters

- `{date time}`

A date and time of the format `{MM/DD/YYYY HH:MM}` or `{MM-DD-YYYY HH:MM}`, where the year includes the century and the hour is 24-hour based.

**Note:** A date and time, which includes embedded white space, must be enclosed in braces `{}`.

- `{delta time}`

A timestep or time duration of the format `{count units}`, where units are “HOURS,” “DAYS,” “WEEKS,” or “MONTHS.”

**Note:** A delta time, which includes embedded white space, must be enclosed in braces `{}`.

## RCL Commands Reference

Following are the RCL commands in alphabetical order.

**Note:** There are also RCL commands specific to RiverSMART. See the RiverSMART documentation for more information on these commands.

## • CloseWorkspace

CloseWorkspace

Closes the model that is open.

**Note:** The CloseWorkspace command does not close RiverWare. RiverWare closes once the execution of the complete batch script is complete. Multiple Openworkspace and Closeworkspace commands within the same script will be executed within a single RiverWare session.

## • ConfigureExcelDataset

ConfigureExcelDataset <Dataset Name> [workbook=...] [sheet=...] [workbookDir=...]

This command changes the workbook, the sheet, or the folder for an Excel Dataset used in a Database DMI.

- <Dataset Name> is an existing Excel dataset in the model. If the dataset does not exist in the model or is not an Excel dataset, the batch script will fail with an error message.

**Caution:** Only Excel datasets using the “Map with Header Text and Sheet Names” are supported by this event. Excel datasets using “Map by Spreadsheet Ranges” are not supported. See “[Slot Mapping—Header Text and Sheet Names Approach](#)” in *Data Management Interface (DMI)* for more information on this approach.

**Caution:** If the model is saved using the SaveWorkspace command, the modified Excel datasets will be saved.

**Note:** The workbook argument takes precedence over the workbookDir argument.

Example

```
ConfigureExcelDataset {My Excel Dataset} {workbook=C:/My Data/My workbook.xlsx}
ConfigureExcelDataset {My Excel Dataset} {sheet=MyNewSheetName}
ConfigureExcelDataset {My Excel Dataset} {workbookDir=C:/Temp/MyTempFolder}
```

## • CreateSlotCache

CreateSlotCache

Creates a slot cache that contains the current values of all series slots on the workspace, in the range of the current controller. See “[Slot Cache](#)” in *User Interface* for a description of the cache.

**Note:** The slot cache is under development. Please contact [riverware-support@colorado.edu](mailto:riverware-support@colorado.edu) for more information and the current status of this feature.

## • DeleteAllOptRestorePoints

DeleteAllOptRestorePoints

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### Batch Mode and RiverWare Command Language

This command deletes all optimization restore points currently in existence. It does not have any arguments. See [“Restore Point” in Optimization](#) for a description of optimization restore points.

#### • **EnableInfoDiag**

`EnableInfoDiag <1|0>`

Enable Informational Diagnostics in the diagnostics manager dialog. When the argument is 1, diagnostics will be enabled. When the argument is zero, then the diagnostics will be disabled.

#### • **ExecuteScript**

`ExecuteScript <Script Name>`

Executes the specified script as defined in the Script Manager. See [“Script Management”](#) on page 1 for details on RiverWare Script Management. The script name should be enclosed in curly braces `{ }` if it contains embedded white space.

Example:

```
ExecuteScript {Run Operations Model}
```

#### • **ExportOptAnalysisAsCSV**

`ExportOptAnalysisAsCSV <file name> [<section name> true|false true|false]`

Exports the Optimization Analysis information as a comma separated values file. This corresponds to the information presented in the Priority-oriented Optimization Solution Analysis Tool. (See [“Priority-oriented Optimization Solution Analysis Tool” in Optimization](#).) The arguments are as follows:

- `<file name>` is the name of the file to write. If the file path contains embedded white space, it should be enclosed in curly braces `{ }`.
- `<section name>` are the (optional) sections to export. Defaults to all sections. The section names are as follows and correspond to combinations of the Optimization sections; see [“Export as CSV” in Optimization](#) for details.

None

GoalSummaries

Solutions

NewConstraints

PriorConstraints

PhysicalConstraints

SlotVariables

NonSlotVariables

SolutionsAndSummaries

PolicyConstraints

PoliciesSolutionsAndSummaries

AllConstraints

AllVariables



All

- The first optional `true|false` specifies whether to skip export of constraint and variable rows associated with fully satisfied constraints or solutions. Defaults to true, i.e., do not export these rows for fully satisfied solutions.  
**Note:** If this argument is present, all previous arguments must also be present.
- The second optional `true|false` specifies whether to skip export of constraint and variable rows associated with the solution to the postponed problem, which is relevant only for runs in which a seed value was used to postpone one or more solutions. Defaults to true, i.e., do not export these rows for postponed problem solutions.  
**Note:** If this argument is present, all previous arguments must also be present.

Example:

```
ExportOptAnalysisAsCSV {C:/MyOutputs/MyOptAnalysis.csv} {PolicyConstraints} {true}
```

### • ExportOptRestorePoint

```
ExportOptRestorePoint <restore point name> <file name>
```

Exports an optimization restore point to a file. See [“Restore Point” in Optimization](#) for information on optimization restore points. The arguments are as follows:

- `<restore point name>` is the name of the restore point to export. If the name contains embedded white space, it should be enclosed in curly braces `{ }`.
- `<file name>` is the name of the file to export the restore point. If the name contains embedded white space, it should be enclosed in curly braces `{ }`. Restore point files are generally given the `.orp` file extension.

The export command will overwrite any existing file with the same name.

Example:

```
ExportOptRestorePoint {Blue River Restore Pt} {C:/RWFiles/BlueRiverRestorePt.orp}
```

### • ExtractMrmData

```
ExtractMrmData {MRM configuration name} keyword=value ...
```

This command modifies the MRM configuration in terms of the RDF output options (See [“RDF Options” in Solution Approaches](#)) and then starts the process to write the data in the current model to the RDF output files.

**This command was implemented for use by the Data Extractor Tool as described in [“Data Extractor Tool” in Solution Approaches](#) and more specifically in [“Performing the Extraction” in Solution Approaches](#). It is not intended to be used outside of this tool, at this time.**

### • GenerateMrmExcel

```
GenerateMrmExcel {MRM configuration name} keyword=value ...
```

This command modifies the MRM configuration in terms of the RDF output Excel options (See [“RDF Options” in Solution Approaches](#)) and then starts the process to convert the RDF file to an Excel workbook.

This command was implemented for use by the Data Extractor Tool as described in “[Data Extractor Tool](#)” in *Solution Approaches* and more specifically in “[Performing the Extraction](#)” in *Solution Approaches*. It is not intended to be used outside of this tool, at this time.

### • **GetRunInfo**

```
GetRunInfo <#RunInfo|#MRM <MRM configuration>>
[!InitDate] [!EndDate]
[!Duration] [!Step] [!Controller]
```

Prints single run information (#RunInfo) or multiple run information (#MRM) to standard output. If #MRM is specified, you also need to specify an MRM configuration <MRM configuration>.

- **!InitDate** prints the run’s initial date.
- **!EndDate** prints the run’s end date.
- **!Duration** prints the run’s duration.
- **!Step** prints the run’s timestep.
- **!Controller** prints the current controller.

If no parameters are specified, all information is printed.

Following are examples of the syntax:

```
GetRunInfo #RunInfo !InitDate
GetRunInfo #RunInfo !EndDate
```

### • **GetSlot**

```
GetSlot <Object.Slot> [<date time>]
```

Returns the series or scalar slot’s value, in user units, as a Tcl variable. **Object.Slot** should be enclosed in curly braces {} if it contains embedded white space.

- **<date time>** is only for series slots and must be enclosed in curly braces {}, as it contains embedded white space - for example, {05-15-2010 12:00}.

The command is used when writing batch mode scripts which make use of the Tcl scripting language. GetSlot returns the value in a Tcl variable, so sample usage includes:

```
set v [GetSlot {Object.Slot} {05-05-2010 12:00}]
setSlot {Object.Slot} {05-05-2010 12:00} $v
```

The Tcl variable can be used in arithmetic expressions, conditionals, etc:

```
set v [expr 0.5 * $v]
if {$v > 1.0} {
  ...
} else {
  ...
}
```

Further scripting help with the TCL language is beyond the scope of this reference but can be found in online searches.

### • HDB\_EnsembleID

HDB\_EnsembleID <HDB dataset name> <Ensemble ID> ...

Sets the Ensemble ID for an HDB dataset that is configured to have its ID selected when MRM is started.

- <HDB dataset name> is the name of the HDB dataset that is configured to have its ID selected when MRM is started. If the name contains embedded white space, it must be enclosed in braces {}.
- <Ensemble ID> is the Ensemble ID to set on the dataset.

IDs can be set for more than one dataset by adding addition pairs of names and IDs.

### • HDB\_ModelRunID

HDB\_ModelRunID <HDB dataset name> <Model Run ID> ...

Sets the model run ID for an HDB dataset that is configured to have its ID selected when its DMI is invoked.

- <HDB dataset name> is the name of the HDB dataset that is configured to have its model run ID selected when its DMI is invoked. If the name contains embedded white space, it must be enclosed in braces {}.
- <Model Run ID> is the Model Run ID to set on the dataset.

IDs can be set for more than one dataset by adding addition pairs of names and IDs.

### • ImportDMI

ImportDMI <file name> [ReplaceInModel | RenameImported]

Import a DMI that was previously exported to a file. The first argument (required) is the name of the file which contains exported DMIs, datasets or name maps. The second argument (optional) specifies how conflicts are resolved when imported entities have the same name as entities in the model:

- <file name> is the name of an exported DMI file to import. If the file path contains embedded white space, it should be enclosed in curly braces {}.
- [ReplaceInModel | RenameImported] - Specify how to handle DMI items that already exist. The options are:
  - ReplaceInModel - the entities in the model are replaced with the imported entity. This is the default.
  - RenameImported - the imported entity is renamed with a suffix IMPORT N

Example:

```
ImportDMI {C:/temp/ArborInputDMI.dmi} ReplaceInModel
```

For more information on importing DMIs, see [“Import” in Data Management Interface \(DMI\)](#).

**Note:** Database DMIs, Control File-Executable and Trace Directory DMIs can be imported with this command.

### • ImportOptRestorePoint

ImportOptRestorePoint <file name>

Imports an optimization restore point that was previously exported to a file. See “[Restore Point](#)” in *Optimization* for information on optimization restore points.

- <file name> is the name of the restore point file to import. If the file path contains embedded white space, it should be enclosed in curly braces {}. Restore point files are generally given the .orp file extension.

The import command will overwrite any existing restore points with the same name as the restore point being imported from the file.

Example:

```
ImportOptRestorePoint {C:/RWFiles/BlueRiverRestorePt.orp}
```

### • ImportScripts

ImportScripts <script file> [importMode=<rename|replace|skip|cancel>]

Import one or more scripts from a file.

- <script file> is the name of the file that was exported from the script manager. It can contain scripts and groups.
- [importMode=<rename|replace|skip|error>] The first three import modes corresponds to the options when importing a script. See “[Importing Scripts](#)” in *Automation Tools*. The mode **error** will stop the RCL script and return an error. The mode **error** is the default.

Example:

```
ImportScripts C:\\temp\\ExportedScript.xml importMode=replace
```

### • InvokeDMI

InvokeDMI <DMI name> [-UserParam=value]

Invokes a DMI.

- <DMI name> is the name of the DMI to invoke. If the name contains embedded white space, it must be enclosed in braces {}.
- [-UserParam=value] sets the user parameter “userParam” to “value”.

Example:

```
InvokeDMI {Operations Data Input DMI}
```

### • InvokeDssDMI

InvokeDssDMI <DMI name> !DssFile <DSS file> !DssFPart <DSS F part>

Invokes a DSS DMI.

- <DMI name> is the name of the DSS DMI to invoke. If the name contains embedded white space, it must be enclosed in braces {}.

- `<DSS file>` is the name of the DSS file. If the file name contains embedded white space, it must be enclosed in braces `{}`.
- `<DSS F part>` is the DSS F part. If the F part contains embedded white space, it must be enclosed in braces `{}`.

### • ListDMI

`ListDMI [!Type <Input|Output>] [!Dataset <dataset type>]<output file>`

Writes the names of the DMIs which match the specified criteria to the output file.

- `[!Type <Input|Output>]` limits the output to either Input or Output DMIs.
- `[!Dataset <dataset type>]` limits the output to DMIs which use only the specified dataset type.
- `<output file>` is the file the DMI list is written to.

Examples:

```
ListDMI C:/MyOutputs/DMIListFile.txt
```

Writes the names of all DMIs to the file `DMIListFile.txt`.

```
ListDMI !Type Input C:/MyOutputs/DMIListFile.txt
```

Writes the names of all Input DMIs to the file `DMIListFile.txt`.

```
ListDMI !Type Input !Dataset DSS C:/MyOutputs/DMIListFile.txt
```

Write the names of all Input DMIs which use only DSS datasets to the file `DMIListFile.txt`.

### • LoadDiagSettings

`LoadDiagSettings <file path>`

Loads the diagnostic settings from the specified file.

- `<file path>` is the path to the diagnostics setting file.

The diagnostic settings file must have been saved from the model file beforehand. See [“Loading and Saving Diagnostic Configurations” in \*Debugging and Analysis\*](#) for information on saving diagnostic settings to a file. As an example, you might have preconfigured diagnostic settings for printing out detailed informational diagnostics for debugging purposes when you run in batch mode.

**Caution:** If the model is saved using the `SaveWorkspace` command, it will be saved with these new diagnostic settings.

Example:

```
LoadDiagSettings C:/MyRiverWareFiles/DebugDetails_DiagSettings.txt
```

### • LoadOptSet

`LoadOptSet <file path>`

Loads the optimization goal set into the workspace.

- `<file path>` is the path to where the optimization goal set resides on disk.

Example:

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### Batch Mode and RiverWare Command Language

```
LoadOptSet C:/MyRiverWareFiles/MyOptGoalSet.opt
```

#### • LoadRules

```
LoadRules <file path>
```

Loads a ruleset into the workspace.

- <file path> is the path to where the ruleset resides on disk.

Example:

```
LoadRules C:/MyRiverWareFiles/MyRuleset.rls
```

#### • OpenGlobalSet

```
openGlobalSet <file path>
```

Opens a global function set.

**Note:** This must be called before LoadRules if the ruleset references functions in the global set.

- <file path> is the path to where the set resides on disk.

Example:

```
openGlobalSet C:/MyRiverWareFiles/MyGlobalFunctions.gfs
```

#### • OpenWorkspace

```
openWorkspace <file path>
```

Opens a model.

- <file path> is the path to where the model resides on disk.

Example:

```
openWorkspace C:/MyRiverWareFiles/BasinOperationsModel.mdl
```

#### • Output

```
Output <outputDeviceName>
```

Generates an output file that is specified in the Output Manager of the model. See [“Output Manager” in Output Utilities and Data Visualization](#) for details about the available output devices and how they can be configured. Plot pages cannot be generated from a RCL script.

- <outputDeviceName> is the name of the output device saved in the Output Manager of the model.

Examples:

```
Output OperationsForecastReport
```

```
Output FloodStorageAllocationChart
```

#### • RequiredInfo

```
RequiredInfo <message>
```

This command adds the specified message to the diagnostic output as a Required Information message. It is a means to allow you to specify any text in the RCL script to be added to the diagnostic output file.

- `<message>` is any user-specified text string. The text must be enclosed in braces `{ }` if it includes spaces.

Example:

```
RequiredInfo {Executing a planning run}
```

Output in the diagnostic output file:

```
_REQINFO_:      "Executing a planning run"
```

## • **RequireVersion**

```
RequireVersion <major version>.<minor version>[.<patch level>]
```

Specifies the RiverWare version required to execute the batch script. If the RiverWare version is earlier than the required version, the command issues an error message and stops the script file's execution.

- `<major version>.<minor version>[.<patch level>]` are the major version, minor version and optional patch level of RiverWare necessary to execute this batch script.

Example

```
RequireVersion 7.3.1
```

If the version of RiverWare is earlier than 7.3.1, the batch script will fail with an error message.

## • **ReorderGoalSet**

```
ReorderGoalSet <order slot>
```

This command allows you to reorder items (goals, groups, or certain statements) within the currently loaded optimization goal specified set. The desired order is specified by the single column order slot, a table slot, each of whose rows corresponds to the name of an item in the goal set. The values in the single column of the order slot are interpreted as integers giving the desired order of that object with respect to other set items referred to in the table. This command is similar to the Reorder RPL Set Script action; see [“Reorder RPL Set”](#) on page 66.

Example:

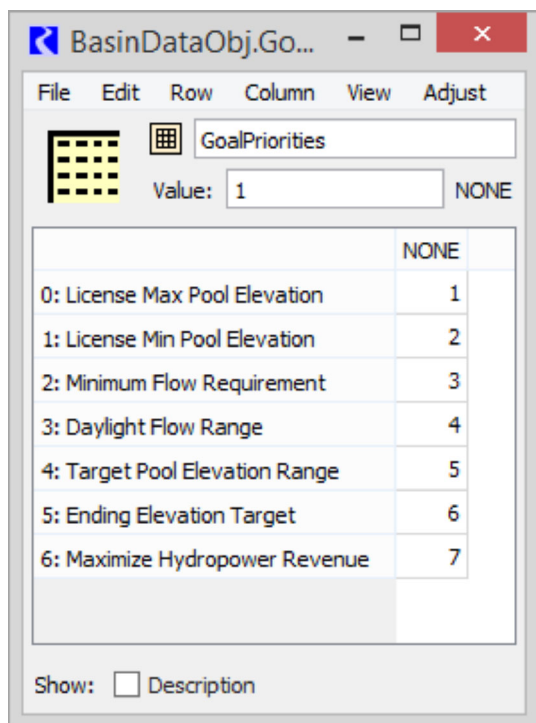
```
ReorderGoalSet BasinDataObj.GoalPriorities
```

The command refers to the table slot shown in [Figure 2.2](#).

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**Figure 2.2** Table slot with priorities to reorder the goal set

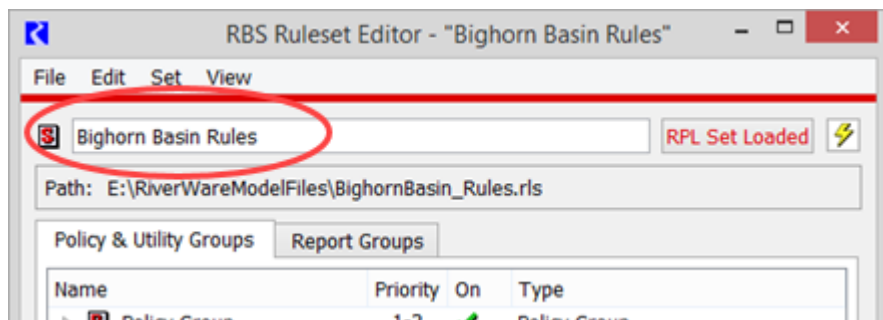


#### • SaveRplSet

SaveRplSet <Set Name> <Application Type> <File Path> [DoNotOverwrite]

This command allows you to save to a file a RPL set that is currently open in the model. If the specified RPL set is not open in the model, the batch script will fail with an error message.

- <Set Name> is the name of the RPL set to be saved. If the set name includes spaces, it must be enclosed in braces {}. This is the name displayed at the top of the RPL Set Editor, *not* the file name.



- <Application Type> refers to the type of RPL set. If the application type includes spaces, it must be enclosed in braces {}. The valid options are  
Global Functions  
Initialization Rules  
Optimization



### Rule Based Simulation

- **<File Path>** is the path to where the RPL set is saved. If the file already exists, the command will overwrite the existing file.
- **[DoNotOverwrite]** is an optional argument. If present, the command will not overwrite an existing file.

Examples:

```
SaveRplSet {Bighorn Basin Rules} {Rule Based Simulation} C:\MyFiles\Bighorn.rls
SaveRplSet {Bighorn Opt Goals} Optimization C:\MyFiles\BighornOpt.opt
```

### • SaveWorkspace

SaveWorkspace [file path]

Saves the model that is currently open.

- **[file path]** is the path to where the model is to be saved on disk. If the file path is not specified, the **OpenWorkspace** file path is used.

Examples:

SaveWorkspace

Saves the model to the same file name from which it was opened

SaveWorkspace C:/MyOutputs/SavedModelFile.mdl

Saves the model to the file specified

### • SetDiagFile

SetDiagFile <diagnostic file>

Sets the location of the diagnostic output file.

- **<diagnostic file>** is the path to where the diagnostic file will be written.

### • SetEnv

SetEnv <variable> <value>

Sets an environment variable for use within the RiverWare run.

See “[Environment Variables](#)” on page 95 for details on setting and using environment variables.

### • SetRunInfo

SetRunInfo <#RunInfo|#MRM <MRM configuration>>

[!InitDate <date time>]

[!StartDate <date time>]

[!EndDate <date time>]

[!Duration <delta time>]

[!Controller <controller type>]

## Chapter 2

### Batch Mode and RiverWare Command Language

Sets single run information (#RunInfo) or multiple run information (#MRM). If #MRM is specified, you also need to specify an MRM configuration <MRM configuration>.

- **!InitDate** sets the run's initial date to <date time> of format {MM-DD-YYYY HH:MM}.
- **!StartDate** sets the run's start date to <date time> of format {MM-DD-YYYY HH:MM}.  
**Note:** If you specify both **!InitDate** and **!StartDate** with the same **SetRunInfo** command, the argument that appears second will take precedence.
- **!EndDate** sets the run's end date to <date time> of format {MM-DD-YYYY HH:MM}.
- **!Duration** sets the run's duration to <delta time> of format {Number TimeUnits}.
- **!Controller** sets the run's controller to <controller type>. The controller type must be one of the controllers available in the Run Control dialog:
  - Simulation
  - Rulebased Simulation
  - Optimization**Note:** Controller types that contain embedded white space must be enclosed in braces {}.

Following are samples of the syntax:

```
SetRunInfo #RunInfo !InitDate {04-21-2011 24:00} !EndDate {04-26-2011 24:00}
SetRunInfo #RunInfo !StartDate {04-22-2011 24:00}
SetRunInfo #RunInfo !EndDate {04-26-2011 24:00}
SetRunInfo #RunInfo !Duration {200 Days}
SetRunInfo #RunInfo !Controller {Rulebased Simulation}
SetRunInfo #MRM "Paleo Runs"
```

#### • **SetSlot**

**SetSlot** <Object.Slot> [<date time>] <value>

Sets the series or scalar slot's value to the value, assumed to be in user scale and units. **Object.Slot** should be enclosed in curly braces {} if it contains embedded white space.

- <date time> is only for SeriesSlots and must be enclosed in curly braces {} as it contains embedded white space - for example, {05-15-2010 12:00}.

See [“GetSlot”](#) on page 100, for details on possible usage.

#### • **SetTrace**

**SetTrace** <1|0>

Enable or disable trace messages from the RCL interpreter. The trace messages are prefaced with “TRACE:” and written to standard output; they can be captured in a file with the **--log <log file>** command line option.

#### • **SlotList**

**SlotList** <output file>

Writes information about all slots in the model to the output file.

- `<output file>` is the file the slot list is written to.

The information is written as comma-separated values to the output file; the following fields are written.

- Inno: line number, always 0 (included for compatibility)
- Section: the containing object's meta-type, one of SimObj, Account, Supply, Exchange or Payback)
- O: the containing object's name
- Otype: the containing object's type, for example Reach
- S: the slot's name
- Stype: the slot's type
- UnitType: the slot's unit type, for example Flow
- UnitScale: the slot's scale
- Units: the slot's unit, for example cfs
- UnitFmt: the slot's format, one of %d %f or %e
- UnitSpec: the slot's display precision
- DataType: the slot's data type, one of OneInit, MultiInit AllInput, AllOutput or Other

### • SlotListDMI

```
SlotListDMI <DMI name> <output file>
```

Writes information about the slots to/from which the DMI imports/exports data; the information is written as comma-separated values to the output file. The information includes:

- The slot's name.
- The slot's priority (determined by its dataset association).
- The slot's begin date "mm-dd-yyyy".
- The slot's begin time "hh:mm:ss".
- The slot's end date "mm-dd-yyyy".
- The slot's end time "hh:mm:ss".
- The dataset associated with the slot .
- The dataset's type, currently DSS or HDB. If the Dataset is DSS, then the slot's DSS path "/A/B/C/D/E/F/" is also provided as another comma separated value.

### • StartController

```
StartController [!MRM <MRM configuration>]  
[firstTrace=N]  
[numTrace=M] ctlFile=file]
```

## Chapter 2

### Batch Mode and RiverWare Command Language

Starts the current RiverWare controller.

- `!MRM` will start the Multiple Run Management controller instead of the single run controller, using the specified MRM configuration. RCL currently does not support many commands for defining multiple runs, so models should already contain the multiple run definitions.

The following commands override the values in the MRM configuration in the model file (`[!MRM <MRM configuration>]` must be used if you wish to specify these parameters):

- `[firstTrace =N]` sets the index sequential initial offset
- `[numTrace=M]` sets the index sequential number of runs
- `[ctlFile=file]` sets the output control file.

#### • SyncObj

```
SyncObj [!Acct] [!ExDiffTS]
[!StartDate <start date>]
[!EndDate <end date>]
```

Synchronize slots to either the Run Control or the specified dates.

- `[!Acct]` includes accounting slots.
- `[!ExDiffTS]` excludes slots whose timestep size differs from the Run Control timestep size.
- `[!StartDate <start date>]` synchronizes slots to the specified start timestep, rather than the Run Control initial timestep.
- `[!EndDate <end date>]` synchronizes slots to the specified end timestep, rather than the Run Control finish timestep.

**Note:** All series slots on the specified objects will be synchronized, except custom slots configured to Prevent Time Synchronization as described in [“Time Series Range—Series Slot” in User Interface](#).

Examples:

`SyncObj`

Synchronizes all series slots to the Run Control initial timestep and finish timestep.

```
SyncObj !ExDiffTS !StartDate {04-23-2013 12:00} !EndDate {04-26-2013 12:00}
```

Synchronizes series slots to the specified start and end timesteps and excludes slots that have a timestep size that differs from the Run Control timestep size.